

# REPRODUCIBLE RAMSEY ANALYSIS ACROSS ASYMPTOTIC AND FINITE REGIMES: RETAINED SPINES, GAUSSIAN RESIDUALS, AND PROOF-CARRYING BARRIERS

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ABSTRACT. We present a unified reproducible account of three logically independent results in Ramsey theory. First, relative to version-pinned regularization, parameterized-book, and spine-compatibility interfaces and a locally certified six-stage off-diagonal rate theorem, an exact-diagonal correlation certificate yields

$$R(k, k) \leq (3.780685290)^{k+o(k)},$$

with certified unrounded base below 3.780685288379640114. The compact inequalities are enclosed by 512-bit ball arithmetic and replayed by four separately structured implementations sharing the same interval-arithmetic trust dependency. The positive-moment and tail transfer used there are proved locally. Second, after fixing the absolute source constant  $K$ , and conditional on the version-pinned lower-interface items (S1), (S2), (S4), and (S5), together with the additional weighted-extension hypothesis (S3), a deterministic nonexchangeable triangular residual ledger supplies, for every sufficiently large fixed  $C$ , a positive source-relative term

$$\hat{H}_*(C) = \frac{1 + o_{C \rightarrow \infty}(1)}{64 \log C}$$

in the fixed-ratio lower-bound exponent. The threshold  $C_0(K)$  is existential. The coefficient  $1/64$  is a cap only within the specified scalar residual class, and no comparison across incompatible terminal normalizations is asserted. Item (S3) is not attributed as a theorem stated by HMS or Lin–Niu. Third, for a content-addressed 100-vertex  $R(3, 18)$  near miss, semantic CNF reconstruction and six checked DRAT refutations prove that its one-sided deletion repair radius under free additions is at least seven. The exact-seven layer remains UNKNOWN; this local theorem does not imply  $R(3, 18) \geq 101$ . The common contribution is an auditable separation of imported interfaces, analytic arguments, interval certificates, proof-carrying finite computations, and unresolved searches, with content-addressed artifacts and tiered reproduction entry points.

[2020]05D10, 05C35, 68V20 Ramsey numbers, computer-assisted proof, interval arithmetic, Gaussian random graph, SAT, DRAT, reproducibility

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## 1. INTRODUCTION AND MAIN RESULTS

For positive integers  $k, \ell$ , the Ramsey number  $R(k, \ell)$  is the least  $n$  such that every red–blue colouring of  $K_n$  contains a red  $K_k$  or a blue  $K_\ell$ . The classical upper and lower exponential scales originate in the work of Erdős–Szekeres and Erdős [ES35, Erd47]. Modern improvements combine delicate analytic inequalities, asymptotic reductions, and computer-assisted certificates. In the finite regime, SAT solvers can explore highly constrained graph spaces, but a solver outcome becomes a theorem only after the graph-to-CNF reduction and a proof trace have both been checked.

This article places three results developed here in one manuscript. They concern different regimes and use different mathematics; their unity is a common claim-to-evidence discipline rather than a claimed common proof mechanism. The first two give conditional, source-relative refinements of asymptotic ledgers, whereas the third establishes a local finite obstruction without improving a tabulated Ramsey number.

**1.1. Diagonal upper bound.** Part I combines a certified six-stage off-diagonal rate, an exact-diagonal correlation inequality, and the version-pinned regularization, parameterized-book, and spine-compatibility interfaces of Yang and Mao [YM26]. The positive-moment and tail transfer is proved locally. Under the source interfaces stated in source interface 4.2 and the

certified rate theorem Theorem 5.1, it proves

$$R(k, k) \leq (3.780685290)^{k+o(k)},$$

with certified unrounded base less than 3.780685288379640114. This is an asymptotic, source-relative, computer-assisted theorem. It supplies neither an effective finite- $k$  threshold nor a global optimum or priority claim.

**1.2. Fixed-ratio lower bound.** Part II assumes the version-pinned items (S1), (S2), (S4), and (S5) imported from the Gaussian construction of Hunter, Milojević and Sudakov and the exact cumulant input of Lin and Niu [HMS25, LN26], together with the separately stated weighted-extension hypothesis (S3). Item (S3) is not attributed as a theorem stated in either source. After fixing the absolute source constant  $K$ , those five explicit items give an existential threshold  $C_0 = C_0(K) > 1$  such that, for each fixed  $C \geq C_0$ ,

$$\liminf_{\ell \rightarrow \infty} \frac{1}{\ell} \log R(\ell, \lfloor C\ell \rfloor) \geq -\frac{1}{2} \log p_C + \frac{B_R(C)}{2} + G_*(C) + \hat{H}_*(C),$$

where

$$\hat{H}_*(C) = \frac{1 + o_{C \rightarrow \infty}(1)}{64 \log C} > 0.$$

The order of limits is part of the statement and proof: after fixing  $K$  and  $C$ , one takes  $\ell \rightarrow \infty$ , then sends the auxiliary cutoff  $w \downarrow \omega_0$ , and only afterwards studies the leading behaviour as  $C \rightarrow \infty$ . The threshold  $C_0(K)$  is non-effective, and the theorem does not assert a strict comparison with every published final constant under a different normalization.

**1.3. Finite proof-carrying barrier.** Part III fixes a labelled, content-addressed graph  $H$  on 100 vertices with 827 edges, no independent set of order 18, and one triangle. If additions outside  $E(H)$  are free and only deletions from  $E(H)$  are charged, semantic reconstruction and six DRAT refutations prove that every graph  $F$  on the same labelled vertex set that is triangle-free and satisfies  $\alpha(F) < 18$  obeys  $|E(H) \setminus E(F)| \geq 7$ . Exact seven remains UNKNOWN. The theorem is local to this seed and metric: it neither constructs a 100-vertex  $(3, 18)$ -Ramsey graph nor implies  $R(3, 18) \geq 101$ .

**1.4. Organization.** Section 2 states the evidence taxonomy and trust boundaries used throughout. Parts I–III give the complete arguments and their result-specific verification records. The unified reproduction contract appears in Section 26; full technical records, proof ledgers, and artifact tables are retained in the appendices. Symbols introduced inside a Part are local to that Part unless explicitly declared otherwise. This convention prevents unrelated analytic functions, Gaussian potentials, and finite graphs from being conflated.

## 2. SOURCE INTERFACES AND EVIDENCE TAXONOMY

The three results have different logical dependencies. We therefore attach an evidence type to every principal statement instead of using “computer-assisted” as a single undifferentiated label.

**Definition 2.1** (Evidence types). The manuscript uses the following five categories.

- (i) A *source interface* is either a version-pinned statement imported from an external or frozen local source or a separately identified extra hypothesis. Its justification is outside the current logical kernel unless reproduced in an appendix.
- (ii) An *analytic theorem* follows from written mathematical arguments once its named source interfaces are granted.
- (iii) An *interval-certified theorem* additionally depends on outward-rounded enclosures covering a stated continuum.
- (iv) A *proof-carrying SAT theorem* depends on a proved encoding, exhaustive case coverage, and independently replayed proof traces.
- (v) A *bounded observation* records a terminated computation without promoting SAT, UNSAT, or UNKNOWN beyond what its certificate establishes.

Table 1: Claims, evidence, and non-implications.

| ID | Claim                                                                          | Evidence and dependency                                                       | Required boundary                                                                            |
|----|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| U0 | Six-stage off-diagonal rate $U = F_6$ , with $e^{U(1)} = 3.780698427796 \dots$ | Repaired book induction and continuum interval certificates.                  | A locally replayed computer-assisted rate theorem, not a proof-assistant formalization.      |
| U1 | Exact-diagonal correlation certificate at the stated rational parameters.      | Analytic ratio, separator and tail arguments plus 512-bit compact enclosures. | Search selected the point; the frozen checker, not the search grid, proves it.               |
| U2 | Conditional diagonal base 3.780685290.                                         | U0, U1, and the pinned retained-spine interfaces.                             | Asymptotic and source-relative; no effective threshold, global optimum, or world-best claim. |

| ID        | Claim                                                                                               | Evidence and dependency                                                                                                                                                                       | Required boundary                                                                                                                                                                                                                                                                         |
|-----------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L0–<br>L3 | Exact-merge rigidity and the controlled-residual addition $\hat{H}_*(C) = (1 + o(1))/(64 \log C)$ . | Written argument conditional on pinned items (S1), (S2), (S4), (S5), plus the additional weighted-extension hypothesis (S3); arithmetic replay checks formulas only and is not the induction. | Item (S3) is not attributed as an HMS/Lin–Niu theorem. Fix the source constant $K$ and sufficiently large $C$ , then take $\ell \rightarrow \infty$ , $w \downarrow \omega_0$ , and only afterwards $C \rightarrow \infty$ ; $1/64$ is a cap only in the specified scalar residual class. |
| F0–<br>F1 | The fixed seed has the stated graph properties and deletion repair radius at least seven.           | Direct bitset checks, branch completeness, CNF semantics, and six replayed DRAT proofs.                                                                                                       | Local seed and one-sided metric; no implication $R(3, 18) \geq 101$ .                                                                                                                                                                                                                     |
| F2        | Three exact-seven runs ended UNKNOWN.                                                               | Bounded solver logs.                                                                                                                                                                          | Neither existence nor nonexistence evidence.                                                                                                                                                                                                                                              |

Shared trust is not independence. The upper-bound replays use separately structured implementations but share Arb or python-flint containment semantics and the same mathematical specification. The finite proof traces are checked independently of the SAT solver, but their theorem still depends on the encoding and case-cover proofs. The lower arithmetic checker replays identities and asymptotic margins; it does not replace the reverse-induction and extraction proof.

Version and hash discipline. All imported source snapshots, frozen parameter files, graph matrices, CNF instances, proof traces, and checkers used as evidence are identified by version or SHA-256 digest. A digest authenticates bytes, not mathematics; the relevant semantic implication is stated separately in each Part.

## Part 1. A retained-spine diagonal upper bound

### 3. DIAGONAL UPPER BOUND: THEOREM AND PROOF ARCHITECTURE

For positive integers  $k, \ell$ , let  $R(k, \ell)$  be the least  $n$  such that every red–blue colouring of  $K_n$  contains either a red  $K_k$  or a blue  $K_\ell$ . The classical argument of Erdős and Szekeres gives  $R(k, k) \leq 4^{k+o(k)}$  [ES35]. Campos, Griffiths, Morris and Sahasrabudhe obtained the first exponential improvement of this

base [CGMS23]. Gupta, Ndiaye, Norin and Wei subsequently reorganized the book method and stated the concrete bound  $R(k, k) \leq 3.8^{k+o(k)}$  [GNNW24]. Yang and Mao introduced a retained-spine refinement in a multicolour setting [YM26].

This part combines two ingredients. The first is a certified, symmetric off-diagonal rate

$$\log R(k, \ell) \leq kU(\ell/k) + o(k), \quad 1 \leq \ell \leq k, \quad (\text{U1.1})$$

where  $U = F_6$  is given explicitly in Section 5. The second is a new correlation certificate designed for the retained-spine transfer. The certificate is analytic except on compact intervals, where every inequality is proved by outward-rounded Arb balls [Joh17].

**3.1. Main result.** We state the result with its imported interfaces visible. The exact interfaces are recorded in Section 4; they are theorem statements from Yang–Mao together with the reviewed rate theorem (U1.1), rather than unstructured assumptions about an implementation.

**Theorem 3.1** (Conditional main theorem). *Assume the version-pinned regularization, parameterized-book, and spine-compatibility interfaces stated in source interface 4.2, and the certified rate theorem Theorem 5.1. The positive-moment and tail implication used to establish the correlation input are proved locally in Section 6. Then*

$$R(k, k) \leq \exp\left((U(1) - 3.4754 \cdot 10^{-6})k + o(k)\right) \leq (3.780685290)^{k+o(k)}.$$

The certified unrounded upper base is less than 3.780685288379640114.

This part is a source-relative proof package: every imported statement is named, every new inequality is proved here, and every numerical step is replayable by separately structured implementations. Those implementations share Arb’s interval-containment semantics as a trust dependency. No finite value of the asymptotic threshold is claimed.

**3.2. The exact-diagonal mechanism.** Fix

$$u_0 = \frac{1235783}{500000}, \quad L = u_0^3, \quad a = e^{-u_0},$$

and define

$$E(z) = \sum_{n \geq 0} \frac{z^n}{(3n)!}, \quad H(z) = 1 + aE(z)^2, \quad G(z) = \frac{H(z) - H(-z)}{2}.$$

The correlation form is

$$F(x, y) = G(x)H(y) + G(y)H(x).$$

Our central observation is that the complete two-dimensional problem reduces to the exact identity

$$F(z, z) = H(z)^2 - H(z)H(-z), \quad z \geq 0,$$



not merely to the larger square difference  $H(z)^2 - H(-z)^2$ . This improves the compact envelope while preserving a simple analytic tail.

**3.3. Proof architecture.** The proof has four logically separate layers.

- (i) Section 5 states the off-diagonal rate and records its exact polynomial and verification boundary.
- (ii) Section 6 proves the ratio/seperator, the two-dimensional reduction, the exact diagonal envelope and the moment/tail budget.
- (iii) Section 7 proves the retained-spine max–min transfer and inserts the certified parameters.
- (iv) Section 8 describes continuum coverage, independent implementations and the content-addressed evidence records.

The complete repaired book induction and the frozen rate record are included in the appendices. Search scripts are not part of the proof; they only chose the exact rational data subsequently verified from scratch.

#### 4. RAMSEY RATES AND IMPORTED INTERFACES

**4.1. Symmetric homogeneous rate.** Let  $U : [0, 1] \rightarrow \mathbb{R}$  be continuous with  $U(0) = 0$ . Define its symmetric homogeneous extension by

$$\widehat{U}(x, y) = \begin{cases} \max\{x, y\} U\left(\frac{\min\{x, y\}}{\max\{x, y\}}\right), & \max\{x, y\} > 0, \\ 0, & x = y = 0. \end{cases} \quad (\text{U2.1})$$

For residual problems only, we extend the notation by  $R(a, b) = 1$  when  $\min\{a, b\} \leq 0$ ; this means that no further page vertices are needed in at least one colour. It is not a new convention for positive Ramsey numbers.

**Lemma 4.1** (Uniform homogeneous extension). *Assume in addition that  $U \geq 0$ . Suppose (U1.1) holds uniformly for  $1 \leq \ell \leq k$ , and the same bound is available after exchanging the two colours. Then, uniformly for  $x, y \in [0, 1]$ ,*

$$\log R(\lfloor xk \rfloor, \lfloor yk \rfloor) \leq k\widehat{U}(x, y) + o(k). \quad (\text{U2.2})$$

*Proof.* The extension  $\widehat{U}$  is continuous on the square (continuity at the origin uses  $U(0) = 0$ ), hence uniformly continuous. Fix  $\epsilon > 0$ , and choose  $K_0$  from the uniform epsilon formulation of (U1.1). For  $x \geq y$ , put  $k' = \lfloor xk \rfloor$  and  $\ell' = \lfloor yk \rfloor$ . If  $k' \geq K_0$  and  $\ell' \geq 1$ , then

$$\log R(k', \ell') \leq k'U(\ell'/k') + \epsilon k' = k\widehat{U}(k'/k, \ell'/k) + \epsilon k.$$

Uniform continuity and the  $O(k^{-1})$  perturbation of the normalized coordinates replace the last  $\widehat{U}$  by  $\widehat{U}(x, y) + o(1)$ , uniformly. If  $k' < K_0$ , only finitely many positive Ramsey numbers occur, so their logarithms are  $O_{K_0}(1) = o(k)$ ; nonnegativity of  $\widehat{U}$  permits the same upper bound. If one floored coordinate is zero, the residual convention gives logarithmic cost zero. The case  $y \geq x$  follows by symmetry. Since  $\epsilon$  is arbitrary, these cases give one uniform  $o(k)$ .  $\square$

**4.2. Yang–Mao interfaces.** The following package is extracted from the v1 source of Yang–Mao [YM26]. Its source member and line map are frozen in Appendix A. We state the complete specialization used later, so that the conditional theorem is falsifiable without guessing what the words “book interface” mean.

For completeness, we fix the precise meaning of the correlation input. We write  $\mathcal{G}_2^{(3)}(\beta, C)$  when the following assertion holds: for every finite set  $\mathcal{X}$ , independent identically distributed  $U, U' \in \mathcal{X}$ , real Hilbert spaces  $\mathcal{H}_1, \mathcal{H}_2$ , and maps  $\sigma_i : \mathcal{X} \rightarrow \mathcal{H}_i$ , if  $Z_i = \langle \sigma_i(U), \sigma_i(U') \rangle$ , then there are  $i \in \{1, 2\}$  and  $\lambda \geq -1$  such that

$$\mathbb{P}(Z_i \geq \lambda, Z_{3-i} \geq -1) \geq \beta \exp(-C(\lambda + 1)^{1/3}). \quad (\text{U2.2a})$$

For  $0 < \beta \leq 1$ ,  $C > 0$ , and parameters  $p, \delta, \lambda_0 > 0$ , put

$$L_\delta = \log(1/\delta), \quad L_p = \log(1/p), \quad \rho_\beta = \log(2/\beta),$$

$$\Pi = \frac{3\delta}{p} + \frac{6L_\delta L_p}{\lambda_0}, \quad q = L_p + \Pi, \quad (\text{U2.3})$$

and

$$\Xi = 2\rho_\beta + 4C\lambda_0^{1/3} + \frac{12CL_\delta}{\lambda_0^{2/3}}. \quad (\text{U2.4})$$

**Source interface 4.2** (Retained-spine source interface). Fix  $0 < \eta$ ,  $0 < p < 1/2 - \eta$ ,  $0 < \delta \leq \min\{p/4, 1/4\}$ ,  $\lambda_0 \geq \max\{2, 6\log(1/\delta)\}$ , and  $\tau > 0$ . Suppose the correlation property  $\mathcal{G}_2^{(3)}(\beta, C)$  holds. The version-pinned Yang–Mao specialization supplies the following conclusions for all sufficiently large  $k$ .

- (a) For every positive integer  $N$ , every red–blue colouring of  $K_N$  admits pairwise disjoint sets  $S_R, S_B, W$  such that  $S_R$  is a red clique,  $S_B$  is a blue clique, every edge from  $S_R$  to  $W$  is red, and every edge from  $S_B$  to  $W$  is blue. Writing  $s_i = |S_i|$ , one has

$$|W| \geq \left(\frac{1+\eta}{2}\right)^{s_R+s_B} N \quad (\text{U2.5})$$

and every  $w \in W$  has at least  $(1/2 - \eta)|W| - 1$  red neighbours and at least that many blue neighbours inside  $W$ .

- (b) Let  $t, m$  be positive integers with  $t \geq \lambda_0 \delta^{-1/2}$ . If every  $w \in W$  has at least  $p|W|$  neighbours of each colour inside  $W$ , and

$$|W| \geq p^{-t} e^{\Pi t} m, \quad |W| \geq 4t e^{2t\Xi}, \quad (\text{U2.6})$$

then the parameterized-book theorem, applied with  $X = Y_R = Y_B = W$ , returns a colour  $i \in \{R, B\}$ , a monochromatic  $t$ -clique  $T \subseteq W$  of colour  $i$ , and  $P_0 \subseteq W \setminus T$ ,  $|P_0| \geq m$ , such that every edge from  $T$  to  $P_0$  has colour  $i$ .

- (c) The compatibility conclusion is the following exact residual statement. If  $i = R$ , then a red  $K_{k-s_R-t}$  in  $P_0$  (when this order is positive) extends with  $S_R \cup T$ , while a blue  $K_{k-s_B}$  extends with  $S_B$ . If  $i = B$ , interchange the colours, so the two residual orders are  $k - s_R$  and  $k - s_B - t$ . A nonpositive residual order means that the retained spines already contain the required  $K_k$ .

For the proof below the exact correlation input is

$$\beta = \frac{330867}{500000}, \quad C = \frac{12366348252219}{1250000000000}. \quad (\text{U2.7})$$

The point of stating source interface 4.2 is auditability. The correlation property in (U2.7) is proved in this paper; the regularization and book compatibility are imported. The source line mapping and pinned archive hash are included in the reproducibility record.

**4.3. A general retained-spine transfer.** Put

$$c_\eta = \log \frac{2}{1 + \eta}.$$

For a normalized spine vector  $\sigma = (\sigma_R, \sigma_B) \in [0, 1]^2$ , define

$$D(\sigma) = c_\eta(\sigma_R + \sigma_B) + \widehat{U}(1 - \sigma_R, 1 - \sigma_B). \quad (\text{U2.8})$$

Here  $q$  and  $\Xi$  are the explicit quantities in (U2.3)–(U2.4). Define

$$M_\tau(\sigma) = \max\{\widehat{U}((1 - \sigma_R - \tau)_+, 1 - \sigma_B), \widehat{U}(1 - \sigma_R, (1 - \sigma_B - \tau)_+)\}, \quad (\text{U2.9})$$

$$P(\sigma) = c_\eta(\sigma_R + \sigma_B) + \tau q + M_\tau(\sigma), \quad (\text{U2.10})$$

$$Q(\sigma) = c_\eta(\sigma_R + \sigma_B) + 2\tau\Xi, \quad (\text{U2.11})$$

$$B(\sigma) = \max\{P(\sigma), Q(\sigma)\}. \quad (\text{U2.12})$$

Finally set

$$C_* = \max_{\sigma \in [0, 1]^2} \min\{D(\sigma), B(\sigma)\}. \quad (\text{U2.13})$$

**Theorem 4.3** (Retained-spine transfer). *Under source interface 4.2 and the uniform rate (U2.2),*

$$R(k, k) \leq \exp(C_* k + o(k)).$$

*Proof.* Let  $N = \lceil e^{(C_* + \varepsilon)k} \rceil$  and apply regularization. If either preliminary spine already has order at least  $k$ , there is nothing to prove. Otherwise write  $s_i = \sigma_i k + O(1)$ . From (U2.5),

$$\log |W| \geq (C_* + \varepsilon)k - c_\eta(s_R + s_B) + o(k). \quad (\text{U2.14})$$

At this  $\sigma$ , choose the smaller of the direct and book branches.

If  $D(\sigma) \leq B(\sigma)$ , then  $D(\sigma) \leq C_*$ . Equations (U2.2), (U2.8) and (U2.14) imply  $|W| \geq R(k - s_R, k - s_B)$ . The resulting red or blue clique extends with the corresponding preliminary spine.

If  $B(\sigma) < D(\sigma)$ , then both  $P(\sigma) \leq C_*$  and  $Q(\sigma) \leq C_*$ . The former supplies the page size needed for either possible book colour, using (U2.2) and (U2.9); the latter supplies the common reservoir threshold. More explicitly, with  $t = \lfloor \tau k \rfloor$  and  $m$  the larger of the two residual Ramsey thresholds, the two strict exponent inequalities give exactly (U2.6). Since  $p < 1/2 - \eta$  and the reservoir is exponential, the additive  $-1$  in the degree conclusion is absorbed; fixed  $\tau > 0$  also gives  $t \geq \lambda_0 \delta^{-1/2}$ . The parameterized-book interface produces the  $t$ -spine and a page set of size at least  $m$ , and compatibility extends the appropriate page clique with the retained spines to a monochromatic  $K_k$ . The fixed  $\varepsilon$ -slack absorbs every uniform  $o(k)$  term.  $\square$

## 5. THE CERTIFIED OFF-DIAGONAL RATE

Set

$$h(\lambda) = (1 + \lambda) \log(1 + \lambda) - \lambda \log \lambda$$

and

$$\begin{aligned} P_6(\lambda) = & -0.2500000000000000\lambda + 0.019891840059418\lambda^2 \\ & -0.012275954247190\lambda^3 + 0.144110816398083\lambda^4 \\ & + 0.006277913420654\lambda^5 - 0.066101623491668\lambda^6 \\ & - 0.002287675993602\lambda^7 - 0.058238059505066\lambda^8 \\ & + 0.030675864198693\lambda^9 + 0.052472201910597\lambda^{10} \\ & + 0.043300454861853\lambda^{11} - 0.042529975074653\lambda^{12} \\ & - 0.055263639426635\lambda^{13} + 0.036695886931268\lambda^{14}. \end{aligned} \quad (\text{U3.1})$$

All displayed decimals in (U3.1) are exact. Define

$$U(\lambda) = F_6(\lambda) = h(\lambda) + P_6(\lambda)e^{-\lambda}. \quad (\text{U3.2})$$

**Theorem 5.1** (Certified rate, conditional on the stated trust boundary). *For  $1 \leq \ell \leq k$ , uniformly in the ratio  $\lambda = \ell/k$ ,*

$$\log R(k, \ell) \leq kU(\lambda) + o(k), \quad (\text{U3.3})$$

*and the same statement holds after exchanging  $k$  and  $\ell$ . Moreover*

$$e^{U(1)} = 3.7806984277961236987518816497522162 \dots \quad (\text{U3.4})$$

*Proof.* The certificate schema, all continuum obligations, and the theorem-to-rate implication are stated and proved in Proposition C.3. The exact-linked replay accepts all six certificate files, proves the elementary first prior in Lemma C.2, and verifies exact target-to-prior equality at the other five handoffs. Its sixth report gives  $F_6(1) = 1.3299087618219930723187812632 \dots$ ; outward-rounded exponentiation gives (U3.4). Outside the version-pinned Yang–Mao interfaces in source interface 4.2, the repaired BookCor induction is proved locally in Appendix B; the remaining machine trust is Arb’s interval-containment semantics.  $\square$

The proof is an exact-linked six-stage descent built on the repaired book-induction lemma. At each stage an interval certificate verifies the prior concavity, both Ramsey-region orientations, the main exponent inequality and the complete small-ratio interval. The first three stages use 65,536 cells; the last three use 131,072 cells. The sixth-stage minimum main slack is  $4.81847 \cdot 10^{-5}$ , while an independently implemented direct exponent check gives a minimum standard-orientation margin  $1.06281 \cdot 10^{-6}$ .

For the reader who wishes to separate mathematics from computation, the logical content is as follows.

- (i) The combinatorial book induction converts a strict interior point of the asymptotic Ramsey region into a one-step rate improvement.
- (ii) The analytic descent lemma is degree-independent: it requires only a concave prior rate and strict inequalities on the continuum.
- (iii) Every prior polynomial equals the preceding target polynomial as an exact decimal tuple; no refitting occurs between stages.
- (iv) Arb intervals enclose the continuum, not a sampled mesh. Search meshes are used only to propose a target.

The complete standalone book proof, certificate-correctness theorem, and stage-six record appear in Appendices B and C. The six JSON certificates and their hashes are part of the machine-readable evidence package.

## 6. THE EXACT-DIAGONAL CORRELATION CERTIFICATE

We now prove the new analytic input. Recall

$$u_0 = \frac{1235783}{500000}, \quad L = u_0^3, \quad a = e^{-u_0},$$

and

$$E(z) = \sum_{n \geq 0} \frac{z^n}{(3n)!}, \quad H(z) = 1 + aE(z)^2, \quad G(z) = \frac{H(z) - H(-z)}{2},$$

$$F(x, y) = G(x)H(y) + G(y)H(x).$$

**6.1. The bad-event separator.** For  $u \geq 0$ , define the cubic filters

$$P(u) = \frac{e^u + 2e^{-u/2} \cos(\sqrt{3}u/2)}{3}, \tag{U4.1}$$

$$N(u) = \frac{e^{-u} + 2e^{u/2} \cos(\sqrt{3}u/2)}{3}. \tag{U4.2}$$

Then  $E(u^3) = P(u)$  and  $E(-u^3) = N(u)$ . Put

$$\sigma_0 = 10^{-8}, \quad T = 2 + 2\sigma_0, \quad \sigma_* = \frac{1 + 2\sigma_0}{2} = \frac{50000001}{100000000}. \tag{U4.3}$$

**Lemma 6.1** (Half-line ratio). *For every  $u \geq u_0$ ,*

$$\frac{1 + aP(u)^2}{1 + aN(u)^2} > T. \tag{U4.4}$$

*Proof.* Let  $R(u) = (1 + aP(u)^2)/(1 + aN(u)^2)$ . Its derivative has the sign of

$$W(u) = PP'(1 + aN^2) - NN'(1 + aP^2). \quad (\text{U4.5})$$

An outward-rounded enclosure proves  $R(u_0) - T > 2.6595 \cdot 10^{-7}$  and  $W > 0$  on 65,536 rational cells covering  $[u_0, 2.9]$ .

For  $u \geq 2.9$ , use

$$P(u) \geq \frac{e^u - 2e^{-u/2}}{3}, \quad |N(u)| \leq \frac{2e^{u/2} + e^{-u}}{3}.$$

After clearing the positive denominator, it suffices to prove positivity of

$$B_T(u) = e^{2u} - 4Te^u - 4e^{u/2} - 4Te^{-u/2} + 4e^{-u} - Te^{-2u}.$$

At  $u = 2.9$ , interval arithmetic proves  $aB_T - 9(T - 1) > 0$  and the separate base gate

$$D(2.9) > 506.45 > 0.$$

Moreover

$$B'_T(u) \geq D(u) := 2e^u(e^u - 2T) - 2e^{u/2} - 4e^{-u},$$

and

$$D'(u) = 4e^{2u} - 4Te^u - e^{u/2} + 4e^{-u} > 0.$$

Indeed, for  $u \geq 2.9$  one has  $e^u - T > 1$ , and therefore

$$D'(u) > 4e^u - e^{u/2} > 0.$$

Thus  $D(u) > 0$  and then  $B'_T(u) > 0$  throughout  $[2.9, \infty)$ . Together with the endpoint gate, this proves that  $B_T$ , and therefore  $R - T$ , remain positive.  $\square$

**Lemma 6.2** (Separator). *If one of  $A_1, A_2$  is less than  $-1$ , then*

$$F(LA_1, LA_2) < -\sigma_*. \quad (\text{U4.6})$$

*Proof.* Let  $x = -t$  with  $t \geq L$ , put  $A = H(-t)$ , and set  $R = H(t)/H(-t)$ . The exact identity

$$F(-t, y) = \frac{A}{2}((2 - R)H(y) - H(-y)) \quad (\text{U4.7})$$

holds for every real  $y$ . By Lemma 6.1,  $R > 2 + 2\sigma_0$ , whereas  $A, H(y), H(-y) \geq 1$ . Thus the right side of (U4.7) is less than  $-(1 + 2\sigma_0)/2 = -\sigma_*$ . Exchange the coordinates if necessary.  $\square$

## 6.2. Complete two-dimensional reduction.

**Lemma 6.3** (Diagonal reduction). *If  $x, y \geq -L$  and  $m = \max\{x, y, 0\}$ , then*

$$F(x, y) \leq F(m, m). \quad (\text{U4.8})$$

*Proof.* The series for  $H$  has nonnegative coefficients, so  $H$ , and therefore its odd part  $G$ , are nondecreasing on the positive half-line. If  $x = -v^3 \in [-L, 0]$ , the root-filter formula gives

$$|E(-v^3)| \leq e^{v/2}, \quad H(x) \leq 1 + e^{-u_0}e^v \leq 2. \quad (\text{U4.9})$$

Assume  $x \leq y$ . If  $y < 0$ , both odd parts are negative and hence  $F(x, y) < 0 = F(0, 0)$ . If  $x < 0 \leq y$ , discard the nonpositive summand and use (U4.9):

$$F(x, y) \leq G(y)H(x) \leq 2G(y) \leq 2G(y)H(y) = F(y, y).$$

If  $0 \leq x \leq y$ , monotonicity bounds both summands by  $G(y)H(y)$ . This exhausts all sign patterns and their boundaries.  $\square$

On the diagonal we retain the exact identity

$$F(z, z) = 2G(z)H(z) = H(z)^2 - H(z)H(-z), \quad z \geq 0. \quad (\text{U4.10})$$

**6.3. Compact interval and analytic tail.** Let  $z = u^3$  and  $w = (u^3 + u_0^3)^{1/3}$ . Define

$$D_0 = \frac{88053}{10^9}.$$

**Lemma 6.4** (Exact diagonal envelope). *For every  $u \geq 0$ ,*

$$\frac{H(u^3)^2 - H(u^3)H(-u^3)}{e^{4w}} < D_0. \quad (\text{U4.11})$$

*Proof.* The compact interval  $[0, 20]$  is covered by 131,072 exact rational cells. On each cell a 512-bit Arb evaluation encloses the left side of (U4.11); the minimum slack is greater than  $4.2687 \cdot 10^{-6}$ .

For  $u \geq 20$ , use  $H(-u^3) \geq 1$  and write  $H(u^3) = 1 + aP(u)^2$ . Then

$$H(u^3)^2 - H(u^3)H(-u^3) \leq aP(u)^2 + a^2P(u)^4.$$

Since

$$|P(u)| \leq \frac{e^u}{3}(1 + 2e^{-3u/2})$$

and  $w \geq u$ , the normalized expression is at most

$$\frac{a}{9}e^{-2u}(1 + 2e^{-3u/2})^2 + \frac{a^2}{81}(1 + 2e^{-3u/2})^4. \quad (\text{U4.12})$$

Both terms decrease. At  $u = 20$ , (U4.12) is below  $8.805216326983 \cdot 10^{-5}$ , leaving strict slack greater than  $8.3673 \cdot 10^{-10}$ ; its limiting value  $a^2/81$  has the same strict ordering. This proves the entire half-line.  $\square$

Combining Lemmas 6.3 and 6.4, if  $A_1, A_2 \geq -1$  and  $M = \max(A_1, A_2)$ , then

$$F(LA_1, LA_2) < D_0 \exp(4u_0(M + 1)^{1/3}). \quad (\text{U4.13})$$

If  $M \leq 0$ , the left side is nonpositive by the sign case in Lemma 6.3, whereas the right side is positive. If  $M > 0$ , take  $u = u_0 M^{1/3}$ ; then the diagonal-reduction parameter is  $m = LM = u^3$  and the  $w$  in Lemma 6.4 equals  $u_0(M + 1)^{1/3}$ , proving the displayed inequality.

**6.4. Exact moment/tail budget.** The final passage from the separator to a correlation tail is short but proof-critical, so we give it in full rather than treating it as an unnamed interface.

**Lemma 6.5** (Positive-moment expectation). *For any finite set  $\mathcal{X}$ , independent identically distributed  $U, U' \in \mathcal{X}$ , real Hilbert spaces  $\mathcal{H}_1, \mathcal{H}_2$ , and maps  $\sigma_i : \mathcal{X} \rightarrow \mathcal{H}_i$ , put  $Z_i = \langle \sigma_i(U), \sigma_i(U') \rangle$ . Then*

$$\mathbb{E}F(LZ_1, LZ_2) \geq 0. \quad (\text{U4.14})$$

*Proof.* The Taylor coefficients of  $E$  and  $H$  are nonnegative, and taking the odd part of  $H$  merely retains its odd-degree coefficients. Hence every bivariate Taylor coefficient of  $F(x, y) = G(x)H(y) + G(y)H(x)$  is nonnegative. For nonnegative integers  $r, s$ , tensorization gives

$$\mathbb{E}(Z_1^r Z_2^s) = \|\mathbb{E}(\sigma_1(U)^{\otimes r} \otimes \sigma_2(U)^{\otimes s})\|^2 \geq 0.$$

Because  $\mathcal{X}$  is finite,  $(Z_1, Z_2)$  has bounded support; the entire Taylor series is therefore absolutely integrable and may be averaged term by term. Summing the nonnegative terms proves (U4.14).  $\square$

Take

$$\beta = \frac{330867}{500000}, \quad \varepsilon = \frac{6893}{10^7}.$$

Exact rational arithmetic gives

$$\sigma_*(1 - \beta) - D_0\beta(1 + 2/\varepsilon) = \frac{39437634699447}{3446500000000000000} > 10^{-5}. \quad (\text{U4.15})$$

**Proposition 6.6** (Certified correlation tail). *The exact separator, envelope, and budget above imply*

$$\mathcal{G}_2^{(3)}\left(\frac{330867}{500000}, 4u_0(1 + \varepsilon)\right).$$

*Proof.* Fix data from the definition (U2.2a), put  $\mathcal{E} = \{Z_1 \geq -1, Z_2 \geq -1\}$ ,  $M = \max\{Z_1, Z_2\}$ ,  $Y = ((M+1)_+)^{1/3}$ , and  $c = 4u_0$ . By Lemmas 6.2 and 6.5 and (U4.14), while (U4.13) controls the good event,

$$0 \leq \mathbb{E}F(LZ_1, LZ_2) < D_0\mathbb{E}(e^{cY}\mathbf{1}_{\mathcal{E}}) - \sigma_*\mathbb{P}(\mathcal{E}^c). \quad (\text{U4.16})$$

If  $\mathbb{P}(\mathcal{E}) \geq \beta$ , choose  $\lambda = -1$  and either coordinate in (U2.2a). Otherwise suppose, for a contradiction, that the claimed correlation conclusion fails for both coordinates at every  $\lambda \geq -1$ . For  $v \geq 0$ , taking  $\lambda = v^3 - 1$  and using the union bound gives

$$\mathbb{P}(\mathcal{E} \cap \{Y \geq v\}) < 2\beta e^{-c(1+\varepsilon)v}. \quad (\text{U4.17})$$

The layer-cake identity and Tonelli's theorem now give

$$\begin{aligned} \mathbb{E}(e^{cY}\mathbf{1}_{\mathcal{E}}) &= \mathbb{P}(\mathcal{E}) + c \int_0^\infty e^{cv} \mathbb{P}(\mathcal{E} \cap \{Y > v\}) dv \\ &< \beta + 2\beta c \int_0^\infty e^{-c\varepsilon v} dv = \beta(1 + 2/\varepsilon). \end{aligned}$$



Substitution into (U4.16) makes its right-hand side at most  $D_0\beta(1 + 2/\varepsilon) - \sigma_*(1 - \beta) < 0$ , by (U4.15). This contradiction proves the proposition.  $\square$

Finally,

$$4u_0(1 + \varepsilon) = \frac{12366348252219}{1250000000000} = 9.8930786017752,$$

which is the exact handoff asserted in (U2.7).

## 7. THE RETAINED-SPINE CERTIFICATE

The exact outer tuple is

$$\begin{aligned} \eta &= 0.02868896, & p &= 0.47130887, & \delta &= 0.000053863, \\ \lambda_0 &= 13233, & \tau &= 0.00069386, & \Delta &= 0.0000034754. \end{aligned} \quad (\text{U5.1})$$

Every terminating decimal in (U5.1) is interpreted as a rational number.

The functions  $D, B$  and  $C_*$  are those of Theorem 4.3. Direct interval evaluation of the full square would introduce unacceptable variable dependence. The certificate instead uses the geometry of the ordered wedge. Concavity of  $U$  fixes the monotonicity of each page branch; the red and blue page expressions reduce to one-dimensional boundary intervals, and the diagonal reservoir reduces to an endpoint inequality. Every reduction is analytic before numerical evaluation.

**Proposition 7.1** (Outer certificate). *For the exact tuple (U5.1) and the correlation input proved in Section 6,*

$$C_* \leq U(1) - \Delta. \quad (\text{U5.2})$$

*The certified lower margins are as follows.*

| <i>Inequality</i>         | <i>lower margin</i>    |
|---------------------------|------------------------|
| <i>degree gate</i>        | $1.5056 \cdot 10^{-9}$ |
| <i>red page</i>           | $5.1006 \cdot 10^{-9}$ |
| <i>blue page</i>          | $3.4825 \cdot 10^{-6}$ |
| <i>diagonal reservoir</i> | $6.0468 \cdot 10^{-6}$ |

*Proof.* Write  $x = \sigma_R$ ,  $y = \sigma_B$ , and by symmetry first assume  $0 \leq x \leq y \leq 1$ . Put

$$A = U(1) - 2c_\eta, \quad E = U'(1) - c_\eta.$$

The certificate proves  $0 < A < E$ . At  $x = y = 1$ , one has  $U(1) - D(1, 1) = A > \Delta$  because  $r_d = \Delta/A < 1$ , so the direct branch already applies. At every other point of the ordered triangle,  $a = 1 - x \geq b = 1 - y$  has  $a > 0$ , and concavity at one gives

$$aU(b/a) \leq aU(1) - (a - b)U'(1),$$

and hence

$$U(1) - D(x, y) \geq xA + (y - x)E. \quad (\text{U5.3})$$

Thus the direct branch applies outside the closed triangle

$$\mathcal{W} = \{0 \leq x \leq y : xA + (y - x)E \leq \Delta\}.$$

Set

$$r_a = \Delta/E, \quad r_d = \Delta/A, \quad s = 1 - A/E.$$

The certified order  $0 < r_a < \tau < r_d < 1$ ,  $r_d + \tau < 1$ , and  $0 < s < 1$  shows that  $\mathcal{W}$  has vertices  $(0, 0)$ ,  $(0, r_a)$ ,  $(r_d, r_d)$ , with sloping side  $y = r_a + sx$ . In particular, throughout  $\mathcal{W}$ ,

$$0 \leq x \leq y \leq r_d, \quad 0 \leq y - x \leq r_a < \tau. \quad (\text{U5.3a})$$

It remains to bound both possible page colours and the reservoir on  $\mathcal{W}$ . For the first page orientation put

$$z = \frac{1 - \tau - x}{1 - y}, \quad P_R = c_\eta(x + y) + \tau q + (1 - y)U(z).$$

The order above and  $r_d + \tau < 1$  give the complete derivative domain

$$0 < 1 - \frac{\tau}{1 - r_d} \leq z \leq \frac{1 - \tau}{1 - r_a} < 1. \quad (\text{U5.3b})$$

Since  $U'' < 0$  on  $(0, 1]$ ,  $U'(z) \geq U'(1) = E + c_\eta > c_\eta$ . Also, for  $\phi(z) = U(z) - zU'(z)$ ,  $\phi'(z) = -zU''(z) > 0$ , and hence  $\phi(z) \leq \phi(1) = c_\eta + A - E < c_\eta$ . Therefore its coordinate derivatives on all of  $\mathcal{W}$  are

$$\partial_x P_R = c_\eta - U'(z) < 0, \quad \partial_y P_R = c_\eta - U(z) + zU'(z) > 0.$$

It therefore suffices to follow the sloping side. Its derivative is

$$\mathcal{B}(z) = c_\eta - U'(z) + s\{c_\eta - U(z) + zU'(z)\},$$

where  $\mathcal{B}'(z) = (sz - 1)U''(z) > 0$ , because  $0 < s, z < 1$ . Along the side,  $z$  decreases from  $z_a = (1 - \tau)/(1 - r_a)$ ; the 512-bit enclosure gives  $\mathcal{B}(z_a) < -3.6952685 \cdot 10^{-4}$ . Hence  $P_R$  is maximized at  $(0, r_a)$ , where

$$P_R(0, r_a) = c_\eta r_a + \tau q + (1 - r_a)U\left(\frac{1 - \tau}{1 - r_a}\right). \quad (\text{U5.4})$$

For the other page orientation, put

$$w = \frac{1 - \tau - y}{1 - x}, \quad P_B(x, y) = c_\eta(x + y) + \tau q + (1 - x)U(w).$$

Here  $0 < w < 1$  on  $\mathcal{W}$ : positivity follows from  $y \leq r_d < 1 - \tau$ , and  $w < 1$  follows from  $y \geq x$  and  $\tau > 0$ . Thus  $\partial_y P_B = c_\eta - U'(w) < 0$ , so the maximum reduces to  $y = x$ . There it equals

$$P_B(x, x) = 2c_\eta x + \tau q + (1 - x)U\left(1 - \frac{\tau}{1 - x}\right).$$

For  $0 \leq x \leq r_d$ , the variable  $v = \tau/(1 - x)$  lies in

$$\tau \leq v \leq \frac{\tau}{1 - r_d} < 1,$$

so every derivative below is evaluated with  $1 - v \in (0, 1)$ . The diagonal derivative is

$$\Gamma(v) = 2c_\eta - U(1 - v) - vU'(1 - v), \quad \Gamma'(v) = vU''(1 - v) < 0.$$

Thus the maximum occurs at  $x = 0$ ; the enclosure gives  $\Gamma(\tau) < -1.8476200 \cdot 10^{-4}$  and

$$P_B(0, 0) = \tau q + U(1 - \tau). \quad (\text{U5.5})$$

Finally,  $Q(x, y) = c_\eta(x+y) + 2\tau\Xi$  is coordinatewise increasing, so its maximum on  $\mathcal{W}$  is

$$Q(r_d, r_d) = 2c_\eta r_d + 2\tau\Xi. \quad (\text{U5.6})$$

Subtracting (U5.4)–(U5.6) from  $U(1) - \Delta$  gives, respectively, the red-page, blue-page, and reservoir margins in the table. The symmetric triangle  $y \leq x$  exchanges the two colours and adds no new inequality. The same run verifies the scalar source gates, including  $\tau(1/2 - \eta - p) > 1.5056 \cdot 10^{-9}$ , and the exact equality of the inner and outer correlation constants. Exact parameters, interval outputs, and checker identities are recorded in Appendix A.  $\square$

*Proof of Theorem 3.1.* By Theorem 4.3 and Proposition 7.1,

$$R(k, k) \leq \exp((U(1) - \Delta)k + o(k)).$$

A 512-bit Arb evaluation gives

$$\begin{aligned} e^{U(1)-\Delta} &< 3.7806852883796401133524 \\ &< 3.780685288379640114 < 3.780685290, \end{aligned}$$

where the first decimal is an outward-rounded upper endpoint. Its distance to the safe nine-decimal value is greater than  $1.6203 \cdot 10^{-9}$ . This proves both printed claims.  $\square$

## 8. CERTIFIED COMPUTATION AND INDEPENDENT REPLAY

The computational proof follows three rules.

**Exact data.** All polynomial coefficients and parameters are parsed from decimal or rational strings, never from binary floating-point literals. Equality of the inner and outer correlation constants is checked in exact rational arithmetic.

**Continuum coverage.** Every compact domain is a union of closed rational intervals. Arb evaluates an inclusion extension on each cell. Analytic monotonicity or derivative signs treat the unbounded tails. A dense floating scan is not proof and is used only during search.

**Independent implementations.** The final exact-diagonal/outer-transfer package contains two author checkers, a non-importing replay, and an adaptive referee replay. The latter two reconstruct the functions and constants instead of importing author code. They use different subdivisions and endpoint reductions. These four final checkers use 512-bit precision. The logically upstream six-stage rate chain and its direct-region replay use 256-bit precision. Every one of these programs uses `python-flint==0.9.0` and therefore

shares the same Arb containment kernel: implementation diversity is not an independent arithmetic kernel or a proof-assistant formalization.

The canonical rate loader rejects duplicate JSON keys, missing or unknown schema fields, non-string numeric data, and non-finite decimal strings. In **-require-paper-chain** mode it additionally checks the six filenames, their order, and their complete file hashes before interval evaluation. The canonical reproduction wrappers also refuse optimized Python mode, in which proof-critical **assert** statements would otherwise be disabled.

The current theorem files have the following SHA-256 identifiers.

| Role                     | SHA-256 prefix   |
|--------------------------|------------------|
| proof specification      | e4ea58def6405936 |
| inner checker            | 17db00d9374ce3ba |
| outer checker            | e2035cbffefcb147 |
| non-importing replay     | fc379e3b861b6905 |
| adaptive replay          | 3c30f72ddc24848f |
| independent proof review | a21402a5dad40490 |

This table, the six complete rate-certificate hashes in Appendix C, and **external\_sources/manifest.json** form the upper-bound evidence index. The compact ratio, exact-diagonal, and outer-wedge cells are deterministic rational partitions generated by the tracked checkers; they are not separate untracked cell files. The top-level **artifacts/MANIFEST.tsv** has a different role: it lists only the large finite CNF/DRAT release payloads.

From the repository root, a lightweight replay of the exact-diagonal and outer-transfer certificates is

```
bash scripts/bootstrap.sh
bash scripts/reproduce_upper.sh
```

The first command creates a Python 3.11 environment with exact dependency versions. The second runs the four final transfer checkers; it deliberately does not replay the six-stage rate chain. The complete upper-bound replay, including all six rate certificates, both Ramsey-region orientations, the combinatorial audit, and all four final transfer checkers, is

```
.venv/bin/python reproduce.py full
```

The unified manuscript is compiled separately with **make paper**.

## 9. SCOPE, TRUST BOUNDARY AND FURTHER WORK

The theorem is asymptotic. The regularization and book interfaces contain eventual estimates, and this paper does not extract an effective threshold for  $k$ . Consequently the result has no direct implication for a specific finite Ramsey number.

The retained-spine parameters were found by bounded numerical search and then frozen. The proof certifies the frozen point; it does not prove global optimality. A better correlation inequality, a different off-diagonal rate, or a richer outer partition may improve the decimal further.

The external trust boundary consists of the explicitly stated Yang–Mao v1 interfaces, the repaired local book/rate theorem recorded in the appendices, and the containment semantics of Arb. Independent implementations reduce the risk of a transcription or dependency error but are not a proof-assistant formalization.

Finally, the numerical comparison in the introduction is made against published or source-stated bounds that are cited explicitly. This draft does not claim priority, novelty against all unpublished work, acceptance, or an unqualified world-best status.

## Part 2. Controlled Gaussian residuals for fixed-ratio lower bounds

### 10. FIXED-RATIO LOWER BOUND: THEOREM AND PROOF ARCHITECTURE

The probabilistic argument of Erdős gives the classical exponential scale for diagonal and fixed-ratio Ramsey lower bounds [Erd47]. Ma, Shen and Xie recently obtained the first exponential improvement for  $R(\ell, C\ell)$  at every fixed  $C > 1$  [MSX25]. Hunter, Milojević and Sudakov gave a Gaussian random-graph construction and stronger quantitative estimates [HMS25]; Lin and Niu sharpened the truncated Gaussian cumulant analysis [LN26]. Truncated Gaussian variance proxies and cumulants also have an independent probabilistic literature [BMA24].

This part works inside the explicit HMS Appendix-B induction. The new ingredient is not history conditioning by itself—that is already present in the cited constructions—but a deterministic nonexchangeable triangular mean ledger. A boundary derivative must normally match a projection weight exactly. We first show that this exact-merge requirement uniquely determines the array and leaves no scalar optimization freedom. We then retain the negative quadratic term from strong concavity and intentionally permit a small deterministic mismatch. Square completion pays this residual and creates a larger net gain.

**10.1. Main theorem.** For  $C > 1$ , let  $p_C \in (0, 1/2)$  be the unique solution of

$$C = \frac{\log p_C}{\log(1 - p_C)}.$$

The parameters  $p, c, D, \omega_0$ , the source quantities  $B_R(C)$  and  $B_B(C)$ , and the exact-CGF term  $G_*(C)$  are defined explicitly in Section 11. The theorem imports the four version-pinned conditional probability, extraction, and blue-ledger items (S1), (S2), (S4), and (S5), together with the separately identified weighted-extension rule (S3), all stated in source interface 11.2. The one-step nonuniform estimate, first-moment conversion, and red–blue comparison are proved locally.

**Theorem 10.1** (Source-relative fixed-ratio bound). *Fix the absolute source constant  $K$  used in (L2.10). Under source interface 11.2, there exists  $C_0 =$*

$C_0(K) > 1$  such that, for each fixed  $C \geq C_0$ ,

$$\liminf_{\ell \rightarrow \infty} \frac{1}{\ell} \log R(\ell, \lfloor C\ell \rfloor) \geq -\frac{1}{2} \log p_C + \frac{B_R(C)}{2} + G_*(C) + \hat{H}_*(C), \quad (\text{L1.1})$$

where  $\hat{H}_*(C) > 0$  and

$$\hat{H}_*(C) = \frac{1 + o(1)}{64 \log C} \quad (C \rightarrow \infty). \quad (\text{L1.2})$$

The public coarsening  $B_R(C)/2 \geq 1/20$  may be substituted in (L1.1).

The quantifiers are important:  $K$  is fixed first, then  $C$  is fixed before  $\ell \rightarrow \infty$ , and the proof does not give a numerical  $C_0(K)$ . The source contains unspecified  $O(\cdot)$  and  $o(\cdot)$  constants, so an effective threshold cannot be extracted from the pinned text alone.

## 10.2. Contributions.

- (i) We prove rigidity of the exact-gradient triangular recursion.
- (ii) We construct an adapted deterministic residual and prove a uniform full-box Hessian bound.
- (iii) We compute the exact multiplicity of the residual and its square penalty.
- (iv) We insert a three-factor Hölder decomposition without counting the old exact-CGF gain twice.
- (v) We propagate the ledger through reverse induction and quantify the perfect-sequence extraction loss.
- (vi) We optimize a narrowly specified scalar residual class and obtain the leading coefficient  $1/64$ .

The result is stated as a source-relative strengthening. A direct claim of improvement over every printed final normalization in Lin–Niu would require both expressions to be derived from one common terminal probability formula; that comparison is not asserted here.

## 11. GAUSSIAN SOURCE LEDGER AND NOTATION

Fix  $C > 1$ . Let  $p_C \in (0, 1/2)$  be the solution of

$$\log p_C = C \log(1 - p_C),$$

and, in the range of  $C$  used below, set

$$p = p_C + \frac{1}{C} < \frac{1}{2}, \quad \Phi(-c) = p, \quad a = \varphi(c), \quad D = \frac{4aC}{1-p}. \quad (\text{L2.1})$$

For the fixed- $C$  limit  $\ell \rightarrow \infty$ , take

$$d = \lceil D^2 \ell^2 \rceil, \quad D_\ell = \frac{\sqrt{d}}{\ell} = D + O_C(\ell^{-2}). \quad (\text{L2.2})$$

The harmless replacement of  $D$  by  $D_\ell \geq D$  in finite- $\ell$  feasibility inequalities will always be left implicit.

For a standard normal variable conditioned on  $Z \leq -b$ , define

$$m(b) = \frac{\varphi(b)}{\Phi(-b)}, \quad v(b) = 1 + b m(b) - m(b)^2, \quad Y_b = Z + m(b), \quad (\text{L2.3})$$

so that  $Y_b$  is centered and has variance  $v(b)$ . Its positive-direction cumulant generating function is

$$K_b(u) = \log \mathbb{E} e^{u Y_b} = \frac{u^2}{2} + u m(b) + \log \Phi(-b - u) - \log \Phi(-b), \quad u \geq 0, \quad (\text{L2.4})$$

and we write

$$\psi_b(u) = \begin{cases} K_b(u)/u^2, & u > 0, \\ v(b)/2, & u = 0. \end{cases} \quad (\text{L2.5})$$

We shall repeatedly use

$$K_b''(u) = v(b + u), \quad K_b(u) = \int_0^u (u - s) v(b + s) ds. \quad (\text{L2.6})$$

**Lemma 11.1** (One-sided Gaussian calculus). *For every  $b \in \mathbb{R}$ ,  $m'(b) = m(b)(m(b) - b) > 0$ ,  $m''(b) \geq 0$ , and  $v(b) = 1 - m'(b)$  is nonincreasing. Consequently,  $0 < v(b) \leq 1$ ,  $K_b''(u) \in [0, 1]$  for  $u \geq 0$ , and  $\psi_b(u)$  is nonincreasing in both  $b$  and  $u$ .*

*Proof.* Differentiating  $m = \varphi/\Phi(-\cdot)$  gives  $m' = m(m - b)$ . The classical inverse-Mills inequality  $m(b) > b$  for  $b > 0$  (and positivity of  $m$  for  $b \leq 0$ ) gives  $m' > 0$ . Moreover  $v = 1 - m'$ . Monotonicity of the variance of an upper-truncated normal in its upper cutoff, applied at the cutoff  $-b$ , says that  $v$  is nonincreasing in  $b$ ; equivalently  $m'' = -v' \geq 0$ . The variance bounds give  $0 < v \leq 1$ . Formula (L2.6) then implies the assertions concerning  $K_b$  and  $\psi_b$ ; for example  $\psi_b(u) = \int_0^1 (1 - t) v(b + ut) dt$ .  $\square$

Put

$$\begin{aligned} m_0 &= m(c) = \frac{a}{p}, & m'_0 &= m'(c), \\ \alpha_R &= 10\sqrt{\log(10/p)}, & \omega_0 &= \frac{\alpha_R^2}{D} = \frac{100 \log(10/p)}{D}, \end{aligned} \quad (\text{L2.7})$$

and

$$\rho = \frac{m_0 m'_0}{D}, \quad \theta = m_0^2, \quad A_0 = m_0^2 m'_0. \quad (\text{L2.8})$$

For fixed  $w > \omega_0$ , define the lower and upper cutoff extrema

$$\begin{aligned} m_w &= m(c - w), & M_w &= m(c + w), \\ j_-(w) &= m'(c - w), & j_+(w) &= m'(c + w). \end{aligned} \quad (\text{L2.9})$$

The distinction between  $m_w$  and  $M_w$  is essential: the former gives the history-uniform exact-CGF deficit, whereas the latter bounds the new linear tilts.

**11.1. The frozen red and blue source ledgers.** Fix one absolute source constant  $K$  that dominates the red  $O(\sqrt{\log(1/p)}/D)$  errors in the pinned HMS induction, and put

$$\vartheta_C = 1 - K \frac{\sqrt{\log(1/p)}}{D} > 0, \quad (\text{L2.10})$$

$$B_R(C) = -\log \frac{p}{p_C} + \vartheta_C \frac{a^3}{3p^3 D}, \quad B_B(C) = \log \frac{1-p_C}{1-p} - \frac{4a^3 C}{3(1-p)^3 D}. \quad (\text{L2.11})$$

The dependence of  $\vartheta_C$ ,  $B_R(C)$ , and the eventual threshold  $C_0$  on this once-fixed  $K$  is suppressed in the notation; equivalently one may write  $\vartheta_{C,K}$ ,  $B_{R,K}(C)$ , and  $C_0(K)$ . The literal red target envelope is

$$\mathcal{H}_C(r) = p^{(r)} \exp \left[ -\vartheta_C \frac{a^3}{p^3 \sqrt{d}} \binom{r}{3} + \frac{a}{p} d^{-1/5} \binom{r}{2} + r \right]. \quad (\text{L2.12})$$

Thus  $B_R/2$  and  $CB_B/2$ , respectively, are the exact limiting red and blue source bonuses relative to  $-\frac{1}{2} \log p_C$ . In particular, these symbols do not denote the conclusion of the present paper.

**11.2. The pre-existing exact-CGF deficit.** Set  $t_0 = m_0 \sqrt{d}$ . For  $1 \leq k \leq \ell$ , set

$$P_{k,d}^0 = \frac{\sqrt{d}}{\sqrt{d} - 4m_0 k}, \quad (\text{L2.13})$$

and, whenever the denominator is positive, define

$$\mathbf{d}_{k,w,d} = \frac{m_0^2 m_w^2}{d} k(k-1)^2 \left[ 1 - P_{k,d}^0 \psi_{c-w} \left( \frac{P_{k,d}^0 m_0 m_w (k-1)}{\sqrt{d}} \right) \right], \quad (\text{L2.14})$$

$$\mathcal{D}_{C,w,d}(r) = \sum_{k=1}^{r-1} \mathbf{d}_{k,w,d}. \quad (\text{L2.15})$$

For  $0 \leq x \leq 1$ , let

$$P_*(x) = \frac{D}{D - 4m_0 x}, \quad u_*(x) = \frac{P_*(x) m_0 m(c - \omega_0) x}{D}. \quad (\text{L2.16})$$

For all sufficiently large  $C$ ,  $D > 8m_0$  and  $P_*(1)v(c - \omega_0) < 2$ ; hence (L2.13)–(L2.17) have positive denominators and nonnegative integrands throughout the range used below. The exact source-relative CGF term used in the theorem is

$$G_*(C) = \frac{m_0^2 m(c - \omega_0)^2}{D^2} \int_0^1 x^3 [1 - P_*(x) \psi_{c-\omega_0}(u_*(x))] dx. \quad (\text{L2.17})$$

Equivalently, if  $G(C, w)$  is obtained from (L2.17) by replacing  $\omega_0$  with  $w$ , then

$$\lim_{\ell \rightarrow \infty} \ell^{-2} \mathcal{D}_{C,w,d}(\ell) = G(C, w), \quad G_*(C) = \lim_{w \downarrow \omega_0} G(C, w). \quad (\text{L2.18})$$



11.3. **The controlled-residual constants.** Set

$$a_- = \min\{A_0, \theta\}, \quad a_+ = \max\left\{\theta, \frac{A_0 + \rho\theta}{1 - \rho}\right\}, \quad (\text{L2.19})$$

$$\mu_w = j_-(w) - j_+(w)^2 \frac{m_0 + \theta/D}{D(1 - \rho)}, \quad (\text{L2.20})$$

and

$$\widehat{L}_w(C) = \frac{4m_0 M_w^2}{D} + \frac{2M_w^2 a_+}{D^2} + \frac{32m_0^2 M_w^2 a_+}{a_- D^2}. \quad (\text{L2.21})$$

Finally put

$$\widehat{B}_w(C) = m_0^2 - \frac{16}{3} - \widehat{L}_w(C), \quad E_w(C) = \widehat{B}_w(C)\theta - \frac{\theta^2}{2\mu_w}, \quad (\text{L2.22})$$

$$\widehat{H}(C, w) = \frac{\widehat{B}_w(C)A_0 + E_w(C)}{12D^2}, \quad \widehat{H}_*(C) = \lim_{w \downarrow \omega_0} \widehat{H}(C, w). \quad (\text{L2.23})$$

11.4. **Probability space, exposure histories, and source events.** For each  $0 \leq r \leq \ell$ , let  $\mathbf{x}_1, \dots, \mathbf{x}_r$  be independent  $N(0, d^{-1}I_d)$  vectors. An edge is blue when  $\langle \mathbf{x}_i, \mathbf{x}_j \rangle \geq -c/\sqrt{d}$  and red otherwise. We use the Bartlett representation  $\mathbf{y}_1, \dots, \mathbf{y}_r$ : the entries  $y_j(i)$ ,  $i < j$ , are independent  $N(0, d^{-1})$ , the diagonal entries are independent with  $y_i(i) \sim \sqrt{\chi_{d-i+1}^2/d}$ , and  $y_j(i) = 0$  for  $i > j$ . This representation has the same joint Gram matrix as the original Gaussian vectors.

Let  $M_i = (y_j(i))_{j=i+1}^r$  be the strict part of column  $i$ , and put

$$\mathcal{D}_r = \sigma(y_1(1), \dots, y_r(r)).$$

Define the exposure filtration

$$\mathcal{F}_s = \mathcal{D}_r \vee \sigma(M_1, \dots, M_s), \quad 0 \leq s \leq r. \quad (\text{L2.24})$$

All conditional assertions below are statements about regular conditional probabilities and hold almost surely. Write  $\pi_s$  for projection onto the first  $s$  Bartlett coordinates and put

$$h_{ij} = \langle \pi_{i-1}(\mathbf{y}_i), \pi_{i-1}(\mathbf{y}_j) \rangle, \quad 1 \leq i < j \leq r. \quad (\text{L2.25})$$

Thus  $h_{ij}$  is  $\mathcal{F}_{i-1}$ -measurable. Let  $E_{ij}$  be the blue-edge event and set

$$I_s = \bigcap_{s < i < j \leq r} \overline{E}_{ij}, \quad \mathcal{C}_s = \bigcap_{s < i < j \leq r} E_{ij}. \quad (\text{L2.26})$$

For the source perfectness events put  $\alpha_B = 10\sqrt{C \log(10/p)}$  and  $\delta = \alpha_B d^{-1/4}$ . The event  $B_r^R$  says that, for every  $i \leq r$ ,

$$\|\mathbf{y}_i\| \in (1 - \delta, 1 + \delta), \quad \|\pi_{i-1}(\mathbf{y}_i)\| \leq \frac{\alpha_R \sqrt{\ell}}{\sqrt{d}}; \quad (\text{L2.27})$$

$B_r^B$  is defined by replacing  $\alpha_R$  with  $\alpha_B$ . We use the probability notation

$$P_{R,r}^* = \mathbb{P}(I_0 \cap B_r^R), \quad P_{R,r} = \mathbb{P}(I_0), \quad P_{B,r}^* = \mathbb{P}(\mathcal{C}_0 \cap B_r^B), \quad P_{B,r} = \mathbb{P}(\mathcal{C}_0). \quad (\text{L2.28})$$

Empty intersections have probability one.

For  $0 \leq s \leq r$ , let

$$\kappa_{r,s}^R = \mathbb{P}(B_r^R \mid \mathcal{F}_s), \quad \mathfrak{H}_{r,s}^R = \{\kappa_{r,s}^R > 0\}. \quad (\text{L2.29})$$

We call an  $\mathcal{F}_s$ -history *red-admissible* when it belongs to  $\mathfrak{H}_{r,s}^R$ . This definition is independent of a choice of coordinates for the conditional kernel. On the complementary histories,  $\mathbb{P}(I_s \cap B_r^R \mid \mathcal{F}_s) = 0$  almost surely. Thus every conditional estimate below only needs to be checked on  $\mathfrak{H}_{r,s}^R$ .

At exposure  $i$ , let  $A_i^R = \bigcap_{j>i} \bar{E}_{ij}$ . Conditional on  $A_i^R$  and  $\mathcal{F}_{i-1}$ , set  $X_j = y_j(i)$ ,  $j > i$ . Its exact upper cutoff is

$$X_j \leq -\frac{b_{ij}}{\sqrt{d}}, \quad b_{ij} = \frac{c + \sqrt{d} h_{ij}}{y_i(i)}. \quad (\text{L2.30})$$

The red-perfect extraction map  $\Gamma_R$  scans  $\mathbf{x}_1, \dots, \mathbf{x}_\ell$  in order and retains an index when adjoining its vector preserves the two inequalities in (L2.27). For a specified  $I \subseteq [\ell]$ , define

$$p_I = \mathbb{P}(I_0 \cap \{\Gamma_R = I\}). \quad (\text{L2.31})$$

This fixes both the event and the retained-order relabeling used later.

For a deterministic nonnegative triangular array  $T = (T_{jq})_{1 \leq j < q \leq r}$ , define the weighted projection potential

$$\mathcal{P}_s^T = \sum_{s < j < q \leq r} T_{jq} \langle \pi_s(\mathbf{y}_j), \pi_s(\mathbf{y}_q) \rangle. \quad (\text{L2.32})$$

For  $i \geq 1$ , its normalized one-step kernel is

$$\begin{aligned} \mathcal{K}_i^T &= \exp\left(\sum_{j>i} T_{ij} h_{ij}\right) \mathbb{P}(A_i^R \mid \mathcal{F}_{i-1}) \\ &\times \mathbb{E}\left[\exp\left\{-\sum_{i < j < q \leq r} T_{jq} X_j X_q\right\} \middle| A_i^R, \mathcal{F}_{i-1}\right]. \end{aligned} \quad (\text{L2.33})$$

**Source interface 11.2** (Version-pinned source and weighted-extension interface). Items (S1), (S2), (S4), and (S5) are imported from the content-addressed HMS v2 and Lin-Niu v2 sources identified in Appendix E. Item (S3) is the exact weighted-extension hypothesis required by this paper; it is not represented as a theorem stated verbatim in either source.

(S1) For every  $r \leq \ell$  and  $1 \leq i < r$ , on every red-admissible  $\mathcal{F}_{i-1}$ -history, the next-column coordinates in (L2.30), conditionally on  $A_i^R$ , are independent  $N(0, d^{-1})$  variables truncated by the displayed upper cutoffs, and

$$|b_{ij} - c - \sqrt{d} h_{ij}| \leq d^{-1/5}, \quad |\sqrt{d} h_{ij}| \leq \omega_0. \quad (\text{L2.34})$$

Thus, for every fixed  $w > \omega_0$ , all cutoffs lie in  $[c - w, c + w]$  once  $\ell \geq \ell_0(C, w)$ .

- (S2) Under the conditioning and history quantifiers in (S1), with  $\zeta_j = \sqrt{d}(X_j - \mathbb{E}[X_j \mid A_i^R, \mathcal{F}_{i-1}])$ , the variables  $(\zeta_j)_{j>i}$  are conditionally independent, centered, and 1-subgaussian, and

$$\log \mathbb{E}[e^{x\zeta_j^2} \mid A_i^R, \mathcal{F}_{i-1}] \leq \frac{16}{3}x, \quad 0 \leq x \leq \frac{1}{8}. \quad (\text{L2.35})$$

For  $k = r - i$ ,  $Q_0 = d/(4t_0k)$ , and centered  $\zeta_j$ , the frozen uniform-quadratic factor obeys

$$\frac{1}{Q_0} \log \mathbb{E} \exp \left\{ -\frac{Q_0 t_0}{d} \sum_{i < j < q \leq r} \zeta_j \zeta_q \right\} \leq \frac{4t_0k}{d}. \quad (\text{L2.36})$$

The positive-direction CGF is exactly (L2.4). On this history define the source baseline tilt

$$u_j^0 = \frac{t_0}{d} \sum_{\substack{q > i \\ q \neq j}} m(b_{iq}), \quad j > i.$$

This is the quantity recalled in (L5.9). In the source pre-Cauchy allocation, the centered-linear proxy is precisely  $\sum_{j>i} (u_j^0)^2$ ; replacing that proxy by the exact CGF produces the same-history deficit (L5.16). For the uniform array  $T_{jq}^0 = t_0$ , the version-pinned source scalar allocation and this exact-CGF replacement give, on every history in  $\mathfrak{H}_{r,i-1}^R$ ,

$$\mathcal{K}_i^{T^0} \leq \frac{\mathcal{H}_C(k+1)}{\mathcal{H}_C(k)} e^{-d_{k,w,d}} \quad (k = r - i). \quad (\text{L2.36a})$$

This scalar inequality, rather than a final Ramsey bound, is the imported baseline. No nonuniform one-step estimate is imported.

- (S3) For every target  $r \leq \ell$ , every deterministic nonnegative triangular  $T$  fixed before exposure, and every deterministic real sequence  $\lambda_1, \dots, \lambda_{r-1}$ , the following weighted reverse-induction rule holds. If, on every history in  $\mathfrak{H}_{r,i-1}^R$ , with  $k = r - i$ ,

$$\mathcal{K}_i^T \leq \frac{\mathcal{H}_C(k+1)}{\mathcal{H}_C(k)} e^{-\lambda_i}, \quad 1 \leq i < r, \quad (\text{L2.37})$$

then, for every  $0 \leq s \leq r$ ,

$$\mathbb{P}(I_s \cap B_r^R \mid \mathcal{F}_s) \leq \mathcal{H}_C(r-s) \exp \left[ -\mathcal{P}_s^T - \sum_{i=s+1}^{r-1} \lambda_i \right]. \quad (\text{L2.38})$$

On  $(\mathfrak{H}_{r,s}^R)^c$ , the left side of (L2.38) is zero, so the same inequality holds almost surely on the entire conditional state space. Only the implication (L2.37)  $\Rightarrow$  (L2.38) is the additional hypothesis in (S3). The following source identity and deterministic projection algebra

record what that implication must preserve; they neither add a second hypothesis nor prove (S3). The source law of total probability is

$$\mathbb{P}(I_{i-1} \cap B_r^R \mid \mathcal{F}_{i-1}) = \mathbb{P}(A_i^R \mid \mathcal{F}_{i-1}) \mathbb{E}[\mathbb{P}(I_i \cap B_r^R \mid \mathcal{F}_i) \mid A_i^R, \mathcal{F}_{i-1}], \quad (\text{L2.39})$$

and the exact projection identities are

$$\begin{aligned} \mathcal{P}_i^T &= \sum_{i < j < q \leq r} T_{jq} \langle \pi_{i-1} \mathbf{y}_j, \pi_{i-1} \mathbf{y}_q \rangle + \sum_{i < j < q \leq r} T_{jq} X_j X_q, \\ \mathcal{P}_{i-1}^T &= \sum_{i < j < q \leq r} T_{jq} \langle \pi_{i-1} \mathbf{y}_j, \pi_{i-1} \mathbf{y}_q \rangle + \sum_{j > i} T_{ij} h_{ij}. \end{aligned} \quad (\text{L2.40})$$

Equations (L2.33), (L2.39), and (L2.40) record the algebra that the weighted extension in (S3) is required to preserve. The rule propagates (L2.38) by reverse induction from  $s = r$ . Sections 13–15 compare the nonuniform kernel with the scalar baseline (L2.36a) and prove (L2.37) for the array used here.

- (S4) If red-perfect extraction deletes  $u$  vertices from a target of size  $\ell$ , then, for each specified retained set  $I$  of size  $\ell - u$ ,

$$p_I \leq P_{R, \ell-u}^* \left( \frac{p}{10} \right)^{10\ell u}, \quad (\text{L2.41})$$

for every  $0 \leq u \leq \ell$ , with  $P_{R,0}^* = \mathcal{H}_C(0) = 1$ . The probability  $P_{R, \ell-u}^*$  uses the retained-order relabeling defined after (L2.31).

- (S5) For each fixed  $C$ , there is a real sequence  $\epsilon_{C, \ell} \rightarrow 0$  such that, with  $q_\ell = \lfloor C\ell \rfloor$ , the unchanged blue induction and its perfect-sequence extraction give

$$P_{B, q_\ell} \leq (1-p)^{\binom{q_\ell}{2}} \exp \left[ \frac{4a^3}{(1-p)^3 \sqrt{d}} \binom{q_\ell}{3} + \epsilon_{C, \ell} \ell^2 \right]. \quad (\text{L2.42})$$

No Ramsey-rate conclusion, no red–blue comparison, and no new residual term is included in (S1), (S2), (S4), or (S5). The scalar source baseline is (L2.36a); the only additional source-side hypothesis is the precisely delimited weighted rule (S3).

The constant  $K$  in (L2.10) is not effective in the pinned source. Consequently all sufficiently-large- $C$  statements below are existential.

## 12. RIGIDITY OF EXACT GRADIENT MERGE

For a target ordered set  $[r]$ , consider deterministic triangular weights  $W_{ij}$  and the connection-plus-mean potential

$$F_i(b) = \sum_{j > i} \log \Phi(-b_j) - \frac{1}{d} \sum_{i < j < q} W_{jq} m(b_j) m(b_q). \quad (\text{L3.1})$$

At the central cutoff,

$$\partial_j F_i(c1) = -m_0 - \frac{m_0 m_0'}{d} \sum_{\substack{q > i \\ q \neq j}} W_{jq}. \quad (\text{L3.2})$$

**Proposition 12.1** (Unique exact-merge array). *Normalize the coefficient of  $\sum_{j>i} \log \Phi(-b_j)$  in (L3.1) to be one. If the linear term in (L3.2), multiplied by  $b_j - c = \sqrt{d} h_{ij}$ , must equal  $-W_{ij} h_{ij}$  at every boundary, then the weights are uniquely determined by*

$$W_{ij} = m_0 \sqrt{d} + \frac{m_0 m'_0}{\sqrt{d}} \sum_{\substack{q>i \\ q \neq j}} W_{jq}. \quad (\text{L3.3})$$

*There is no further normalized scalar degree of freedom.*

*Proof.* The coefficient comparison gives (L3.3). Process the first endpoint in decreasing order. Every weight on the right side has already been defined, so the system is triangular and has a unique solution.  $\square$

*Remark 12.2* (Global presentation scale). Multiplying the entire inequality by  $\kappa > 0$  is a presentation equivalence, not a new normalized array. After dividing the connection term back to coefficient one, the normalized array still satisfies (L3.3). If one instead calls  $W^{\text{eff}} = \kappa W$  the effective coefficient without renormalizing, then its affine recursion has base term  $\kappa m_0 \sqrt{d}$ , not  $m_0 \sqrt{d}$ . Thus a normalized solution and a  $\kappa$ -scaled effective coefficient are never identified. No assertion is made about edge-dependent rescalings.

The feedback operator in (L3.3) has row norm at most  $\rho = m_0 m'_0 / D = o(1)$ . Its accumulated increment is

$$2A_0 \binom{r}{4} [1 + O(\rho)]. \quad (\text{L3.4})$$

With the source centered coefficient  $m_0^2 [1 + o(1)]$ , the exact-merge ledger contributes the previously reviewed term  $(1 + o(1)) / (96 \log C)$ . Exact feedback evaluation can change only the next order.

### 13. CONTROLLED RESIDUAL AND STRONG CONCAVITY

Recall  $t_0 = m_0 \sqrt{d}$  from Section 11, and set

$$\gamma_d = \frac{m_0 m'_0}{\sqrt{d}}, \quad e_{ij} = \theta(i - 1). \quad (\text{L4.1})$$

In decreasing order of the first endpoint define

$$T_{ij} = t_0 + \gamma_d \sum_{\substack{q>i \\ q \neq j}} T_{jq} + e_{ij}. \quad (\text{L4.2})$$

The array is deterministic and adapted: when column  $i$  is exposed, every future weight in (L4.2) has already been fixed.

With  $F_i$  as in (L3.1), now using  $T$ , we obtain the exact residual identity

$$\partial_j F_i(c\mathbf{1}) = -\frac{T_{ij} - e_{ij}}{\sqrt{d}}. \quad (\text{L4.3})$$

Thus the mismatch is explicit and deterministic.

**Lemma 13.1** (Uniform Hessian bound). *For all sufficiently large fixed  $C$ , every  $b \in [c - w, c + w]^{V_i}$  satisfies*

$$-\nabla^2 F_i(b) \succeq \mu_w I, \quad (\text{L4.4})$$

where  $\mu_w > 0$  is defined in (L2.20).

*Proof.* The recursion gives

$$T_{\max} \leq \frac{t_0 + \theta r}{1 - \rho}, \quad \frac{1}{d} \max_j \sum_{q \neq j} T_{jq} \leq \frac{m_0 + \theta/D}{D(1 - \rho)}.$$

The diagonal of  $-\nabla^2 F_i$  is at least  $j_-(w)$ . The absolute off-diagonal row sum is at most the second term subtracted in (L2.20), multiplied by  $j_+(w)^2$ . Gershgorin's theorem proves (L4.4).  $\square$

Write

$$\delta_j = b_j - c = \sqrt{d} h_{ij} + \varepsilon_{ij}, \quad |\varepsilon_{ij}| \leq d^{-1/5}.$$

Strong concavity, (L4.3), and

$$\frac{e_{ij} \delta_j}{\sqrt{d}} - \frac{\mu_w \delta_j^2}{2} \leq \frac{e_{ij}^2}{2\mu_w d} \quad (\text{L4.5})$$

give

$$F_i(b) \leq F_i(c\mathbf{1}) - \sum_{j>i} T_{ij} h_{ij} + \sum_{j>i} \frac{e_{ij}^2}{2\mu_w d} + \frac{d^{-1/5}}{\sqrt{d}} \sum_{j>i} T_{ij}. \quad (\text{L4.6})$$

The full boundary coefficient merges into the projection potential, while the failure of exact gradient merge is paid by the square term.

#### 14. DETERMINISTIC MASS AND THE CENTERED-MOMENT LEDGER

Write  $\Delta_{ij} = T_{ij} - t_0$ . The recursion implies

$$A_0(r - j - 1) + \theta(j - 1) \leq \Delta_{jq} \leq \frac{A_0 + \rho\theta}{1 - \rho}(r - j - 1) + \theta(j - 1). \quad (\text{L5.1})$$

Consequently

$$a_-(r - 2) \leq \Delta_{jq} \leq a_+(r - 2). \quad (\text{L5.2})$$

For  $V_i = \{i + 1, \dots, r\}$ , of size  $k = r - i$ , put

$$S_i = \sum_{\{j,q\} \subset V_i} \Delta_{jq}, \quad R_i = \max_{j \in V_i} \sum_{q \neq j} \Delta_{jq}.$$

Then

$$S_i \geq a_-(r - 2) \binom{k}{2}, \quad R_i \leq a_+(r - 2)(k - 1), \quad \frac{R_i}{S_i} \leq \frac{2a_+}{a_-k}. \quad (\text{L5.3})$$

**Lemma 14.1** (Exact accumulated multiplicities). *Let*

$$K_r = \sum_{i=1}^{r-1} (r-i)(i-1)^2 = \binom{r}{3} + 2\binom{r}{4}. \quad (\text{L5.4})$$

*Then*

$$\sum_{i=1}^{r-1} S_i \geq 2A_0 \binom{r}{4} + \theta K_r, \quad \sum_{1 \leq i < j \leq r} e_{ij}^2 = \theta^2 K_r. \quad (\text{L5.5})$$

*Proof.* Sum the two terms of the lower bound in (L5.1) with their exposure multiplicities. The future-degree term counts every ordered quadruple twice, giving  $2\binom{r}{4}$ ; the residual term is exactly the sum in (L5.4). The second identity follows directly from  $e_{ij} = \theta(i-1)$ .  $\square$

**14.1. Exact conditional factorization.** Fix a red-admissible  $\mathcal{F}_{i-1}$ -history for target  $r$  at exposure step  $i$ , and condition further on  $A_i^R$ . For  $j \in V_i$ , let  $X_j$  be the coordinate in (L2.30), write  $b_j = b_{ij}$ , put  $m_j = m(b_j)$ , and standardize its centered part by

$$\zeta_j = \sqrt{d}(X_j - \mathbb{E}X_j) = \sqrt{d}X_j + m_j. \quad (\text{L5.6})$$

By (S2), the  $\zeta_j$  are conditionally independent, centered and 1-subgaussian. Expanding the weighted quadratic form gives the exact identity

$$\begin{aligned} \log \mathbb{E} \exp \left\{ - \sum_{\{j,q\} \subset V_i} T_{jq} X_j X_q \right\} \\ = -\frac{1}{d} \sum_{\{j,q\} \subset V_i} T_{jq} m_j m_q + \Lambda_i, \end{aligned} \quad (\text{L5.7})$$

$$\Lambda_i = \log \mathbb{E} \exp \left\{ \begin{aligned} & \sum_{j \in V_i} (u_j^0 + \delta u_j) \zeta_j \\ & - \frac{t_0}{d} \sum_{j < q} \zeta_j \zeta_q - \frac{1}{d} \sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \end{aligned} \right\}. \quad (\text{L5.8})$$

where

$$u_j^0 = \frac{t_0}{d} \sum_{q \neq j} m_q = \frac{m_0}{\sqrt{d}} \sum_{q \neq j} m_q, \quad \delta u_j = \frac{1}{d} \sum_{q \neq j} \Delta_{jq} m_q. \quad (\text{L5.9})$$

All tilts in (L5.9) are nonnegative.

**14.2. Three-factor Hölder and the square-MGF cost.** For  $k \geq 2$ , choose

$$Q_0 = \frac{d}{4t_0k}, \quad Q_1 = \frac{d}{4R_i}, \quad P^{-1} = 1 - Q_0^{-1} - Q_1^{-1}. \quad (\text{L5.10})$$

If  $R_i = 0$ , the increment factor is absent and we set  $Q_1 = \infty$ . For all sufficiently large  $C$ ,

$$Q_0^{-1} + Q_1^{-1} \leq \frac{4m_0}{D} + \frac{4a_+}{D^2} < \frac{1}{2}, \quad (\text{L5.11})$$

so  $Q_0, Q_1 > 1$  and  $1 < P \leq 2$ . Hölder's inequality applied to the three random factors in (L5.8) yields

$$\begin{aligned} \Lambda_i &\leq \frac{1}{P} \sum_j K_{b_j}(P(u_j^0 + \delta u_j)) \\ &\quad + \frac{1}{Q_0} \log \mathbb{E} \exp \left\{ -\frac{Q_0 t_0}{d} \sum_{j < q} \zeta_j \zeta_q \right\} \\ &\quad + \frac{1}{Q_1} \log \mathbb{E} \exp \left\{ -\frac{Q_1}{d} \sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \right\}. \end{aligned} \quad (\text{L5.12})$$

The middle line is the unchanged HMS uniform-quadratic factor and costs at most  $4t_0 k/d$ .

For the last line, write  $R_j = \sum_{q \neq j} \Delta_{jq}$ . Nonnegativity of  $\Delta$  gives

$$-\sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \leq \frac{1}{2} \sum_j R_j \zeta_j^2. \quad (\text{L5.13})$$

Because  $Q_1 R_j / (2d) \leq 1/8$ , the pinned square-MGF estimate

$$\log \mathbb{E} e^{x \zeta_j^2} \leq \frac{16}{3} x, \quad 0 \leq x \leq \frac{1}{8}, \quad (\text{L5.14})$$

and conditional independence imply

$$\begin{aligned} \frac{1}{Q_1} \log \mathbb{E} \exp \left\{ -\frac{Q_1}{d} \sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \right\} &\leq \frac{1}{Q_1} \sum_j \frac{16}{3} \frac{Q_1 R_j}{2d} \\ &= \frac{16}{3} \frac{S_i}{d}, \end{aligned} \quad (\text{L5.15})$$

where  $\sum_j R_j = 2S_i$ . This derives the constant  $16/3$ .

**14.3. The same-history CGF bridge.** Let  $P^0 = P_{k,d}^0$ , so that  $(P^0)^{-1} = 1 - Q_0^{-1}$ . The old optimized uniform ledger on this same conditional history has deficit

$$\mathfrak{d}_i(b) = \sum_{j \in V_i} \left[ (u_j^0)^2 - \frac{1}{P^0} K_{b_j}(P^0 u_j^0) \right]. \quad (\text{L5.16})$$

Equations (L2.6), the cutoff window, and  $\sum_{q \neq j} m_q \geq (k-1)m_w$  give the deterministic lower bound

$$\mathfrak{d}_i(b) \geq \mathfrak{d}_{k,w,d}. \quad (\text{L5.17})$$

Indeed, after writing  $K_b(P^0 u)/P^0 = P^0 u^2 \psi_b(P^0 u)$ , this is exactly the monotonicity comparison used in (L2.14). Thus the term whose accumulated limit is  $G_*(C)$  is identified before any new factor is introduced.

It remains to compare the new linear factor with the second term in (L5.16). Since  $K_b'' \in [0, 1]$ , for  $u, \delta \geq 0$  and  $1 < P \leq 2$ ,

$$\frac{K_b(P(u + \delta)) - K_b(Pu)}{P} \leq 2u\delta + \delta^2. \quad (\text{L5.18})$$



Moreover, differentiating  $K_b(Pu)/P$  and using (L2.6) gives

$$0 \leq \frac{\partial}{\partial P} \frac{K_b(Pu)}{P} \leq u^2. \quad (\text{L5.19})$$

As

$$P - P^0 = PP^0 \left( \frac{1}{P^0} - \frac{1}{P} \right) = \frac{PP^0}{Q_1} \leq \frac{16R_i}{d}, \quad (\text{L5.20})$$

we obtain

$$\begin{aligned} & \frac{1}{P} \sum_j K_{b_j}(P(u_j^0 + \delta u_j)) - \frac{1}{P^0} \sum_j K_{b_j}(P^0 u_j^0) \\ & \leq 2 \sum_j u_j^0 \delta u_j + \sum_j (\delta u_j)^2 + \frac{16R_i}{d} \sum_j (u_j^0)^2. \end{aligned} \quad (\text{L5.21})$$

Each term is now bounded without probabilistic input. From (L5.3), (L5.9),  $m_j \leq M_w$ , and  $k/\sqrt{d} \leq 1/D$ ,

$$2 \sum_j u_j^0 \delta u_j \leq \frac{4m_0 M_w^2 S_i}{D} \frac{1}{d}, \quad (\text{L5.22})$$

$$\sum_j (\delta u_j)^2 \leq \frac{2M_w^2 a_+ S_i}{D^2} \frac{1}{d}, \quad (\text{L5.23})$$

$$\frac{16R_i}{d} \sum_j (u_j^0)^2 \leq \frac{32m_0^2 M_w^2 a_+ S_i}{a_- D^2} \frac{1}{d}. \quad (\text{L5.24})$$

For completeness, (L5.22) uses  $\sum_j \delta u_j \leq 2M_w S_i/d$ ; (L5.23) uses  $\max_j \delta u_j \leq M_w R_i/d$ ; and (L5.24) uses

$$\sum_j (u_j^0)^2 \leq \frac{2m_0^2 M_w^2 S_i}{a_-} \frac{1}{d}, \quad (\text{L5.25})$$

which follows from the first inequality in (L5.3) and  $r - 2 \geq k - 1$ . The sum of (L5.22)–(L5.24) is precisely  $\hat{L}_w(C) S_i/d$ .

Finally, at the central cutoff the deterministic  $\Delta$ -part of (L5.7) is  $-m_0^2 S_i/d$ . Combining it with (L5.15) and (L5.21)–(L5.24) leaves

$$-\hat{B}_w(C) \frac{S_i}{d}. \quad (\text{L5.26})$$

Equations (L5.16)–(L5.17) charge the old exact-CGF deficit once; (L5.26) charges only the new increment factor and the change  $P^0 \rightarrow P$ ; and (L4.6) pays the boundary residual. Hence the three ledger entries are disjoint on the same conditional history.

## 15. REVERSE INDUCTION AND PERFECT-SEQUENCE EXTRACTION

By (L7.7)–(L7.11), after increasing the existential threshold  $C_0(K)$  and restricting  $w$  to a sufficiently small right-neighborhood of  $\omega_0$ , we have  $\hat{B}_w(C) > 0$ ,  $E_w(C) > 0$ , and  $\mu_w > 0$ . We fix this range throughout the

section; in particular, every monotonicity assertion below has its sign justified before it is used.

Define the new accumulated deterministic deficit

$$\mathcal{N}_{C,w,d}(r) = \frac{1}{d} \left[ 2\widehat{B}_w A_0 \binom{r}{4} + E_w K_r \right]. \quad (\text{L6.1})$$

At an exposure step with  $k = r - i$  future vertices, the same-history calculation of Section 14, followed by the pathwise boundary estimate (L4.6), proves the one-step kernel bound (L2.37) with

$$\lambda_i = \mathbf{d}_{k,w,d} + \widehat{B}_w(C) \frac{S_i}{d} - \sum_{j>i} \frac{e_{ij}^2}{2\mu_w d} - \frac{d^{-1/5}}{\sqrt{d}} \sum_{j>i} T_{ij}. \quad (\text{L6.2})$$

Every coefficient is deterministic and fixed before column  $i$  is sampled. Equations (L2.33) and (L2.40) retain every future coefficient  $T_{jq}$ , while (L4.6) supplies each new boundary edge  $ij$  with its full coefficient  $T_{ij}$ . Thus the exact implication (S3) applies without an exchangeability assumption.

Apply (L2.38) at  $s = 0$ , average over  $\mathcal{F}_0$ , and then use Lemma 14.1. Since  $\mathcal{P}_0^T = 0$ , this gives

$$P_{R,r}^* \leq \mathcal{H}_C(r) \exp[-\mathcal{D}_{C,w,d}(r) - \mathcal{N}_{C,w,d}(r) + J_{r,d}], \quad (\text{L6.3})$$

where

$$0 \leq J_{r,d} \leq \frac{m_0 + \theta/D}{1 - \rho} d^{-1/5} \binom{r}{2} = o_C(r^2). \quad (\text{L6.4})$$

Here  $J_{r,d}$  denotes the displayed deterministic upper envelope, so  $J_{q,d} \leq J_{\ell,d}$  for  $q \leq \ell$ . The base cases  $r = 0, 1, 2$  follow from the empty-event convention and the same one-step identity; no asymptotic estimate is needed there.

**15.1. Uniform restoration after deleting vertices.** Let extraction retain  $q = \ell - u$  vertices. Relabel them in retained order and rebuild the deterministic array for target size  $q$ . Since  $q/\sqrt{d} \leq \ell/\sqrt{d}$ , all row-sum, Hessian and Hölder feasibility inequalities only improve.

First, the bracket in (L2.14) lies in  $[0, 1]$ , and  $m_w \leq m_0$ . Consequently

$$\begin{aligned} 0 &\leq \mathcal{D}_{C,w,d}(\ell) - \mathcal{D}_{C,w,d}(\ell - u) \\ &\leq \frac{m_0^4}{d} \sum_{k=\ell-u}^{\ell-1} k(k-1)^2 \leq \frac{m_0^4}{D^2} \ell u. \end{aligned} \quad (\text{L6.5})$$

For sufficiently large  $C$ ,  $m_0^4/D^2 = (1 + o(1))/(8 \log C) < 1$ , so the right side is at most  $\ell u$ .

Next,

$$\binom{\ell}{4} - \binom{\ell-u}{4} \leq u \binom{\ell}{3}, \quad \binom{\ell}{3} - \binom{\ell-u}{3} \leq u \binom{\ell}{2}. \quad (\text{L6.6})$$

Using  $K_r = \binom{r}{3} + 2\binom{r}{4}$ , (L6.6) yields

$$\begin{aligned} 0 &\leq \mathcal{N}_{C,w,d}(\ell) - \mathcal{N}_{C,w,d}(\ell - u) \\ &\leq \frac{1}{d} \left[ 2\hat{B}_w A_0 u \binom{\ell}{3} + E_w \left( u \binom{\ell}{2} + 2u \binom{\ell}{3} \right) \right] \\ &\leq \left[ \frac{\hat{B}_w A_0 + E_w}{3D^2} + \frac{E_w}{2D^2\ell} \right] \ell u. \end{aligned} \quad (\text{L6.7})$$

The last bracket is  $O(1/\log C) + O_C(\ell^{-1}) < 1$  for sufficiently large  $C$  and then sufficiently large  $\ell$ , so (L6.7) also costs at most  $\ell u$ .

Finally, the ratio between the source envelopes at sizes  $\ell - u$  and  $\ell$  costs at most

$$[\log(1/p) + 1] \ell u. \quad (\text{L6.8})$$

Indeed,  $\binom{\ell}{2} - \binom{\ell-u}{2} \leq \ell u$  pays the missing  $p$ -factors, and  $\binom{\ell}{3} - \binom{\ell-u}{3} \leq \ell^2 u/2$ , together with  $\vartheta_C a^3/(p^3 D) \leq 2$  for all sufficiently large  $C$ , pays the cubic term. The positive  $d^{-1/5} \binom{r}{2}$  and  $r$  terms in  $\mathcal{H}_C(r)$ , as well as (L6.4), are monotone in  $r$  and require no restoration charge.

**15.2. The binomial subset sum.** Apply the pinned extraction inequality (L2.41), then (L6.3) at  $q = \ell - u$ , and finally (L6.5)–(L6.8). For every specified retained set  $I$  and every  $1 \leq u \leq \ell$ ,

$$\begin{aligned} p_I &\leq P_{R,\ell-u}^* \left( \frac{p}{10} \right)^{10\ell u} \\ &\leq \mathcal{H}_C(\ell) \exp[-\mathcal{D}_{C,w,d}(\ell) - \mathcal{N}_{C,w,d}(\ell) + J_{\ell,d}] e^{-\eta_C \ell u}, \end{aligned} \quad (\text{L6.9})$$

where the explicit residual exponent

$$\eta_C = 10 \log(10/p) - \log(1/p) - 3 > 0 \quad (\text{L6.10})$$

pays, respectively, the source-envelope, old-CGF, and new-residual restoration charges. Summing over all possible deleted sets is now an exact binomial calculation:

$$\sum_{u=1}^{\ell} \binom{\ell}{u} e^{-\eta_C \ell u} = (1 + e^{-\eta_C \ell})^{\ell} - 1 = o(1). \quad (\text{L6.11})$$

Including the  $u = 0$  perfect term proves the unconditional red probability bound

$$P_{R,\ell} \leq \mathcal{H}_C(\ell) \exp[-\mathcal{D}_{C,w,d}(\ell) - \mathcal{N}_{C,w,d}(\ell) + J_{\ell,d}] (1 + o(1)). \quad (\text{L6.12})$$

This proves extraction of the complete new exponent, rather than merely asserting that a source failure factor is large enough.

## 16. FIRST MOMENTS, RED-BLUE CROSSING AND ASYMPTOTICS

**16.1. The two explicit first-moment thresholds.** Fix  $C$  and  $w > \omega_0$ , and let  $\ell \rightarrow \infty$ . From (L2.12), (L2.18), (L6.1), and (L6.12),

$$\limsup_{\ell \rightarrow \infty} \frac{1}{\ell^2} \log P_{R,\ell} \leq \frac{1}{2} \log p - \vartheta_C \frac{a^3}{6p^3 D} - G(C, w) - \widehat{H}(C, w). \quad (\text{L7.1})$$

The discretization term vanishes because  $J_{\ell,d} = o_C(\ell^2)$ . Let  $n = \lfloor e^{\sigma \ell} \rfloor$ . Since  $\binom{n}{\ell} \leq n^\ell$ , the expected number of red  $K_\ell$ 's tends to zero whenever

$$\begin{aligned} \sigma &< -\frac{1}{2} \log p + \vartheta_C \frac{a^3}{6p^3 D} + G(C, w) + \widehat{H}(C, w) \\ &= -\frac{1}{2} \log p_C + \frac{B_R(C)}{2} + G(C, w) + \widehat{H}(C, w). \end{aligned} \quad (\text{L7.2})$$

This proves the red threshold directly from the probability ledger.

For  $q_\ell = \lfloor C\ell \rfloor$ , (L2.42) and  $q_\ell/\ell = C + O(\ell^{-1})$  give

$$\limsup_{\ell \rightarrow \infty} \frac{1}{\ell^2} \log P_{B,q_\ell} \leq \frac{C^2}{2} \log(1-p) + \frac{2a^3 C^3}{3(1-p)^3 D}. \quad (\text{L7.3})$$

The bound  $\binom{n}{q_\ell} \leq n^{q_\ell}$  therefore makes the expected number of blue  $K_{q_\ell}$ 's tend to zero whenever

$$\sigma < -\frac{C}{2} \log(1-p) - \frac{2a^3 C^2}{3(1-p)^3 D} = -\frac{1}{2} \log p_C + \frac{CB_B(C)}{2}. \quad (\text{L7.4})$$

No blue improvement is used. A union bound over red and blue vertex sets therefore produces a coloring with neither forbidden clique for every  $\sigma$  strictly below the minimum of the right sides of (L7.2) and (L7.4).

**16.2. The crossing is separated by a constant.** The pinned Gaussian asymptotics, all in the order  $C \rightarrow \infty$ , are

$$\begin{aligned} \frac{p}{p_C} &= 1 + O((\log C)^{-1}), & \frac{a^3}{p^3 D} &= \frac{1}{2} + o(1), & \vartheta_C &= 1 - o(1), \\ B_R(C) &= \frac{1}{6} + o(1), & CB_B(C) &= 1 + o(1). \end{aligned} \quad (\text{L7.5})$$

The exact integral (L2.17) also gives

$$G_*(C) = \frac{1 + o(1)}{32 \log C} = o(1). \quad (\text{L7.6})$$

Indeed  $m(c - \omega_0)/m_0 \rightarrow 1$ ,  $P_*(x) \rightarrow 1$  uniformly, and the bracket in (L2.17) tends uniformly to one, while  $m_0^4/D^2 = (1 + o(1))/(8 \log C)$ .

The controlled-residual constants satisfy

$$\begin{aligned} \rho &= o(1), & m'_0 &\rightarrow 1, & m_0 m''(c) &= o(1), \\ M_w/m_0 &\rightarrow 1, & j_\pm(w) &\rightarrow 1, & \mu_w &\rightarrow 1, \end{aligned} \quad (\text{L7.7})$$

where the right limit  $w \downarrow \omega_0$  is taken for each fixed  $C$  before  $C \rightarrow \infty$ , and

$$\hat{L}_w = o(m_0^2), \quad \hat{B}_w = m_0^2[1 + o(1)]. \quad (\text{L7.8})$$

Thus every positivity and feasibility condition used above holds for all sufficiently large fixed  $C$ , uniformly for  $w$  in a sufficiently small right-neighborhood of  $\omega_0$ .

Equations (L7.5)–(L7.8) show

$$\frac{CB_B(C)}{2} - \left[ \frac{B_R(C)}{2} + G_*(C) + \hat{H}_*(C) \right] = \frac{5}{12} + o(1) > 0. \quad (\text{L7.9})$$

This positive limiting gap, rather than the mere fact that the added terms are  $o(1)$ , proves that red remains the bottleneck. Taking the supremum of (L7.2) over  $w > \omega_0$ , using continuity as  $w \downarrow \omega_0$ , and then using (L7.9) proves Theorem 10.1.

**16.3. The new leading coefficient.** Finally,

$$m_0^2 = (2 + o(1)) \log C, \quad D^2 = (32 + o(1))(\log C)^3, \quad (\text{L7.10})$$

and hence

$$\hat{B}_w A_0 = (4 + o(1))(\log C)^2, \quad E_w = (2 + o(1))(\log C)^2. \quad (\text{L7.11})$$

Substitution into (L2.23) gives

$$\hat{H}_*(C) = \frac{(6 + o(1))(\log C)^2}{12(32 + o(1))(\log C)^3} = \frac{1 + o(1)}{64 \log C}. \quad (\text{L7.12})$$

The exact-merge predecessor is  $H_*(C) = (1 + o(1))/(96 \log C)$ , so the new term is eventually strictly larger. Since  $B_R(C)/2 \rightarrow 1/12$ , the public coarsening  $B_R(C)/2 \geq 1/20$  also follows after increasing the existential threshold.

**16.4. A precisely delimited scalar method cap.**

**Definition 16.1** (Admissible scalar residual family). Fix constants  $M_0, M_{\text{ext}} > 0$ . An  $(M_0, M_{\text{ext}})$ -admissible family  $\mathcal{E}$  assigns, for every sufficiently large  $C$ , every  $w$  in a right-neighborhood of  $\omega_0$ , and every  $0 \leq r \leq \ell$ , deterministic data

$$(e^{C,w,\ell,r}, T^{C,w,\ell,r}, \beta_{C,w,\ell,r}, \mu_{C,w,\ell,r}).$$

The same constants  $M_0, M_{\text{ext}}$  apply to every member of the family and to every  $C, w, \ell, r$ . At  $r = 0, 1$ , the arrays are empty and we set  $\beta = 0, \mu = 1$ , and  $\mathcal{G}_{\mathcal{E}} = 0$ .

(A1) The increments are fixed before exposure and

$$0 \leq e_{ij} \leq M_0 m_0^2 r \quad (1 \leq i < j \leq r). \quad (\text{L7.13})$$

(A2) The array is the unique solution of the generalized recursion

$$T_{ij} = t_0 + \gamma_d \sum_{\substack{q > i \\ q \neq j}} T_{jq} + e_{ij}, \quad \Delta_{ij} = T_{ij} - t_0 \geq 0. \quad (\text{L7.14})$$

Thus neither the recursion nor the admissible class refers to the particular choice  $e_{ij} = \theta(i-1)$ .

(A3) Put

$$S_i = \sum_{\{j,q\} \subset V_i} \Delta_{jq}, \quad R_i = \max_{j \in V_i} \sum_{q \neq j} \Delta_{jq}, \quad k = |V_i|.$$

The three-factor split is feasible at every step:

$$\frac{4t_0k}{d} + \frac{4R_i}{d} \leq \frac{1}{2}. \quad (\text{L7.15})$$

For  $r \geq 2$ , writing  $F_i^T$  for (L3.1) with coefficient array  $T$ , the two scalar coefficients obey the exact constraints

$$0 < \mu_{C,w,\ell,r} \leq \inf_{1 \leq i < r} \inf_{b \in [c-w, c+w] V_i} \lambda_{\min}(-\nabla^2 F_i^T(b)), \quad 0 \leq \beta_{C,w,\ell,r} \leq m_0^2. \quad (\text{L7.16})$$

With these coefficients, define

$$\lambda_i^\varepsilon = \mathbf{d}_{r-i,w,d} + \beta_{C,w,\ell,r} \frac{S_i}{d} - \sum_{j>i} \frac{e_{ij}^2}{2\mu_{C,w,\ell,r}d} - \frac{d^{-1/5}}{\sqrt{d}} \sum_{j>i} T_{ij}. \quad (\text{L7.17})$$

The family must certify, on every history in  $\mathfrak{H}_{r,i-1}^R$ , the one-step inequality

$$\mathcal{K}_i^T \leq \frac{\mathcal{H}_C(r-i+1)}{\mathcal{H}_C(r-i)} e^{-\lambda_i^\varepsilon}. \quad (\text{L7.18})$$

Thus this scalar class credits exactly the displayed  $\beta S_i/d$  term, pays the displayed scalar square penalty, and discards every favorable term not listed in (L7.17).

(A4) Relative to the exact weighted rule (S3), the family's credited deterministic contribution is the normalized functional

$$\mathcal{G}_\mathcal{E}(C, w, \ell, r) = \frac{1}{d} \left[ \beta_{C,w,\ell,r} \sum_{i=1}^{r-1} S_i - \frac{1}{2\mu_{C,w,\ell,r}} \sum_{1 \leq i < j \leq r} e_{ij}^2 \right]. \quad (\text{L7.19})$$

This definition excludes adaptive weights, matrix-valued completion, and unlisted source credits.

(A5) After deleting  $u$  vertices, relabeling in retained order, and rebuilding the family member, with  $x_+ = \max\{x, 0\}$ ,

$$0 \leq [\mathcal{G}_\mathcal{E}(C, w, \ell, \ell) - \mathcal{G}_\mathcal{E}(C, w, \ell, \ell - u)]_+ \leq \frac{M_{\text{ext}}}{\log C} \ell u \quad (\text{L7.20})$$

for every  $0 \leq u \leq \ell$ . Thus extraction is a quantified condition, not a reference to the optimizer-specific estimate.

Define

$$\begin{aligned} \text{Lead}_C(\mathcal{E}) &= \limsup_{w \downarrow \omega_0} \limsup_{\ell \rightarrow \infty} \frac{\mathcal{G}_\mathcal{E}(C, w, \ell, \ell)}{\ell^2}, \\ \mathfrak{M}_{M_0, M_{\text{ext}}}(C) &= \sup_{\mathcal{E}} \text{Lead}_C(\mathcal{E}), \end{aligned} \quad (\text{L7.21})$$

where the supremum is over  $(M_0, M_{\text{ext}})$ -admissible families. The right-neighborhood in the first limsup may depend on  $C$ , but not on  $\ell$  or  $r$ .

**Proposition 16.2** (Cap within Definition 16.1). *For every fixed  $M_0, M_{\text{ext}}$ , the leading functional in (L7.21) satisfies*

$$\limsup_{C \rightarrow \infty} 64 \log C \mathfrak{M}_{M_0, M_{\text{ext}}}(C) \leq 1. \quad (\text{L7.22})$$

*There are fixed absolute constants  $M_0^*, M_{\text{ext}}^*$  for which the family  $e_{ij} = \theta(i-1)$ ,  $\beta = \hat{B}_w(C)$ , and  $\mu = \mu_w$  is admissible and satisfies*

$$\lim_{C \rightarrow \infty} 64 \log C \text{Lead}_C(\mathcal{E}^*) = 1. \quad (\text{L7.23})$$

*Thus  $1/64$  is both the cap and an attained leading coefficient within the precisely delimited scalar class, for fixed admissibility constants large enough to contain  $\mathcal{E}^*$ .*

*Proof.* Fix an admissible family and abbreviate its two scalar coefficients by  $\beta$  and  $\mu$ . The exact constraint (L7.16) gives

$$\beta \leq m_0^2. \quad (\text{L7.24})$$

At  $b = c\mathbf{1}$ , the row condition (L7.15) and the diagonal formula (L-A.A.1) give, for every  $j \in V_i$ ,

$$\mu \leq \lambda_{\min}(-\nabla^2 F_i^T(c\mathbf{1})) \leq (-\nabla^2 F_i^T(c\mathbf{1}))_{jj} \leq m'_0 + \frac{1}{8}m_0 m''(c) = 1 + o(1). \quad (\text{L7.25})$$

Here  $t_0 k + R_i \leq d/8$ ,  $m'_0 \rightarrow 1$ , and  $m_0 m''(c) = o(1)$ . Hence every valid common scalar completion constant satisfies, uniformly over the class,

$$\mu \leq 1 + o(1). \quad (\text{L7.26})$$

Let  $\mathcal{L}$  be the feedback operator in (L7.14). Its row norm is at most  $\gamma_d r \leq m_0 m'_0 / D = \rho = o(1)$ . Writing

$$g_{ij} = A_0(r - i - 1),$$

the recursion for  $\Delta$  is  $\Delta = g + \mathcal{L}\Delta + e$ . Therefore

$$\Delta = (I - \mathcal{L})^{-1}g + e + O(\rho m_0^2 r) \quad (\text{L7.27})$$

uniformly on every edge, with an implicit constant depending only on  $M_0$ , by (L7.13). The exact exposure multiplicity is

$$\sum_{i=1}^{r-1} S_i = \sum_{1 \leq j < q \leq r} (j-1) \Delta_{jq}. \quad (\text{L7.28})$$

The first term of (L7.27) contributes

$$2A_0 \binom{r}{4} [1 + O(\rho)] \quad (\text{L7.29})$$

to (L7.28). The last term changes the numerator in (L7.19) by at most  $O_{M_0}(\rho m_0^4 r^4) = o(m_0^4 r^4)$ .

For the direct residual term, scalar square completion is pointwise:

$$\beta(j-1)e_{jq} - \frac{e_{jq}^2}{2\mu} \leq \frac{\mu\beta^2}{2}(j-1)^2. \quad (\text{L7.30})$$

Summing (L7.30) gives  $\mu\beta^2 K_r/2$ . Set  $r = \ell$ , divide (L7.19) by  $\ell^2$ , and use  $\binom{\ell}{4}/\ell^4 \rightarrow 1/24$ ,  $K_\ell/\ell^4 \rightarrow 1/12$ , and  $d/\ell^2 \rightarrow D^2$ . Equations (L7.24)–(L7.30) yield

$$\text{Lead}_C(\mathcal{E}) \leq \frac{\beta A_0}{12D^2} + \frac{\mu\beta^2}{24D^2} + o\left(\frac{m_0^4}{D^2}\right) \leq \frac{m_0^4}{8D^2}[1 + o(1)] = \frac{1 + o(1)}{64 \log C}. \quad (\text{L7.31})$$

The error is uniform for fixed  $M_0, M_{\text{ext}}$ , so taking the supremum and then the limsup proves (L7.22).

For  $e_{ij} = \theta(i-1)$ , equations (L5.5), (L7.7), and (L7.8) make (L7.30) asymptotically tight. Equations (L5.11), (L6.2), and (L4.4) verify (L7.15)–(L7.18), and (L7.13) holds with  $M_0^* = 1$ . To verify (L7.20), write  $\Delta^{(r)}$  for the array rebuilt at target  $r$ , and put  $r_- = \ell - u$ . Subtracting the two positive recursions (L7.14), and using  $\|\mathcal{L}\|_\infty \leq \rho < 1/2$ , gives absolute constants  $K_1, K_2$  such that

$$\max_{1 \leq j < s \leq r_-} \left| \Delta_{js}^{(\ell)} - \Delta_{js}^{(r_-)} \right| \leq K_1 m_0^2 u, \quad \max_{j < s \leq \ell} \Delta_{js}^{(\ell)} \leq K_2 m_0^2 \ell. \quad (\text{L7.32})$$

Indeed, the second bound is (L5.1). If  $U$  denotes the first maximum, subtraction of (L7.14) gives  $U \leq A_0 u + \rho U + \gamma_d u K_2 m_0^2 \ell$ . Now  $A_0 = O(m_0^2)$ ,  $\gamma_d \ell \leq \rho$ , and  $\rho < 1/2$  give the first bound. The multiplicity identity (L7.28) and (L5.4) now give

$$\left| \sum_i S_i^{(\ell)} - \sum_i S_i^{(r_-)} \right| \leq K_3 m_0^2 \ell^3 u, \quad |K_\ell - K_{r_-}| \leq K_4 \ell^3 u \quad (\text{L7.33})$$

for absolute  $K_3, K_4$ . Since  $\beta \leq m_0^2$ ,  $\mu_w \rightarrow 1$ , and  $m_0^4/D^2 = O(1/\log C)$ , (L7.33) proves (L7.20) for some absolute  $M_{\text{ext}}^*$ . Finally, the functional is exactly (L2.23) up to  $o(1/\log C)$ , and (L7.12) gives the asserted limit (L7.23).  $\square$

The proposition does not quantify over outcome-adaptive weights, matrix-valued square completion, higher-order potentials, a sharper source quadratic MGF, or a different extraction invariant.

## 17. REPRODUCIBILITY AND CLAIM BOUNDARY

The frozen pre-canonical route note, arithmetic checker, and its independent source-level review have SHA-256 prefixes 2234a769, 616040b9 and ef6ddbce, respectively. Their historical status labels do not override this manuscript's canonical theorem statement or proof-status ledger. The checker replays the recursion, multiplicity identities, Hessian samples, exact-CGF increments, finite margins at four large  $C$ -values and convergence to the leading constant. It is an diagnostic arithmetic witness, not a substitute for the reverse-induction proof.



From the repository root run

```
bash scripts/bootstrap.sh
bash scripts/reproduce_lower.sh
```

to reproduce the bridge, exact-CGF predecessor, nonexchangeable ledger and controlled-residual arithmetic calculations with pinned dependencies. A passing arithmetic replay is not labelled a theorem proof.

The theorem is source-relative, fixed- $C$ , asymptotic in  $\ell$ , and non-effective in  $C_0(K)$ . It supplies no new blue-side construction and no finite Ramsey number. The current manuscript does not claim a common- normalization strict improvement over the final printed Lin–Niu theorem, global optimality, priority, publication, or world-best status.

Logical proof status. As a standalone theorem without source interface 11.2, (L1.1) is PROVABLE AFTER WEAKENING / EXTRA ASSUMPTION, not an unconditional claim. Under the explicit items (S1)–(S5), the weighted one-step comparison, residual contribution, extraction algebra, and crossing argument are PROVABLE AS STATED. In particular, the manuscript does not replace the weighted source interface by numerical evidence.

### Part 3. A proof-carrying finite Ramsey barrier

#### 18. FINITE RAMSEY BARRIER: PROBLEM AND PROOF ARCHITECTURE

Write  $R(3, k)$  for the least  $n$  such that every graph on  $n$  vertices contains either a triangle or an independent set of order  $k$ . The current dynamic survey records

$$100 \leq R(3, 18) \leq 120;$$

see Radziszowski’s continuously maintained table [Rad26]. A triangle-free graph on 99 vertices with independence number 17 certifies the lower endpoint. Extending such a graph by one vertex produces a natural, but logically delicate, local search around the next possible certificate.

This part studies one labelled 100-vertex extension  $H$ . The graph has no independent 18-set but has a unique triangle. It is therefore a near miss, not a Ramsey graph. We ask a deliberately local question: how many *original* edges must be deleted before the triangle can be removed while arbitrary original nonedges are allowed to change into edges? This asymmetric edit model makes additions completely free; consequently, a lower bound in this model is stronger than a lower bound obtained by fixing the nonedges or by restricting the total symmetric-difference distance.

The principal result is a proof-carrying lower bound of seven on that local deletion radius. The argument has four independently inspectable layers.

- (i) A direct graph check establishes the seed identity, its unique triangle, and the absence of an independent 18-set.
- (ii) The unique triangle gives an exhaustive three-way branch split. A previous proof-carrying package excludes total deletion budget at most five, reducing the new layer to exact budget six.

- (iii) Each exact-six branch is translated into a finite CNF *relaxation* of the graph problem. All structural clauses are reconstructed from the seed; the independent-set clauses are validated as universally sound.
- (iv) CaDiCaL emits a DRAT refutation for each branch, and a separately pinned `drat-trim` executable verifies every refutation. Solver status strings are not used as proof evidence.

This layering is important. An unsatisfiable CNF proves the graph claim only after formula soundness and branch completeness have been established. A solver’s report of unsatisfiability proves the CNF claim only after an independently checkable proof trace has been accepted. We record both implications explicitly.

The computation is also informative about what remains open. We tested the next deletion layer, exact budget seven, with one bounded solver call in each branch. Every call reached its wall-time limit before producing a separated model or a proof. We record these endpoints as UNKNOWN. They are not evidence either for or against the existence of a seven-deletion repair. Claim boundary. The theorem concerns one fixed, labelled seed and one one-sided edit metric. It neither constructs a 100-vertex  $(3, 18)$ -Ramsey graph nor rules out such graphs elsewhere in the search space. Thus it does not improve either endpoint of the globally tabulated interval for  $R(3, 18)$ . The value of the result is instead methodological and local: it gives a complete, proof-carrying obstruction around a particularly close seed, with all negative conclusions separated from incomplete search.

## 19. THE FROZEN SEED AND THE PRECISE STATEMENT

All vertices are labelled by

$$V = \{0, 1, \dots, 99\}.$$

The input graph  $H$  is the adjacency matrix in the artifact

`r3_18_n100_nearmiss.txt`.

Its SHA-256 identity is

`e6527601f72ebb2d3aed11adcdd2ffb0c6aab5e326d81d439c10ee468844e85e`.

Content addressing is part of the mathematical statement: replacing the matrix, even by an isomorphic relabelling, changes the frozen input unless the new identity is separately certified.

**Lemma 19.1** (Seed facts). *The graph  $H$  has 827 edges, satisfies  $\alpha(H) < 18$ , and has exactly one triangle, namely*

$$T = \{97, 98, 99\}.$$

*Its first 99 vertices induce a graph  $G_0$  with 809 edges, no triangle, and  $\alpha(G_0) = 17$ .*

*Computational proof.* The matrix parser checks that the input is a symmetric  $100 \times 100$  binary matrix with zero diagonal. Exhaustive enumeration of the  $\binom{100}{3} = 161700$  triples finds only  $T$ . A bitset maximum-clique search on the complement returns no clique of order 18, which is equivalent to  $\alpha(H) < 18$ . The  $99 \times 99$  principal block agrees entry-for-entry with the separately frozen matrix  $G_0$ , whose SHA-256 is

3e9d16e29111f3a2f25ae3b992235804c2612de2f6ada5570901d17ceedfca45.

The same verifier finds no independent 18-set in  $G_0$  and returns an independent 17-set. It also finds no triangle in  $G_0$ . The new vertex has neighbourhood

$$N_H(99) = \{33, 34, \dots, 48, 97, 98\},$$

and the only edge induced by this neighbourhood is  $(97, 98)$ , providing a second direct check of uniqueness. The negative clique routine was compared with literal subset enumeration on 8640 small random graph/target pairs; all decisions and returned witnesses agreed. Reproduction commands are given in Section 24 and Appendix G.  $\square$

Evidence grade and logical role. The assertion  $\alpha(H) < 18$  is an exact deterministic trusted-code result: the frozen matrix and inspectable bitset search reproduce it, and exhaustive small-instance comparisons test the pruning invariant. It is not backed by a DRAT/LRAT trace or a proof-assistant certificate. This assertion describes why  $H$  is a near miss, but it is not a logical premise of Theorem 19.3. The deletion-barrier proof uses the fixed edge set and seed triangle of  $H$ , while the independence-number hypothesis is imposed on the putative final graph  $F$ . Thus the implication proved below does not infer anything from  $\alpha(H) < 18$ .

The graph  $G_0$  reproduces the already tabulated lower bound  $R(3, 18) \geq 100$ . The graph  $H$ , by contrast, contains a triangle and therefore does not certify  $R(3, 18) \geq 101$ .

**Definition 19.2** (Free-addition deletion distance). For a graph  $F$  on  $V$ , define

$$d_H(F) = |E(H) \setminus E(F)|.$$

Only missing input edges are charged. Every pair in  $\binom{V}{2} \setminus E(H)$  may independently be added to  $F$  at zero cost. Let

$$\mathcal{F} = \{F \text{ on } V : F \text{ is triangle-free and } \alpha(F) < 18\}$$

and define

$$\rho(H) = \inf\{d_H(F) : F \in \mathcal{F}\},$$

with the convention  $\inf \emptyset = \infty$ .

**Theorem 19.3** (Fixed-seed deletion barrier). *For the content-addressed graph  $H$  above,*

$$\rho(H) \geq 7.$$

Equivalently, no  $F \in \mathcal{F}$  satisfies  $d_H(F) \leq 6$ , even when arbitrary subsets of the 4123 original nonedges are added.

The convention in the definition matters: Theorem 19.3 is compatible both with  $\rho(H) = 7$  and with a larger or infinite radius. We do not determine which occurs.

## 20. REDUCTION TO SIX PROOF-CARRYING BRANCH FORMULAS

Let

$$f_0 = (97, 98), \quad f_1 = (97, 99), \quad f_2 = (98, 99)$$

be the three edges of the unique seed triangle. We first state the checked dependency at the preceding deletion layer.

**Proposition 20.1** (Budget-five dependency). *There is no  $F \in \mathcal{F}$  with  $d_H(F) \leq 5$ .*

*Proof-carrying computation.* Every triangle-free  $F$  omits at least one  $f_i$ . In branch  $i$ , the unit  $\neg x_{f_i}$  is imposed and a sequential counter permits at most four further deletions among the other 826 input edges. All triangle clauses and a finite bank of universally valid independent-18 hitting clauses are imposed. The resulting three CNFs have respectively 169382, 184967, and 177799 clauses. All three formulas were reconstructed clause-by-clause from  $H$ , and all three DRAT refutations were replayed with a freshly compiled checker from the pinned source revision. The full identities are in Table 7; the formula-soundness argument is the same as Proposition 21.2. Therefore each branch is impossible, and the three branches cover every candidate of total deletion budget at most five.  $\square$

By Proposition 20.1, a counterexample to Theorem 19.3 would have exactly six deleted input edges. Fix any one absent edge  $f_i$  of the seed triangle. The other five deletions are counted among  $E(H) \setminus \{f_i\}$ . Notice that the other two triangle edges are not fixed present; if either is also deleted, it simply consumes one of these five residual deletions. Thus the branches cover all candidates and may overlap.

For each  $i \in \{0, 1, 2\}$ , let  $\Phi_i$  be the formula defined precisely in Section 21. It contains the fixed negative unit for  $f_i$ , an exact-five counter on the residual seed edges, all triangle clauses, and a branch-specific bank of independent-set hitting clauses.

**Lemma 20.2** (Exact-six branch cover). *If  $F \in \mathcal{F}$  and  $d_H(F) = 6$ , then the edge assignment of  $F$  extends to a satisfying assignment of at least one of  $\Phi_0, \Phi_1, \Phi_2$ .*

*Proof.* Because  $F$  is triangle-free, at least one edge  $f_i$  of  $T$  is absent. Choose such an  $i$ . The chosen unit clause holds. Exactly five members of  $E(H) \setminus \{f_i\}$  are absent, so the primary edge assignment extends to the auxiliary variables of the exact-five counter. Triangle-freeness satisfies every triangle clause. Finally,  $\alpha(F) < 18$  implies that each installed 18-set contains a present

pair, so every installed hitting clause holds. Hence  $F$ 's primary assignment extends to a model of  $\Phi_i$ .  $\square$

**Proposition 20.3** (Certified branch refutations). *Each formula  $\Phi_0, \Phi_1, \Phi_2$  is unsatisfiable.*

*Proof certificate.* The three frozen DIMACS files have the identities and dimensions in Table 2. The corresponding DRAT traces were accepted by the pinned `drat-trim` checker, each reporting the exact status line `s VERIFIED`. Before replay, independent semantic checkers reconstructed the entire triangle prefix, cardinality block, fixed unit, and every 153-literal hitting clause; they also checked the compressed and raw identities. Details are given in Section 22 and Appendix G.  $\square$

*Proof of Theorem 19.3.* Suppose  $F \in \mathcal{F}$  and  $d_H(F) \leq 6$ . Proposition 20.1 implies  $d_H(F) = 6$ . By Lemma 20.2,  $F$ 's edge assignment extends to a model of at least one  $\Phi_i$ . This contradicts Proposition 20.3. Therefore no such  $F$  exists and  $\rho(H) \geq 7$ .  $\square$

## 21. CNF ENCODING AND ITS SOUNDNESS

**21.1. Primary variables and structural clauses.** For every unordered pair  $e = \{u, v\} \in \binom{V}{2}$ , a primary variable  $x_e$  records whether  $e \in E(F)$ . The 4950 primary variables are numbered in lexicographic pair order. Crucially, variables exist for all original nonedges; those variables are absent from the deletion counter and are not fixed false.

Triangle-freeness is encoded without separation: for every  $a < b < c$ , the formula contains

$$\neg x_{ab} \vee \neg x_{ac} \vee \neg x_{bc}.$$

There are exactly  $\binom{100}{3} = 161700$  such clauses.

For every installed set  $S \in \binom{V}{18}$ , the formula contains the positive hitting clause

$$C_S = \bigvee_{e \in \binom{S}{2}} x_e. \tag{F4.1}$$

It has  $\binom{18}{2} = 153$  distinct primary literals. Every graph with independence number below 18 satisfies  $C_S$ . We install only a finite selected bank  $\mathcal{B}_i$ , not all  $\binom{100}{18}$  clauses. Consequently,  $\Phi_i$  is a relaxation of the exact Ramsey branch. This is the useful direction: unsatisfiability of the relaxation implies unsatisfiability of the exact branch.

**21.2. Exact deletion counter.** In branch  $i$ , let

$$L_i = (\neg x_e : e \in E(H) \setminus \{f_i\})$$

be the ordered list obtained by restricting the lexicographic pair order to the seed edges other than  $f_i$ . The list has 826 literals. A PySAT sequential-counter equality encoding implements

$$\sum_{\ell \in L_i} [\ell \text{ is true}] = 5. \quad (\text{F4.2})$$

We use the standard sequential counter of Sinz [Sin05]. The concrete generator is exposed by PySAT [IMaMS18]; its semantics is not used as an unchecked black box.

For completeness, define the irredundant sequential schema explicitly. Given signed literals  $z_0, \dots, z_{n-1}$  and  $1 \leq q \leq n-2$ , introduce  $y_{r,j}$  for  $0 \leq r < q$  and  $0 \leq j < n-q$ , and let  $\text{SC}_{\leq q}(z)$  contain

$$\begin{aligned} \neg z_j \vee y_{0,j} & \quad (0 \leq j < n-q), \\ \neg y_{r,j} \vee y_{r,j+1} & \quad (0 \leq r < q, 0 \leq j < n-q-1), \\ \neg z_{j+r+1} \vee \neg y_{r,j} \vee y_{r+1,j} & \quad (0 \leq r < q-1, 0 \leq j < n-q), \\ \neg z_{j+q} \vee \neg y_{q-1,j} & \quad (0 \leq j < n-q). \end{aligned}$$

**Lemma 21.1** (Sequential-counter projection semantics). *For every assignment of the primary variables,*

$$\exists y \text{ SC}_{\leq q}(z, y) \iff \sum_{j=0}^{n-1} [z_j \text{ is true}] \leq q. \quad (\text{F4.3})$$

Consequently, for  $n = 826$ ,

$$\text{SC}_{\leq 821}(\neg z) \wedge \text{SC}_{\leq 5}(z)$$

has an auxiliary extension if and only if exactly five members of  $z$  are true. The budget-five block  $\text{SC}_{\leq 4}(z)$  has an auxiliary extension if and only if at most four members are true.

*Proof.* Suppose first that at most  $q$  input literals are true. Assign  $y_{r,j} = 1$  exactly when at least  $r+1$  of  $z_0, \dots, z_{j+r}$  are true. The first clause family records the first threshold crossing. The second propagates an attained threshold to a longer prefix. In the third family, truth of both  $z_{j+r+1}$  and  $y_{r,j}$  raises the prefix count from at least  $r+1$  to at least  $r+2$ . The last family cannot be violated, because such a violation would exhibit  $q+1$  true inputs. This gives the required auxiliary extension.

Conversely, induction on  $r$  shows that the clauses force  $y_{r,j} = 1$  whenever at least  $r+1$  of  $z_0, \dots, z_{j+r}$  are true. For  $r = 0$ , use the first clause at the position of a true input and then the propagation clauses. For the induction step, let  $p$  be the position of the  $(r+1)$ -st true input and put  $j' = p - r$ . The induction hypothesis forces  $y_{r-1,j'}$ ; the third family, together with  $z_p$ , forces  $y_{r,j'}$ , and propagation reaches  $y_{r,j}$ . If  $q+1$  inputs were true, let  $p$  be the position of the  $(q+1)$ -st and put  $j = p - q$ . The preceding argument forces  $y_{q-1,j}$ , while the last family forbids its conjunction with  $z_p$ , a contradiction. This proves (F4.3). Applying (F4.3) to  $z$  and to its 826 complements gives

respectively the upper and lower sides of the equality claim; the at-most-four claim is the direct case  $q = 4$ .  $\square$

In the exact-five DIMACS block, the complement counter at bound 821 comes first and has 4105 auxiliary variables and 7394 clauses; the positive counter at bound 5 has another 4105 auxiliaries and 9026 clauses. Thus the equality block has 8210 auxiliary variables and 16420 clauses, ending at variable 13160. The checker `routes/finite/check_r3_18_seqcounter.py` does not import or call PySAT’s cardinality generator. It independently generates the four clause families, compares all six actual counter blocks clause-for-clause in their canonical literal order, checks their digests and variable ranges, and constructs the prefix-threshold auxiliary assignment from the proof above. Exhaustive small-instance tests enumerate both primary and auxiliary assignments and verify both directions of the projected at-most and equality semantics.

Thus

$$\Phi_i = \left( \bigwedge_{a < b < c} (\neg x_{ab} \vee \neg x_{ac} \vee \neg x_{bc}) \right) \wedge \text{Card}_{=5}(L_i) \wedge \neg x_{f_i} \wedge \bigwedge_{S \in \mathcal{B}_i} C_S. \quad (\text{F4.4})$$

TABLE 2. Exact-six branch dimensions. All branches have 4950 primary variables, 8210 counter auxiliaries, 161700 triangle clauses, 16420 counter clauses, and one fixed unit.

| branch | fixed absent edge | hitting clauses | variables | total clauses |
|--------|-------------------|-----------------|-----------|---------------|
| 0      | (97, 98)          | 251771          | 13160     | 429892        |
| 1      | (97, 99)          | 63943           | 13160     | 242064        |
| 2      | (98, 99)          | 5422            | 13160     | 183543        |

**Proposition 21.2** (CNF soundness). *For each  $i$ , every graph  $F \in \mathcal{F}$  with  $d_H(F) = 6$  and  $f_i \notin E(F)$  induces a primary assignment that extends to a model of  $\Phi_i$ .*

*Proof.* The triangle block holds by triangle-freeness. The fixed unit holds by the branch premise. Since  $f_i$  is one of exactly six missing input edges, exactly five literals in  $L_i$  are true; Lemma 21.1 supplies an auxiliary assignment. Each  $S \in \mathcal{B}_i$  contains a present pair because  $\alpha(F) < 18$ , so every  $C_S$  holds. No condition was imposed on an original nonedge except through triangle and hitting clauses, hence arbitrary additions remain represented.  $\square$

**21.3. Sound reuse of the branch-0 cut union.** Branch 0 uses 251771 distinct 18-sets deduplicated from seven checked sources. The ordered mask digest is

`f10690b826b86eb03567a2ffa2fb553801fae32af20cfe4337118bddf4e41afa.`

TABLE 3. Audited construction of the branch-0 universal cut union.

| source                        | source masks | new masks | cumulative |
|-------------------------------|--------------|-----------|------------|
| branch-0 discovery checkpoint | 183133       | 183133    | 183133     |
| branch-0 earlier bank         | 141149       | 4096      | 187229     |
| branch-1 bank                 | 63943        | 52515     | 239744     |
| branch-2 bank                 | 5422         | 3931      | 243675     |
| budget-five branch 0          | 287          | 286       | 243961     |
| budget-five branch 1          | 15872        | 4262      | 248223     |
| budget-five branch 2          | 8704         | 3548      | 251771     |

TABLE 4. Exact-six proof artifacts and replay scale. Hashes are abbreviated here and given in full in Table 8.

| branch | raw DRAT bytes | proof lines | compressed bytes | replay   |
|--------|----------------|-------------|------------------|----------|
| 0      | 3493847713     | 6649939     | 656144450        | verified |
| 1      | 2137671968     | 5590150     | 433552903        | verified |
| 2      | 759008359      | 2837771     | 152784369        | verified |

Only vertex-set masks were transferred. No learned clause, branch assumption, or solver conclusion was reused. Since every clause  $C_S$  in (F4.1) follows solely from  $\alpha(F) < 18$ , it is valid in every branch. The source counts and cumulative deduplication totals appear in Table 3.

## 22. PROOF PRODUCTION, CHECKING, AND TRUST

**22.1. From finite formulas to checked refutations.** The final proof-producing runs used CaDiCaL [BFF<sup>+</sup>24] at source commit

c60730422e758ef1cebe7aeddff2dda31c996bf04.

An exit code 20 and the text `UNSATISFIABLE` were treated only as requests to check the emitted proofs. The final mathematical evidence is the successful replay of the DRAT traces [WHH14, Heu16] by the official `drat-trim` source at commit

2e3b2dc0ecf938addbd779d42877b6ed69d9a985.

The principal recorded executable has SHA-256

31d9e8e04bc76a65f0a18ea212ac44af4cf7bc398629495cec093ced772c50f4.

All three checks terminated with exit code zero and reported an exact `s VERIFIED` status line.

For branch 0, a fresh replay in a temporary directory took 273.448 seconds inside the checker and reported 135152646 resolution steps in the retained core. Branch 1 reported 209897525 resolution steps in its independently checked run. These timings are descriptive, not logical premises. The source commit and successful replay are the portable evidence; the executable hash



records the historical build and is not required to match a fresh binary compiled on another platform.

**22.2. Semantic reconstruction.** The proof checker establishes unsatisfiability of the bytes in each DIMACS file. Separate semantic checkers establish that those bytes are the intended relaxations. For every branch they independently:

- (1) parse the frozen seed and regenerate the lexicographic pair-variable map;
- (2) compare all 161700 triangle clauses in order;
- (3) regenerate the production PySAT counter and, in a separate checker that does not import **CardEnc**, independently generate the sequential-counter schema and compare its variables and clauses in order;
- (4) check the branch unit and the exact clause partition;
- (5) reconstruct every hitting clause as all 153 pair variables of one 18-vertex mask;
- (6) verify the compressed and raw sizes and SHA-256 identities; and
- (7) verify the budget-five dependency and the three-edge cover.

Branch 0 additionally reparses all seven cut sources and reproduces the deduplicated union. Branches 1 and 2 likewise reconstruct their complete stored cut banks. An earlier interrupted branch-1 trace fails gzip integrity, remains quarantined, and is excluded from every manifest and conclusion.

**22.3. Trust boundary.** The claim is proof-carrying, not formally verified in a proof assistant. Its earliest machine dependencies are:

- the bitset independent-set verifier used for the descriptive seed fact  $\alpha(H) < 18$ , which is not a premise of the deletion-barrier implication;
- the **drat-trim** proof kernel, together with its compiler and execution hardware; and
- the independently inspectable scripts that map matrices, sequential schemas, and masks to DIMACS.

The audit narrows this boundary through independent matrix parsing, exhaustive triangle enumeration, small-instance cross-checks of the clique routine, exhaustive small-instance projection tests for the independent counter schema, constructive auxiliary assignments, complete clause reconstruction, content hashes, gzip integrity tests, and proof replay with a freshly compiled checker. PySAT remains production provenance, but Lemma 21.1 and the **CardEnc**-free comparison mean that its counter semantics is no longer assumed merely because PySAT emitted the bytes. A future LRAT conversion and formally verified LRAT checker could reduce the trusted kernel further, but is not required for the conventional proof-carrying SAT standard used here.

### 23. THE UNRESOLVED EXACT-SEVEN LAYER

The theorem implies only  $\rho(H) \geq 7$ . To distinguish this lower bound from an exact value, we report the first bounded experiment at deletion budget seven.

In branch  $i$ , the fixed unit  $\neg x_{f_i}$  accounts for one deletion and an equality counter requires exactly six of the other 826 input edges to be absent. All 4123 original nonedges remain free. The universal bank of 251771 hitting clauses from branch 0 is valid in every branch and was preloaded. Every initial formula therefore had 4950 primary variables, 9840 counter auxiliaries, 161700 triangle clauses, 19680 counter clauses, one fixed unit, and 251771 hitting clauses, for a total of 433152 clauses.

*Computational observation 23.1* (Bounded endpoints, not a theorem). One solver call was launched per branch with a parent-enforced 300-second discovery wall. Branch 0 used CaDiCaL, branch 1 Glucose 4.2, and branch 2 MapleChrono. Each remained inside its first SAT call until the wall expired. No branch produced a model that reached independent-set separation, and no branch produced a checked refutation. The endpoint of all three branches is therefore UNKNOWN.

The zero recorded separation iterations means that no SAT model returned before interruption; it does not mean the formula is unsatisfiable. The recoverable state consists of the completely reconstructed initial formula and universal cut bank. Internal CDCL state was not serialized. Exact hashes of the frozen result and checkpoint records are listed in Appendix H.

Accordingly, none of the following is established: a repair with exactly seven deletions; nonexistence of such a repair; the exact value of  $\rho(H)$ ; or a 100-vertex (3, 18)-Ramsey graph. A machine-readable Boolean field whose default value is false is likewise not evidence of nonexistence unless the companion establishment flag and a checked proof support it.

### 24. REPRODUCIBILITY AND ARTIFACT PUBLICATION

**24.1. Two verification tiers.** The public source repository [Xin26] separates lightweight tracked checks from large proof replay. Its accompanying evidence release is designated at <https://github.com/jackal092927/ramsey-bounds-reproducible/releases/tag/evidence-2026-08-30> and is valid only if it contains exactly the twelve compressed CNF/DRAT payloads listed in `artifacts/MANIFEST.tsv`. Publication, immutability, credential-free download, and final-tag replay are release gates recorded separately; they are not premises of the graph theorem.

Tier 1: source and semantics. Python 3.11.15 and the locked environment reproduce graph verification, unit tests, the PySAT-independent sequential-counter schemas and their frozen block digests, and exact-seven frozen-state checks. From the repository root, the lightweight entry point is

```
bash scripts/bootstrap.sh
.venv/bin/python reproduce.py quick
```

The proof-critical Python packages are pinned to `python-flint==0.9.0`, `mpmath==1.4.1`, `python-sat==1.9.dev13`, and `six==1.17.0`. The finite route uses PySAT; the other entries are repository-wide proof dependencies. Tier 2: large proof replay. Once published, the release is downloaded without GitHub credentials by

```
bash scripts/download_release_artifacts.sh
```

The helper first requires the complete release asset-name set to equal the manifest exactly and then checks the content before decompression:

```
.venv/bin/python scripts/verify_artifacts.py \
  --directory /path/to/release-assets --exact
```

The exact mode also rejects unmanifested files, so a stale or mixed download directory cannot masquerade as the immutable release payload. Build `drat-trim` from the pinned source commit with `scripts/build_drat_trim.sh`; a platform-specific executable hash is recorded but not used as a cross-platform acceptance condition. The one integrated command

```
.venv/bin/python reproduce.py finite-heavy \
  --artifact-dir /path/to/release-assets \
  --drat-trim /absolute/path/to/drat-trim
```

materializes a temporary overlay containing tracked semantic metadata and the twelve release payloads. It compares the independent sequential schemas with all six actual counter blocks, reconstructs all six formulas, replays all six proofs, and audits the exact-seven state. A proof is accepted only when the checker exits zero and prints an exact `s VERIFIED` status line.

**24.2. Why large proofs are release assets.** The six exact-six compressed files occupy about 1.26 GB, while the three raw DRAT traces occupy about 6.39 GB. Adding the smaller budget-five proofs to ordinary Git history would still make cloning needlessly expensive. The repository therefore tracks code, matrices, cut banks, semantic records, tests, the paper, and a content-addressed manifest. CNF and DRAT payloads are designated for distribution as immutable GitHub Release assets, keyed by SHA-256 and byte count. No DOI-bearing archival mirror is claimed. This design keeps ordinary clones reviewable while preserving bit-for-bit replay.

The release is complete only if it includes both the three budget-five proof pairs used by Proposition 20.1 and the three exact-six proof pairs used by Proposition 20.3. Their identities are listed in Tables 7 and 8. Incomplete, corrupt, and exploratory traces are explicitly excluded.

**24.3. Reproducibility contract.** A reproduction should report four logically distinct outcomes:

- (1) the seed and artifact hashes match;
- (2) semantic reconstruction passes;
- (3) each required DRAT replay is verified; and
- (4) the exact-seven endpoint remains UNKNOWN, unless a new, separately reviewed certificate supersedes it.

Passing only a solver run, only a hash check, or only unit tests does not by itself reproduce the full theorem.

## 25. SCOPE, LIMITATIONS, AND CONCLUSION

We established a complete local statement: every triangle-free graph with independence number below 18 on the labelled vertex set of  $H$  deletes at least seven of  $H$ 's 827 edges, even if additions among all original nonedges are free. The proof is carried by six checked SAT refutations: three for budget at most five and three for the exact-six layer.

The conclusion does not transfer automatically to an isomorphism class, to a different extension neighbourhood, or to a symmetric edit metric. It does not rule out a remote 100-vertex construction, and it does not improve the global bound on  $R(3, 18)$ . The exact-seven layer remains open. These are mathematical scope restrictions, not merely caveats about running time.

The main methodological lesson is that a computational near miss can support a rigorous negative result without being oversold as a Ramsey-number improvement. A content-addressed seed, a proof of case coverage, independently reconstructed CNF semantics, and replayable UNSAT proofs together establish a precise local barrier. Keeping bounded UNKNOWN runs outside the theorem makes the remaining target equally precise: either exhibit a seven-deletion repair and verify it directly, or produce proof-carrying refutations of all exact-seven branches.

## 26. UNIFIED REPRODUCIBILITY AND PLANNED PUBLIC ARTIFACT RELEASE

The public source repository at <https://github.com/jackal092927/ramsey-bounds-reproducible> exposes one canonical paper and a tiered verification entry point. The evidence package is designated for a future release at <https://github.com/jackal092927/ramsey-bounds-reproducible/releases/tag/evidence-2026-08-30> and is valid only if it contains exactly the twelve compressed CNF/DRAT payloads listed in `artifacts/MANIFEST.tsv`. At the date of this draft, the final tag and immutable twelve-asset Release have not been published; credential-free asset verification and final-tag replay therefore remain pending gates outside the mathematical argument.

The requested ChatGPT Pro second-opinion review has completed. Every recoverable concrete objection was checked against the canonical upper, lower, and finite sources and either repaired or explicitly dispositioned. This AI-assisted author-side review is not external human peer review and does not change any theorem's stated trust boundary. A clean checkout with Python 3.11.15, exactly `uv 0.10.2`, and the locked dependencies is prepared by

```
bash scripts/bootstrap.sh
```

The bootstrap fails closed when that `uv` version is absent and has no unhashed package-manager fallback. The following tiers intentionally make different guarantees.

| Tier                | Coverage                                                                                                                   | Logical scope                                                                           |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>quick</b>        | Four current upper checkers, the current lower arithmetic replay, finite unit/seed checks, and tracked graph certificates. | Fast integrity check; it does not replay the large finite DRAT proofs.                  |
| <b>full</b>         | <b>quick</b> , the complete long upper interval chain and the full lower checker suite.                                    | Replays the computer-assisted upper layer and the lower arithmetic witnesses.           |
| <b>finite-heavy</b> | Hash verification, semantic CNF reconstruction, and six DRAT replays with the pinned checker.                              | Replays the proof-carrying finite theorem for the fixed seed.                           |
| <b>paper</b>        | Deterministic materialization and compilation of this single manuscript.                                                   | Checks source integration and typesetting, not the mathematical certificates by itself. |
| <b>sources</b>      | Networked downloads of the three pinned arXiv members and five upstream graph matrices.                                    | Verifies source identities; it is separate from the offline theorem replay.             |

---

The fast and full tiers are invoked as

```
.venv/bin/python reproduce.py quick
.venv/bin/python reproduce.py full
.venv/bin/python reproduce.py sources
```

Large compressed CNF/DRAT pairs are designated for distribution as immutable, manifest-verified Release assets rather than ordinary Git objects. Once the Release is published, download and authenticate every entry of `artifacts/MANIFEST.tsv` without GitHub credentials by

```
bash scripts/download_release_artifacts.sh
```

Then build the source-pinned checker without imposing a cross-platform binary hash:

```
bash scripts/build_drat_trim.sh /absolute/new/checker-directory
```

The checker executable hash is recorded as current-run provenance. The proof-carrying tier is

```
.venv/bin/python reproduce.py finite-heavy \
  --artifact-dir artifacts/downloads \
  --drat-trim /absolute/path/to/drat-trim
```

The canonical PDF is generated by `make paper`; `make papers` is retained as an alias. Historical part manuscripts under `papers/upper`, `papers/lower`,

and `papers/finite` are source components and provenance records, not separate publication outputs.

The build fixes `SOURCE_DATE_EPOCH`. Two clean builds made with the recorded pdf $\text{\TeX}$  1.40.24 ( $\text{\TeX}$  Live 2022) and Latexmk 4.77 toolchain were byte-identical. Byte identity across different  $\text{\TeX}$  distributions, font packages, or engines is not claimed.

The download helper first requires the release asset-name set to equal the manifest names exactly. The release manifest then authenticates compressed artifacts before decompression. The finite-heavy verifier then reconstructs the structural clauses and independent-set hitting clauses, checks the exact branch counters, and replays the proof traces. This ordering prevents a valid proof of an unrelated or malformed CNF from being mistaken for evidence about the graph theorem.

## 27. CONCLUSIONS, NON-IMPLICATIONS, AND OPEN PROBLEMS

The three Parts give complementary information about Ramsey bounds without pretending that their techniques or normalizations coincide. The diagonal upper result isolates an exact-diagonal correlation gain inside a retained-spine transfer. The fixed-ratio lower result shows how a controlled deterministic residual can be paid by full-box strong concavity. The finite result turns a near miss into a precise, proof-carrying local obstruction.

Three restrictions remain part of the conclusions. First, the upper theorem is source-relative and asymptotic, with no effective finite- $k$  threshold. Second, the lower theorem is conditional on the pinned items (S1), (S2), (S4), and (S5), plus the additional weighted-extension hypothesis (S3), which is not attributed as a theorem stated by HMS or Lin–Niu. After the source constant  $K$  is fixed, it fixes sufficiently large  $C$  before taking  $\ell \rightarrow \infty$ , then sends  $w \downarrow \omega_0$  before studying  $C \rightarrow \infty$ . It has no effective  $C_0(K)$  and makes no cross-normalization priority claim. Third, the finite theorem does not imply  $R(3, 18) \geq 101$ , and the exact-seven layer remains UNKNOWN.

The most direct next steps are correspondingly separate: strengthen the correlation or outer partition in Part I; seek adaptive or higher-order residual potentials beyond the scalar class of Part II; and either exhibit a seven-deletion repair or produce checked exact-seven refutations in Part III. Any future improvement should retain the same separation between discovery, human-readable implication, independently replayable certificate, and open computational state.

Completion of the ChatGPT Pro second-opinion review does not close the separate public-artifact gate: the final tag, immutable Release, and credential-free final-tag replay remain pending at the date of this draft.

## APPENDIX A. EXACT CERTIFICATE DATA

The proof specification is the immutable file `routes/upper/EXACT_DIAGONAL_NEXT_CANDIDATE.md`. Its SHA-256 is

e4ea58def640593690a7545e111c3b38f1bfcf8a5735fe7985600481e0bf36d4.

The exact constants are

$$\begin{aligned} u_0 &= 1235783/500000, & \sigma_0 &= 10^{-8}, & D_0 &= 88053/10^9, \\ \beta &= 330867/500000, & \varepsilon &= 6893/10^7, \\ C &= 12366348252219/1250000000000. \end{aligned}$$

The compact ratio proof uses 65,536 cells on  $[u_0, 2.9]$ ; the exact diagonal uses 131,072 cells on  $[0, 20]$ . Both tails are proved by explicit monotone envelopes. The outer checker verifies the full ordered wedge and the final decimal rounding.

The repository integrity checker validates all six frozen proof files before running any expensive enclosure. In particular, a successful numerical run against a modified proof specification is not accepted as a reproduction of this theorem.

**A.1. Pinned source interface.** The imported source member is `main.tex` from Yang–Mao v1, arXiv:2608.01962v1. Its SHA-256 is

155b7104ec5b6935a576ae9f2b161976a966b0b46bd2b69153c0934ca688da2a.

The exact mapping for the conclusions actually imported in source interface 4.2 is:

| source lines | imported conclusion                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------|
| 318–332      | preliminary colour spines, common reservoir, reservoir-size lower bound, and minimum colour degrees |
| 1146–1565    | parameterized monochromatic-book theorem and proof with the exact $\rho_\beta, \Pi, \Xi$ constants  |
| 1569–1618    | compatibility of preliminary spines with the final book                                             |

For source lineage, but not as additional imported assumptions, the analytic antecedents are:

| source lines     | locally used lineage                                                                                                             |
|------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 370–418          | positive tensor-moment lemma, reproved in the specialized form Lemma 6.5                                                         |
| 426–791, 793–983 | generic root-filter, separator, and expectation–tail argument; the stronger exact-diagonal specialization is proved in Section 6 |
| 989–1009         | definition and parameter domain of $\mathcal{G}_2^{(3)}(\beta, C)$ , restated in (U2.2a)                                         |
| 1017–1142        | density-increment lemma internal to the pinned source proof of the imported book theorem                                         |

The source archive is downloaded on demand and checked against this digest; it is not silently replaced by a later arXiv version. The manuscript

proves the stronger local correlation input (U2.7); only the regularization, parameterized-book, and compatibility conclusions in the first table are imported.

**A.2. Outer-wedge certificate data.** All outer parameters are exact terminating rationals. Substitution in (U2.3)–(U2.4) gives outward-rounded enclosures

$$q \in [0.7559369286692751849342481567047, \\ 0.7559369286692751849342481567048],$$

$$\Xi \in [940.3014597301789400852772355749, \\ 940.3014597301789400852772355750].$$

The proof-critical lower or upper endpoints are:

| quantity                     | certified endpoint                    |
|------------------------------|---------------------------------------|
| $\sup_{(0,1]} U''$           | $< -0.3602795262399473$               |
| degree slack                 | $> 1.5056762000000000 \cdot 10^{-9}$  |
| red boundary derivative      | $< -3.6952685632128072 \cdot 10^{-4}$ |
| red page margin              | $> 5.1006707822905727 \cdot 10^{-9}$  |
| blue diagonal derivative     | $< -1.8476200284472833 \cdot 10^{-4}$ |
| blue page margin             | $> 3.4825658438202073 \cdot 10^{-6}$  |
| reservoir margin             | $> 6.0468779601096049 \cdot 10^{-6}$  |
| base upper endpoint          | $< 3.7806852883796401133524$          |
| safe-decimal rounding margin | $> 1.6203598866476283 \cdot 10^{-9}$  |

The primary outer checker has SHA-256

`e2035cbffefcb147141fcee4831cac2af085f26d9183091ee20d90acc89ac87b,`

the non-importing replay has SHA-256

`fc379e3b861b69054aadaa80ebc3c791cb8358f22c9b0aa01070c56aa131c26c,`

and the adaptive referee replay has SHA-256

`3c30f72ddc24848f2b72c278f35b5d8bd6296ae8a40b045d30c5b7e087bd7150.`

They use different subdivision and endpoint-reduction structures, but all three trust python-flint/Arb containment and the same exact mathematical specification; they are not described as independent formal proofs.

## APPENDIX B. STANDALONE PROOF OF THE GNNW BOOK LEMMA

Date: 2026-08-12

Source being repaired: Gupta–Ndiaye–Norin–Wei, arXiv:2407.19026v1, Lemmas 7–11 and Theorem 12.



**B.1. Claim boundary.** This note proves the combinatorial input used by the certified diagonal Ramsey descent without invoking GNNW Lemma 11 or Theorem 12 as black boxes. It retains their definitions of a candidate and of the rate region, and it uses only elementary Ramsey recursion, convexity, averaging, and induction.

Let a **candidate** be a pair of nonempty disjoint vertex sets  $(X, Y)$  in a red–blue complete graph. Its red cross-density is

$$d(X, Y) = \frac{e_R(X, Y)}{|X||Y|}.$$

It is  $(k, \ell, t)$ -good if  $X \cup Y$  contains a red  $K_k$ , or  $X$  contains a blue  $K_t$ , or  $Y$  contains a blue  $K_\ell$ .

Let  $\mathcal{R}$  be the closure of the set of pairs  $(x, y) \in (0, 1)^2$  for which, for all sufficiently large  $k + \ell$ ,

$$R(k, \ell) \leq x^{-k}y^{-\ell},$$

and let  $\mathcal{R}_*$  be its interior.

**B.1.1. Theorem A (book induction).** Suppose  $0 < \mu_0, x_0, y_0, p < 1$ ,

$$x_0 < p^{1/(1-\mu_0)}(1 - \mu_0), \quad (x_0, y_0) \in \mathcal{R}_*. \quad (\text{U-B.A.1})$$

There is  $L_0$  such that, for all positive integers  $k, \ell, t$  with  $\ell \geq L_0$ , every candidate of density at least  $p$  and satisfying

$$|X||Y| \geq x_0^{-k}y_0^{-\ell}\mu_0^{-t} \quad (\text{U-B.A.2})$$

is  $(k, \ell, t)$ -good.

**B.1.2. Corollary B (balanced-book bound).** Suppose  $0 < \mu, x, y, p < 1$ ,

$$x < p^{1/(1-\mu)}(1 - \mu), \quad (x, y) \in \mathcal{R}_*. \quad (\text{U-B.B.1})$$

For all sufficiently large  $\ell$ , every red–blue coloring of  $K_N$  of red density at least  $p$ , where

$$N \geq x^{-k/2}(\mu y)^{-\ell/2}, \quad (\text{U-B.B.2})$$

contains a red  $K_k$  or a blue  $K_\ell$ .

The statements hold for all  $p, \mu \in (0, 1)$ ; no relation  $p > \mu$  is needed.

## B.2. Four elementary inputs.

**B.2.1. Lemma 1 (an interior point gives its own eventual bound).** If  $(u, v) \in \mathcal{R}_*$ , then, for all sufficiently large  $a + b$ ,

$$R(a, b) \leq u^{-a}v^{-b}. \quad (\text{U-B.1.1})$$

**Proof.** Choose  $\eta > 0$  so that  $(u + 2\eta, v + 2\eta) \in \mathcal{R}$ . Since  $\mathcal{R}$  is the closure of the eventual-bound set, there is a point  $(u', v')$  in that set with  $u' > u$  and  $v' > v$ . Its defining bound is stronger than (U-B.1.1).  $\square$

B.2.2. *Lemma 2 (density convexity).* For every candidate,

$$\sum_{v \in X} d(X, N_R(v) \cap Y) |N_R(v) \cap Y| \geq e_R(X, Y) d(X, Y), \quad (\text{U-B.2.1})$$

where an empty-neighborhood summand is interpreted as zero.

**Proof.** Put  $d_y = |N_R(y) \cap X|$ . Multiplying the left side by  $|X|$  gives  $\sum_{y \in Y} d_y^2$ . Cauchy-Schwarz bounds this below by  $|Y|^{-1} (\sum_y d_y)^2 = e_R(X, Y)^2 / |Y|$ , which is  $|X|$  times the right side.  $\square$

B.2.3. *Lemma 3 (blue-book extraction).* Let  $0 < \nu < 1$ , and let  $b, k, m$  be positive integers with  $m \geq 5\nu^{-1}b^2$  and  $b \leq m$ . Suppose  $|X| \geq 5m^2$  and at least  $R(k, m)$  vertices  $v \in X$  have blue degree at least  $\nu|X|$  inside  $X$ . Then  $X$  contains a red  $K_k$ , or it contains disjoint sets  $S, T$  such that  $S$  is a blue clique, every edge from  $S$  to  $T$  is blue, and

$$|S| = b, \quad |T| \geq \frac{\nu^b}{2} |X|. \quad (\text{U-B.3.1})$$

**Proof.** The high-blue-degree vertices contain a red  $K_k$  or a blue  $K_m$ . In the latter case call its vertex set  $U$ . Put  $d_z = |N_B(z) \cap U|$  for  $z \in X \setminus U$  and let  $D$  be their average. Counting ordered blue pairs from  $U$  to  $X \setminus U$  gives

$$D \geq \frac{m(\nu|X| - m)}{|X| - m} = \nu m - \frac{(1 - \nu)m^2}{|X| - m} > \nu m - \frac{1}{4}, \quad (\text{U-B.3.2})$$

where  $|X| \geq 5m^2$  and  $m \geq 1$  were used in the last inequality. Let  $q = \lfloor D \rfloor$ . The discrete convexity of  $j \mapsto \binom{j}{b}$  on nonnegative integers (with value zero for  $j < b$ ) shows that a uniform  $b$ -subset  $S$  of  $U$  satisfies

$$\mathbb{E}|N_B(S) \cap (X \setminus U)| = \binom{m}{b}^{-1} \sum_{z \in X \setminus U} \binom{d_z}{b} \geq \binom{m}{b}^{-1} \binom{q}{b} |X \setminus U|. \quad (\text{U-B.3.3})$$

Since  $q > D - 1 > \nu m - 5/4$  and  $m \geq 5\nu^{-1}b^2$ , every  $0 \leq i < b$  satisfies

$$\frac{q - i}{m - i} > \nu - \frac{(1 - \nu)i + 5/4}{m - i} \geq \nu \left( 1 - \frac{b + 1/4}{5b^2} \right).$$

For the second inequality, the fraction being subtracted increases with  $i$ , and at  $i = b - 1$  it is at most the claimed value because

$$\frac{(1 - \nu)(b - 1) + 5/4}{5b^2/\nu - (b - 1)} \leq \frac{\nu(b + 1/4)}{5b^2};$$

after cross-multiplication, the difference between the right numerator and the left numerator is  $\nu(b - 1)(1 - (b + 1/4)/(5b^2)) \geq 0$ .

(If  $q < b$ , the last strict lower bound would be positive at  $i = q$ , an impossibility, so the binomial ratio is well-defined.) Hence

$$\frac{\binom{q}{b}}{\binom{m}{b}} \geq \nu^b \left(1 - \frac{b+1/4}{5b^2}\right)^b \geq \frac{3}{4}\nu^b. \quad (\text{U-B.3.4})$$

The last elementary bound is weakest at  $b = 1$ , where its left factor is  $3/4$ ; for  $b \geq 2$ , Bernoulli's inequality gives at least  $1 - (b+1/4)/(5b) > 3/4$ . Finally  $|X \setminus U| \geq (4/5)|X|$ , so (U-B.3.3)–(U-B.3.4) exceed  $3\nu^b|X|/5 > \nu^b|X|/2$ . Some  $S$  therefore has a common blue page  $T$  satisfying (U-B.3.1).  $\square$

**B.2.4. Lemma 4 (the limit needed to choose the moment).** For every  $p, \mu \in (0, 1)$ ,

$$\lim_{r \rightarrow \infty} (p^{1/r} - \mu)^r (1 - \mu)^{1-r} = p^{1/(1-\mu)} (1 - \mu), \quad (\text{U-B.4.1})$$

where  $r$  runs through sufficiently large positive integers, for which the base is positive.

**Proof.** Since  $p^{1/r} \rightarrow 1 > \mu$ , positivity is eventual, and

$$r \log \left(1 + \frac{p^{1/r} - 1}{1 - \mu}\right) \rightarrow \frac{\log p}{1 - \mu}.$$

Exponentiation and the remaining factor  $1 - \mu$  prove (U-B.4.1).  $\square$

**B.3. Proof of Theorem A.** The strict inequality in (U-B.A.1) implies  $x_0 + \mu_0 < 1$ . By Lemma 4 choose an integer  $r > 1$  such that

$$x_0 < (p^{1/r} - \mu_0)^r (1 - \mu_0)^{1-r}. \quad (\text{U-B.5.1})$$

After fixing  $r$ , choose  $\varepsilon > 0$  sufficiently small that

$$\begin{aligned} p &\geq 2\varepsilon, & (1 + \varepsilon)(\mu_0 + \varepsilon) &\leq 1, & x_0 + \mu_0 + 2\varepsilon &< 1, \\ & & (x_0 + 2\varepsilon, y_0 + 2\varepsilon) &\in \mathcal{R}_*, \\ \mu_0 + 2\varepsilon &\leq \frac{\mu_0 + \varepsilon}{(\mu_0 + \varepsilon) + (x_0 + \varepsilon)}, & (p - \varepsilon)^{1/r} &> \mu_0 + 3\varepsilon, \\ x_0 &\leq ((p - \varepsilon)^{1/r} - \mu_0 - 3\varepsilon)^r (1 - \mu_0 - \varepsilon)^{1-r} - \varepsilon. \end{aligned} \quad (\text{U-B.5.2})$$

All requirements follow by openness, continuity, (U-B.5.1), and the strict inequality  $x_0 + \mu_0 < 1$ . Set

$$x = x_0 + \varepsilon, \quad y = y_0 + \varepsilon, \quad \mu = \mu_0 + \varepsilon, \quad \delta_n = \frac{\varepsilon}{n}. \quad (\text{U-B.5.3})$$

We prove the following stronger assertion by induction on  $n = k + t$ , with  $\ell \geq L_0$  fixed. If

$$d(X, Y) \geq p - \delta_n \quad (\text{U-B.5.4})$$

and

$$(d(X, Y) + \delta_n - p)^r |X||Y| \geq x^{-k} y^{-\ell} \mu^{-t}, \quad (\text{U-B.5.5})$$

then  $(X, Y)$  is  $(k, \ell, t)$ -good. The cases  $k = 1$  or  $t = 1$  are immediate.

First note that (U-B.A.2) and  $d(X, Y) \geq p$  imply (U-B.5.5): because  $x_0 \leq x/(1 + \varepsilon)$ ,  $y_0 \leq y/(1 + \varepsilon)$ , and  $\mu_0 \leq \mu/(1 + \varepsilon)$ , it is enough that

$$\frac{\varepsilon^r}{n^r}(1 + \varepsilon)^{n+\ell} \geq 1. \quad (\text{U-B.5.6})$$

A single sufficiently large  $L_0$  makes (U-B.5.6) true for every  $n \geq 2$ .

**B.3.1. Regularization and the size of  $X$ .** Delete from  $X$  any vertex having fewer than  $(p - \delta_n)|Y|$  red neighbors in  $Y$ . Both the nonnegative excess

$$e_R(X, Y) - (p - \delta_n)|X||Y|$$

and its normalization by  $|X||Y|$  do not decrease, so the left side of (U-B.5.5) does not decrease. The positive right side prevents deletion of all vertices. We may consequently assume

$$d(X', Y) \geq p - \delta_n \quad \text{for every nonempty } X' \subseteq X. \quad (\text{U-B.5.7})$$

Lemma 1 and the interior condition in (U-B.5.2) imply, for sufficiently large  $L_0$ , that if

$$|Y| \geq (x + \varepsilon)^{-k}(y + \varepsilon)^{-\ell},$$

then  $Y$  already contains a red  $K_k$  or a blue  $K_\ell$ . Otherwise, using  $d + \delta_n - p \leq 1$  in (U-B.5.5),

$$|X| \geq \left(\frac{x + \varepsilon}{x}\right)^k \left(\frac{y + \varepsilon}{y}\right)^\ell \mu^{-t} \geq (1 + \varepsilon)^{n+\ell}. \quad (\text{U-B.5.8})$$

**B.3.2. The big-blue branch.** Put

$$b = \left\lceil \frac{2r \log n - r \log \varepsilon + \log 2}{\log(1 + \varepsilon)} \right\rceil, \quad m = \lceil 5\mu^{-1}b^2 \rceil, \quad w = R(k, m). \quad (\text{U-B.5.9})$$

Here  $m = O_{r,\varepsilon}(\log^2 n)$ ,  $m \geq b$ , and  $w = R(k, m) \leq \binom{k+m-2}{m-1} \leq (k+m)^m \leq \exp(O_{r,\varepsilon}(\log^3 n))$ . Since (U-B.5.8) is exponential in  $n + \ell$ , increasing  $L_0$  once makes, uniformly for every  $n$ ,

$$|X| \geq 5m^2, \quad w \leq \frac{\varepsilon(p - \varepsilon)}{n^3}|X|. \quad (\text{U-B.5.10})$$

Let

$$W = \{v \in X : |N_B(v) \cap X| \geq (\mu + \varepsilon)|X|\}.$$

If  $|W| \geq w$ , Lemma 3, with  $\nu = \mu + \varepsilon$ , produces a red  $K_k$  or a blue book with base  $S$  of size  $b$  and page  $T$  of size at least  $(\mu + \varepsilon)^b|X|/2$ . If  $b \geq t$ , its base already contains a blue  $K_t$ . Suppose  $b < t$ . From (U-B.5.7),  $d(T, Y) \geq p - \delta_n$ , and hence

$$d(T, Y) + \delta_{n-b} - p \geq \delta_{n-1} - \delta_n \geq \frac{\varepsilon}{n^2}.$$

Using (U-B.5.5),  $d + \delta_n - p \leq 1$ , and the definition of  $b$  gives

$$\begin{aligned} & (d(T, Y) + \delta_{n-b} - p)^r |T| |Y| \\ & \geq \frac{1}{2} \left( \frac{\varepsilon}{n^2} \right)^r (\mu + \varepsilon)^b |X| |Y| \\ & \geq \frac{1}{2} \left( \frac{\varepsilon}{n^2} \right)^r \left( \frac{\mu + \varepsilon}{\mu} \right)^b x^{-k} y^{-\ell} \mu^{-t+b} \\ & \geq x^{-k} y^{-\ell} \mu^{-t+b}. \end{aligned} \tag{U-B.5.11}$$

Induction applied to  $(T, Y)$  with parameter  $t - b$  finishes this branch: a blue  $K_{t-b}$  in  $T$  joins the blue base  $S$ .

We may therefore assume

$$|W| < w. \tag{U-B.5.12}$$

**B.3.3. The red-step/density-increment branch.** Lemma 2 and (U-B.5.10)–(U-B.5.12) show that the contribution from  $W$  is at most

$$w|Y| \leq \frac{\varepsilon}{n^3} e_R(X, Y).$$

Consequently some  $v \in X \setminus W$  satisfies

$$d(X, N_R(v) \cap Y) \geq d(X, Y) - \frac{\varepsilon}{n^3}. \tag{U-B.5.13}$$

Write

$$X_R = N_R(v) \cap X, \quad X_B = N_B(v) \cap X, \quad Y' = N_R(v) \cap Y, \quad p' = p - \delta_{n-1},$$

and

$$\alpha = d(X, Y') - p', \quad \alpha_R = d(X_R, Y') - p', \quad \alpha_B = d(X_B, Y') - p'.$$

An empty  $X_R$  or  $X_B$  is assigned zero weighted contribution and its induction branch is skipped; its corresponding  $\alpha_R$  or  $\alpha_B$  need not be defined. By (U-B.5.7),  $|Y'| \geq (p - \varepsilon)|Y|$ . Also, (U-B.5.4) and (U-B.5.13) give

$$\alpha \geq \delta_{n-1} - \delta_n - \frac{\varepsilon}{n^3} \geq \frac{\varepsilon}{n^2} > 0. \tag{U-B.5.14}$$

Since all edges from  $v$  to  $Y'$  are red,

$$\frac{\alpha_R}{\alpha} \frac{|X_R|}{|X|} + \frac{\alpha_B}{\alpha} \frac{|X_B|}{|X|} + \frac{1}{\alpha|X|} \geq 1. \tag{U-B.5.15}$$

If  $\alpha_R \geq 0$  and

$$\alpha_R^r |X_R| \geq \frac{x\alpha^r |X|}{p - \varepsilon}, \quad (\text{U-B.5.16})$$

then, using  $|Y'| \geq (p - \varepsilon)|Y|$ , (U-B.5.13), and

$$d(X, Y') - p' \geq d(X, Y) + \delta_n - p,$$

condition (U-B.5.16) and (U-B.5.5) show that  $(X_R, Y')$  satisfies the inductive moment inequality for  $(k - 1, \ell, t)$ . Its goodness implies that of  $(X, Y)$  because  $v$  is red to  $X_R \cup Y'$ . The analogous condition, including its necessary sign hypothesis, is

$$\alpha_B \geq 0, \quad \alpha_B^r |X_B| \geq \frac{\mu\alpha^r |X|}{p - \varepsilon} \quad (\text{U-B.5.17})$$

gives the inductive branch  $(k, \ell, t - 1)$ , since  $v$  is blue to  $X_B$ . Assume both branches fail. Set

$$a = |X_R|/|X|, \quad c = |X_B|/|X|, \quad q = 1 - 1/r.$$

Using the strict failures of (U-B.5.16)–(U-B.5.17) in (U-B.5.15) gives

$$x^{1/r} a^q + \mu^{1/r} c^q + \frac{(p - \varepsilon)^{1/r}}{\alpha |X|} > (p - \varepsilon)^{1/r}. \quad (\text{U-B.5.18})$$

Now  $a + c = 1 - |X|^{-1} < 1$  and  $c \leq \mu + \varepsilon$ . The function

$$f(z) = x^{1/r}(1 - z)^q + \mu^{1/r} z^q$$

is increasing on  $0 \leq z \leq \mu/(\mu + x)$ . The special smallness condition in (U-B.5.2) is exactly

$$\mu + \varepsilon \leq \frac{\mu}{\mu + x}.$$

Therefore

$$x^{1/r} a^q + \mu^{1/r} c^q \leq f(c) \leq f(\mu + \varepsilon) \leq x^{1/r}(1 - \mu)^q + \mu + \varepsilon. \quad (\text{U-B.5.19})$$

Equations (U-B.5.8) and (U-B.5.14), after one final uniform increase of  $L_0$ , give

$$\frac{(p - \varepsilon)^{1/r}}{\alpha |X|} \leq \frac{n^2}{\varepsilon(1 + \varepsilon)^{n+\ell}} \leq \varepsilon. \quad (\text{U-B.5.20})$$

Substituting (U-B.5.19)–(U-B.5.20) into the strict inequality (U-B.5.18) yields

$$x^{1/r}(1 - \mu)^{1-1/r} + \mu + 2\varepsilon > (p - \varepsilon)^{1/r}.$$

The positivity condition in (U-B.5.2) allows the rearrangement

$$x > ((p - \varepsilon)^{1/r} - \mu - 2\varepsilon)^r (1 - \mu)^{1-r}. \quad (\text{U-B.5.21})$$

But  $x = x_0 + \varepsilon$  and  $\mu = \mu_0 + \varepsilon$ , so (U-B.5.21) strictly contradicts the final inequality of (U-B.5.2). The induction, and hence Theorem A, is complete.  $\square$

**B.4. Proof of Corollary B.** By openness choose  $y_0 > y$  with  $(x, y_0) \in \mathcal{R}_*$ . Take a uniformly random equitable bipartition of the  $N$  vertices. Each edge has the same probability of crossing, so some partition  $(X, Y)$  has cross-red density at least the global red density  $p$ .

For all sufficiently large  $\ell$ ,

$$(y_0/y)^{\ell/2} \geq 3.$$

Thus (U-B.B.2) makes both equitable parts at least  $x^{-k/2}(\mu y_0)^{-\ell/2}$ , and hence

$$|X||Y| \geq x^{-k} y_0^{-\ell} \mu^{-\ell}.$$

Apply Theorem A with  $t = \ell$ . A red  $K_k$  in  $X \cup Y$ , a blue  $K_\ell$  in  $X$ , or a blue  $K_\ell$  in  $Y$  gives the desired conclusion.  $\square$

**B.5. Source defects neutralized by this proof.** The standalone proof explicitly repairs all of the following source-level issues rather than silently importing them:

1. Lemma 10 does not require  $p > \mu$ .
2. The moment order  $r$  is chosen before  $\varepsilon$ .
3. The northeast rate point is chosen in  $\mathcal{R}_*$ , and Lemma 1 supplies the exact eventual bound used later.
4. The printed comparison for  $y_0$  uses  $x$  in place of  $y$ .
5. The case  $b \geq t$  and empty  $X_R, X_B$  are treated separately.
6. The normalized sizes satisfy  $a + c < 1$ , not the dimensionally false printed inequality.
7. The concavity calculation supplies an upper bound in (U-B.5.19), and the strictness in (U-B.5.18) is preserved to obtain a genuine contradiction.

The only non-symbolic dependency left in the diagonal descent is the stated semantics of the Arb interval implementation used to certify its numerical inequalities. A post-repair adversarial replay of this standalone proof and a generated-copy fidelity audit have both passed. Independent external human review and a DOI-bearing archival deposit remain pending.

## APPENDIX C. SIXTH CHAINED UPPER-BOUND DESCENT: BOUNDED SEARCH LOG

**C.1. Certificate semantics and correctness.** We first state the theorem-to-JSON implication used by all six links. For a finite exact-decimal coefficient

tuple  $c = (c_1, \dots, c_d)$ , write

$$F_c(\lambda) = (1 + \lambda) \log(1 + \lambda) - \lambda \log \lambda + e^{-\lambda} \sum_{j=1}^d c_j \lambda^j. \quad (\text{U-C.C.1})$$

A file with schema `corrected-two-sided-ramsey-v1` contains exact tuples  $c^{\text{prior}}, c^{\text{target}}$ , an exact splice  $\lambda_*$ , and a finite ordered list of records  $(\lambda_i^-, \lambda_i^+, M_i, Y_i)$ . Acceptance means that outward-rounded interval evaluation establishes all of the following obligations.

- (i) The records form an exact partition of  $[\lambda_*, 1]$ , with  $0 < M_i, Y_i < 1$ , and the prior rate is strictly concave on  $(0, 1]$ .
- (ii) On every record and every  $\lambda$  in its closed interval, put

$$X = (1 - M_i)(1 - e^{-F'_{c^{\text{target}}}(\lambda)})^{1/(1-M_i)}, \quad a = -\log X, \quad b = -\log Y_i.$$

The verifier proves  $F_{c^{\text{target}}} > 0$ ,  $F'_{c^{\text{target}}} > 0$ , and both prior-region inequalities

$$a + \mu b > F_{c^{\text{prior}}}(\mu), \quad b + \mu a > F_{c^{\text{prior}}}(\mu) \quad (0 < \mu \leq 1). \quad (\text{U-C.C.2})$$

Strict concavity reduces the two suprema in  $\mu$  to their endpoint and unique stationary-point candidates; every candidate is enclosed.

- (iii) On the same complete record interval the main descent slack satisfies

$$F_{c^{\text{target}}}(\lambda) + \frac{1}{2} \{\log X + \lambda \log M_i + \lambda \log Y_i\} > 0. \quad (\text{U-C.C.3})$$

- (iv) On the open interval  $(0, \lambda_*]$ , the verifier takes  $M = \lambda e^{-\lambda}$ , defines  $X$  by the same formula and  $Y = 1 - X$ , and proves positivity of  $F, F'$  and a strictly positive lower bound for the left side of (U-C.C.3) divided by  $\lambda$ . Bounds  $q_- \lambda \leq e^{-F'} \leq q_+ \lambda$  cancel the two  $\lambda \log \lambda$  singularities, so this is an analytic tail proof rather than extrapolation from  $\lambda_*$ .
- (v) At the first link, the prior is exactly the rate in Lemma C.2. At every later link,  $c^{\text{prior}}$  agrees entry by entry, as exact decimal data, with the preceding accepted target tuple.

We next close the theorem-to-certificate bridge, including the two steps that are easiest to lose in a numerical presentation: the density dichotomy and the induction in the smaller clique order.

**Lemma C.1** (Finite-net rate descent). *Let  $G : [0, 1] \rightarrow \mathbb{R}$  be continuous, twice continuously differentiable on  $(0, 1]$ , and satisfy  $G(0) = 0$ ,  $G, G' > 0$ . Suppose  $G(\lambda) = H(\lambda) + O(\lambda)$  at zero, where  $H(\lambda) = (1 + \lambda) \log(1 + \lambda) - \lambda \log \lambda$ . For every  $\lambda \in (0, 1]$ , suppose there are  $M, X, Y \in (0, 1)$  with*

$$X = (1 - M)(1 - e^{-G'(\lambda)})^{1/(1-M)} \quad (\text{U-C.C.4})$$

and

$$G(\lambda) + \frac{1}{2} \{\log X + \lambda \log M + \lambda \log Y\} > 0. \quad (\text{U-C.C.5})$$



Assume that on every compact  $I \subset (0, 1]$ ,  $\inf_I G' > 0$ ,  $\sup_I M < 1$ , and, for every sufficiently small  $\delta > 0$ , the point

$$p_{\lambda, \delta} = 1 - e^{-G'(\lambda) + 2\delta}, \quad x_{\lambda, \delta} = e^{-\delta} (1 - M) p_{\lambda, \delta}^{1/(1-M)}, \quad (\text{U-C.C.6})$$

paired with  $Y$ , lies in  $\mathcal{R}_*$ . Then, uniformly for  $1 \leq \ell \leq k$ ,

$$\log R(k, \ell) \leq kG(\ell/k) + o(k). \quad (\text{U-C.C.7})$$

*Proof.* Fix  $\epsilon > 0$ . The Erdős–Szekeres recursion and Stirling’s formula give  $\log R(k, \ell) \leq kH(\ell/k) + o(k)$ , uniformly as  $\ell/k \rightarrow 0$ . Since  $H - G = O(\lambda)$ , choose  $\rho > 0$  so that the desired bound with error  $\epsilon k$  already holds when  $\ell \leq \rho k$ . It remains to consider  $I = [\rho, 1]$ .

The hypotheses in the lemma make the logarithm of  $x_{\lambda, \delta}/X(\lambda)$  converge uniformly to zero as  $\delta \downarrow 0$ . Choose one  $\delta > 0$  such that, throughout  $I$ ,

$$e^{-\epsilon} X(\lambda) \leq x_{\lambda, \delta} < X(\lambda). \quad (\text{U-C.C.8})$$

Uniform continuity of  $G, G'$  supplies a finite descending grid  $\{\lambda_j\}$  such that every  $r \in I$  has a  $\lambda_j \in [r, r + h]$  with

$$|G(\lambda_j) - G(r)| \leq \epsilon/2, \quad |G'(\lambda_j) - G'(r)| \leq \delta. \quad (\text{U-C.C.9})$$

Apply the balanced-book corollary of Appendix B at each grid point with  $p = p_{\lambda_j, \delta}$ ,  $\mu = M(\lambda_j)$ ,  $x = x_{\lambda_j, \delta}$ , and  $y = Y(\lambda_j)$ . The strict book hypothesis follows from (U-C.C.6), and the assumed interior condition supplies the other hypothesis. Since the grid is finite, the largest of the resulting thresholds is finite.

Now let  $N \geq e^{k(G(r) + \epsilon)}$ , where  $r = \ell/k \in I$ . If the red-edge density is at least  $1 - e^{-G'(r) + \delta}$ , it is at least  $p_{\lambda_j, \delta}$  by (U-C.C.9). Equations (U-C.C.5), (U-C.C.8), (U-C.C.9),  $\lambda_j k \geq \ell$ , and  $\log(MY) < 0$  give

$$\begin{aligned} N &\geq e^{k(G(\lambda_j) + \epsilon/2)} \\ &\geq (e^{-\epsilon} X(\lambda_j))^{-k/2} (M(\lambda_j) Y(\lambda_j))^{-\ell/2} \\ &\geq x_{\lambda_j, \delta}^{-k/2} (M(\lambda_j) Y(\lambda_j))^{-\ell/2}. \end{aligned}$$

The balanced-book corollary therefore gives a red  $K_k$  or a blue  $K_\ell$ .

Otherwise the average blue degree is greater than  $e^{-G'(r) + \delta}(N - 1)$ , so some vertex has, for all sufficiently large  $k$ , at least

$$\exp\{kG(r) + \epsilon k - G'(r) + \delta/2\} \quad (\text{U-C.C.10})$$

blue neighbours. Induct on  $\ell$ , starting with the already settled range  $\ell \leq \rho k$ . For  $r - 1/k \geq \rho/2$ , the mean-value theorem gives

$$k\{G(r) - G(r - 1/k)\} = G'(\xi), \quad G'(r) - G'(\xi) \leq k^{-1} \sup_{[\rho/2, 1]} |G''| \leq \delta/2$$

once  $k$  is large. Thus (U-C.C.10) is at least the induction threshold  $e^{k(G((\ell-1)/k) + \epsilon)}$ . Its blue neighbourhood contains a red  $K_k$ , or a blue  $K_{\ell-1}$  which extends with the chosen vertex. This completes the induction and the uniform  $\epsilon$ -form of (U-C.C.7).  $\square$

**Lemma C.2** (Elementary first prior). *Let*

$$c^{(0)} = \left(-\frac{1}{4}, \frac{2}{25}, \frac{2}{25}\right) \quad \text{and} \quad F_0(\lambda) = h(\lambda) + e^{-\lambda} \left(-\frac{1}{4}\lambda + \frac{2}{25}\lambda^2 + \frac{2}{25}\lambda^3\right).$$

*Then, uniformly for  $1 \leq \ell \leq k$ ,*

$$\log R(k, \ell) \leq kF_0(\ell/k) + o(k),$$

*and the colour-swapped statement holds as well.*

*Proof.* The first JSON file stores the tuple  $c^{(0)}$  above and the exact splice

$$\lambda_* = 0.00500000000000000001.$$

For every  $0 < \lambda \leq 1$ , set

$$M = \lambda e^{-\lambda}, \quad X = (1 - M)(1 - e^{-F'_0(\lambda)})^{1/(1-M)}, \quad Y = 1 - X.$$

The elementary Ramsey recursion gives  $(X, Y) \in \mathcal{R}$ . For the strictly smaller first coordinate  $x_{\lambda, \delta} < X$  used in Lemma C.1, choose  $z$  with  $x_{\lambda, \delta} < z < X$ . Then  $(z, 1 - z) \in \mathcal{R}$ , while  $x_{\lambda, \delta} < z$  and  $Y = 1 - X < 1 - z$ ; hence  $(x_{\lambda, \delta}, Y) \in \mathcal{R}_*$ .

On  $(0, \lambda_*]$ , the analytic small-ratio routine proves  $F_0, F'_0 > 0$ , bounds  $e^{-F'_0}$  above and below by positive multiples of  $\lambda$ , and returns an outward-rounded Arb enclosure whose lower endpoint satisfies

$$\frac{1}{\lambda} \left[ F_0(\lambda) + \frac{1}{2} \{ \log X + \lambda \log M + \lambda \log Y \} \right] > 0.00330.$$

This is the same cancellation argument recorded in obligation (iv), now applied to the printed polynomial  $c^{(0)}$ ; it covers the open endpoint at zero analytically.

The interval  $[\lambda_*, 1]$  is a union of 4,096 closed rational base cells. At 256-bit precision, with recursive bisection if an inclusion is not immediately decisive, outward-rounded Arb inclusions prove on every cell that  $F_0, F'_0 > 0$ ,  $0 < M, X, Y < 1$ , and

$$F_0(\lambda) + \frac{1}{2} \{ \log X + \lambda \log M + \lambda \log Y \} > 10^{-6}.$$

Thus all hypotheses of Lemma C.1 hold for  $G = F_0$ ; also  $F_0 - h = O(\lambda)$  at zero. That lemma gives the claimed uniform prior rate, without using any later target as an assumption.  $\square$

**Proposition C.3** (Rate-certificate correctness). *Assume the repaired book induction in Appendix B and Arb's containment semantics. If the first certificate satisfies (i)–(v), its target obeys the uniform Ramsey-rate bound (U3.3). If a finite chain is accepted with the exact handoffs in (v), every target in the chain obeys that bound; in particular the sixth target is  $F_6$ .*

*Proof.* On a closed record interval, (U-C.C.2) has strict positive margin in both colour orientations. Together with the already proved uniform prior rate, this allows  $X, Y_i$  to be enlarged slightly while retaining both exponent bounds; hence  $(X, Y_i) \in \mathcal{R}_*$ . Decreasing the first coordinate to  $x_{\lambda, \delta}$  preserves interior

membership. The finitely many records also give  $\sup M_i < 1$ , and the verifier proves  $F, F' > 0$  and (U-C.C.3), which are exactly (U-C.C.4)–(U-C.C.5).

On  $(0, \lambda_*]$ , the elementary witness has  $Y = 1 - X$ . Since  $x_{\lambda, \delta} < X$ , choose  $z$  strictly between them. The elementary recursion places  $(z, 1 - z)$  in  $\mathcal{R}$ , while  $x_{\lambda, \delta} < z$  and  $Y < 1 - z$ ; therefore the perturbed point lies in  $\mathcal{R}_*$ . Obligation (iv) proves the remaining analytic hypotheses on the whole open interval, including a positive normalized main slack rather than a sampled extrapolation. Because every target has the form  $H(\lambda) + e^{-\lambda}P(\lambda)$  with  $P(0) = 0$ , its difference from  $H$  is  $O(\lambda)$ . Thus Lemma C.1 proves one accepted link.

For the first link, Lemma C.2 proves the prior. Exact entrywise target-to-prior equality then permits ordinary induction through the six links, without a rounded or refitted intermediate rate.  $\square$

The six accepted certificate identities, in order, are

1. `certificate-higher-order-quintic-v1.json`, 2664bc421cd0cb7489289caa283a3a3f22830580f7c21bf0e6cbe092910bc277;
2. `certificate-higher-order-quintic-chain-v2.json`, b5b595b4dc9d1bdc2b5714f68fef99ec1c566bbd612d35ca19100d173d41c4d;
3. `certificate-higher-order-sextic-chain-v3.json`, 2052952c3af98074d5442fb736c7a2952146e92051a43fab84e75f099d9e00d7;
4. `certificate-higher-order-octic-chain-v4.json`, 09253d757516dc258396f82c09a442aaaf9d499640c1cc99b5767e6d51d35942;
5. `certificate-higher-order-decic-chain-v5.json`, 4ec4dbbe190a08134efe68c3a76ce4b4fb3aabdad230af703c19a28a03ecf9a9;
6. `certificate-higher-order-tetradecic-chain-v6.json`, 8689e3f80f363ac4b2dada31d6d71d682288eac24ca5e128e22aecffcd4bde8.

The canonical replay invokes `verify_chain_arb.py` with `--require-paper-chain`. Before doing any interval work, that mode requires these six filenames in this order, recomputes and compares all six SHA-256 values, and requires the final target tuple to equal the fourteen terminating decimals displayed below, entry by entry.

**C.2. Claim.** Let

$$H(\lambda) = (1 + \lambda) \log(1 + \lambda) - \lambda \log \lambda$$

and interpret every displayed terminating decimal exactly. Define

$$\begin{aligned}
P_6(\lambda) = & -0.250000000000000\lambda + 0.019891840059418\lambda^2 \\
& -0.012275954247190\lambda^3 + 0.144110816398083\lambda^4 \\
& +0.006277913420654\lambda^5 - 0.066101623491668\lambda^6 \\
& -0.002287675993602\lambda^7 - 0.058238059505066\lambda^8 \\
& +0.030675864198693\lambda^9 + 0.052472201910597\lambda^{10} \\
& +0.043300454861853\lambda^{11} - 0.042529975074653\lambda^{12} \\
& -0.055263639426635\lambda^{13} + 0.036695886931268\lambda^{14}.
\end{aligned}$$

For  $F_6(\lambda) = H(\lambda) + P_6(\lambda)e^{-\lambda}$ , the exact six-stage chain and the locally closed combinatorial descent prove the local computer-assisted theorem

$$R(k, k) \leq (3.7806984277961236987518816497522162 \dots)^{k+o(k)}.$$

The status is **PROVABLE AS STATED** under the same local proof boundary as the fifth-stage theorem. This is not a published result, an explicit finite- $k$  bound, or a proof-assistant formalization. Independent external human review of this sixth link remains pending.

**C.3. Frozen protocol (declared before target search).** This is a bounded attempt to add **one** descent after the frozen fifth-stage certificate. It will not continue to a seventh stage.

The only admissible prior is the exact ten-entry target tuple in **certificate-higher-order-decic-chain-v5.json**, whose SHA-256 is **4ec4dbbe190a08134efe68c3a76ce4b4fb3aabdad230af703c19a28a03ecf9a9**. No rounded, refitted, or independently retyped prior is admissible. Before search, both existing 256-bit Arb implementations must prove this exact prior strictly concave on all of  $(0, 1]$ .

The construction search is limited to target degrees 10 through 14. A candidate will be frozen only if all of the following predeclared gates hold:

1. its base is at least **0.000001** below the frozen fifth-stage base;
2. dense floating replay plus local scalar refinement finds unbuffered continuous slack at least **0.0001500000000000**;
3. the exact certificate has at most 131,072 cells and passes the complete six-stage **verify\_chain\_arb.py** replay;
4. the separately written **verify\_region\_direct\_arb.py** gives both exact Arb exponent margins at least **0.0000005**;
5. exact **Decimal** comparison proves the P5 target tuple equals the P6 prior tuple entry by entry; and
6. **audit\_tests.py** passes unchanged.

A certificate that merely has positive margins but misses one of these gates will be reported as a frontier/FAIL, not as a new Ramsey bound. Floating

optimization and interpolation are construction aids only and have no role in the proof.

**C.4. Pre-search prior audit.** The certificate hash was recomputed before optimization and equals the frozen hash above. On its exact target decimals the two independent 256-bit Arb implementations returned

```
verify_arb.prove_prior_concavity:      P5'' <=
↪ -0.35642515827888227
verify_region_direct_arb.prove_concavity: P5'' <=
↪ -0.3574394324533535
```

Thus the strict-concavity gate passed before any sixth-stage search. The frozen prior base, computed in Arb rather than binary floating point, is  
3.7807566104095593763172345562324210302278058291660...

**C.5. Bounded degree-10–14 frontier.** The first bounded pass used the exact P5 tuple above, a requested construction margin of 0.00015, and no target degree outside 10–14. Its diagnostic frontier was

| target degree | diagnostic base  | sampled        |                              |
|---------------|------------------|----------------|------------------------------|
|               |                  | minimum slack  | disposition                  |
| 10            | 3.78075661040956 | 0.000148091758 | no accepted<br>feasible step |
| 11            | 3.78075661040956 | 0.000148091758 | no accepted<br>feasible step |
| 12            | 3.78075661040956 | 0.000148091758 | no accepted<br>feasible step |
| 13            | 3.78069741660317 | 0.000150109936 | frontier only                |
| 14, raw       | 3.78069363690183 | 0.000150096701 | rejected after<br>dense leak |

The raw degree-14 point looked best on the construction mesh, but a denser mesh found slack only

0.00014795745008933103 near  $\lambda = 0.3386221294$ ,

so it failed the predeclared gate and was not certified. No degree was added and no search range was enlarged. A single guarded degree-14 point was then selected within the already declared range. A broad 20,001-point scan (geometric near the splice and linear thereafter), followed by bounded scalar refinement around every sampled local minimum, returned

```
minimum unbuffered floating slack:
0.00015473769852614172 at  $\lambda \approx 0.33943703594667307$ 
```

next three refined local minima:

```
0.00015519556902354736 at  $\lambda \approx 0.9859202132743512$ 
0.00015519607550973370 at  $\lambda \approx 0.6439420227559093$ 
0.00015519870501323751 at  $\lambda \approx 0.8750494757168542$ 
```

```

minimum sampled F6: 0.030260793830868504 at lambda = 0.005
minimum sampled F6': 0.7658393690544814 at lambda = 1
floating base: 3.780698427796123

```

Thus the unique frozen candidate cleared both the 0.00015 continuous construction floor and the 0.000001 base-improvement floor. The eventual exact Arb base improvement over P5 is  
0.0000581826134356775653529064802047549...

These floating data only selected the candidate. In particular, they are not used to cover the continuum in the proof.

## C.6. Exact replay and claim boundary.

C.6.1. *Frozen certificate and exact handoff.* Only the guarded degree-14 target displayed in the Claim was frozen:

```

file: certificate-higher-order-tetradecic-chain-v6.json
SHA-256:
↔ 8689e3f80f363ac4b2dada31d6d71d68228eac24ca5e128e22aecffcd4bde8
cells: 131072
stored left endpoint: 0.0050000000000000001
stored right endpoint: 1
region buffer: 0.000080000000000000007
constructor predicted worst buffered slack: 0.00011568648667514836

```

The certificate's ten-entry prior tuple was compared with the P5 target tuple using arbitrary-precision `Decimal`, not binary floating point:

```
P5 target == P6 prior: True
```

Thus the sixth link has no rounded or refitted handoff.

C.6.2. *Complete six-stage 256-bit Arb replay.* `verify_chain_arb.py` replayed the elementary prior and all six exact links and returned

```
PASS: verified 6-stage Ramsey-rate certificate chain
```

The new stage's strict lower margins and endpoint value were

```

target small slack/lambda:
0.0037785734384423962376319094335140413...

```

```

standard Ramsey-region margin:
7.464714995786726e-05 at cell 8

```

```

swapped Ramsey-region margin:
5.766703065384951e-05 at cell 131071

```

```

large-regime main slack:
4.818471567626107e-05 at cell 131071

```

```

F6(1):
1.3299087618219930723187812632036105226...

```

```
exp(F6(1)):
3.7806984277961236987518816497522162752...
```

Every proof-critical margin is strictly positive. The analytic small regime and the 131,072 exact cells cover all  $0 < \lambda \leq 1$ .

C.6.3. *Separately written direct region replay.* `verify_region_direct_arb.py` does not import the primary verifier and directly minimizes both continuous exponent slacks over the complete prior-ratio interval. It returned

```
PASS: independent direct two-sided Ramsey-region replay
prior P5'' worst certified upper:      -0.35743943245335352
worst standard direct exponent slack:   1.0628096485437162e-06 at
↪ cell 0
worst swapped direct exponent slack:    5.766703065384951e-05 at
↪ cell 131071
```

Both independently replayed margins exceed the predeclared  $5e-7$  floor. `audit_tests.py` also passed the derivative-sign regression, arbitrary-degree polynomial-derivative regression, and known Horizon union-gap regression.

Consequently every predeclared gate passed and the boxed local asymptotic bound follows. The search stops here: no seventh-stage search was performed.

C.7. **Reproduction commands.** Run from `routes/upper` with the repository environment that supplies `python-flint`:

```
../../../../venv/bin/python verify_chain_arb.py --require-paper-chain \
certificate-higher-order-quintic-v1.json \
certificate-higher-order-quintic-chain-v2.json \
certificate-higher-order-sextic-chain-v3.json \
certificate-higher-order-octic-chain-v4.json \
certificate-higher-order-decic-chain-v5.json \
certificate-higher-order-tetradecic-chain-v6.json

../../../../venv/bin/python verify_region_direct_arb.py \
certificate-higher-order-tetradecic-chain-v6.json

../../../../venv/bin/python audit_tests.py
```

C.8. **Claim boundary.**

1. The sixth certificate proves one finite descent only; it supplies no convergence theorem for an infinite chain.
2. The inherited asymptotic descent has non-effective  $o(k)$  terms, so this is not an explicit finite- $k$  inequality.
3. The numerical proof imports Arb containment semantics through `python-flint` at 256-bit precision.
4. The local combinatorial proof and interval replay have not been formalized in a proof assistant, and this sixth link has not yet received independent external human review or a DOI-bearing archival deposit.

#### APPENDIX D. DETAILED ONE-STEP LEDGER ESTIMATES

This appendix records the estimates used in Sections 13 and 14 in a form independent of the exploratory calculation log. Throughout,  $C$  is fixed and sufficiently large,  $w > \omega_0$ , and the target set exposed after column  $i$  is  $V_i = \{i + 1, \dots, r\}$ , of size  $k = r - i$ .

##### D.1. The Hessian on the entire cutoff box. Recall

$$F_i(b) = \sum_{j \in V_i} \log \Phi(-b_j) - \frac{1}{d} \sum_{\{j,q\} \subset V_i} T_{jq} m(b_j) m(b_q).$$

Since  $(\log \Phi(-x))' = -m(x)$ , the entries of the Hessian are

$$\partial_{jj}^2 F_i(b) = -m'(b_j) - \frac{m''(b_j)}{d} \sum_{q \in V_i \setminus \{j\}} T_{jq} m(b_q), \quad (\text{L-A.A.1})$$

$$\partial_{jq}^2 F_i(b) = -\frac{T_{jq}}{d} m'(b_j) m'(b_q), \quad j \neq q. \quad (\text{L-A.A.2})$$

The inverse Mills ratio satisfies  $m'' \geq 0$ . Hence the interaction term in (L-A.A.1) can only strengthen the diagonal of  $-\nabla^2 F_i$ . On  $[c - w, c + w]^{V_i}$ , monotonicity gives

$$m'(b_j) \geq j_-(w), \quad |m'(b_j) m'(b_q)| \leq j_+(w)^2. \quad (\text{L-A.A.3})$$

The residual recursion and  $r/\sqrt{d} \leq 1/D$  imply

$$T_{\max} \leq \frac{t_0 + \theta r}{1 - \rho}, \quad (\text{L-A.A.4})$$

$$\frac{1}{d} \max_j \sum_{q \neq j} T_{jq} \leq \frac{m_0 + \theta/D}{D(1 - \rho)}. \quad (\text{L-A.A.5})$$

Combining (L-A.A.1)–(L-A.A.5) with Gershgorin's theorem proves

$$-\nabla^2 F_i(b) \succeq \left[ j_-(w) - j_+(w)^2 \frac{m_0 + \theta/D}{D(1 - \rho)} \right] I = \mu_w I. \quad (\text{L-A.A.6})$$

This is a full-box statement, not an expansion valid only at the central cutoff.

##### D.2. Boundary merge with a controlled residual. The gradient at $c\mathbf{1}$ is

$$\partial_j F_i(c\mathbf{1}) = -m_0 - \frac{m_0 m'_0}{d} \sum_{q \in V_i \setminus \{j\}} T_{jq}. \quad (\text{L-A.A.7})$$

The defining recursion

$$T_{ij} = m_0 \sqrt{d} + \frac{m_0 m'_0}{\sqrt{d}} \sum_{q \in V_i \setminus \{j\}} T_{jq} + e_{ij} \quad (\text{L-A.A.8})$$

therefore gives

$$\partial_j F_i(c\mathbf{1}) = -\frac{T_{ij} - e_{ij}}{\sqrt{d}}. \quad (\text{L-A.A.9})$$



For  $\delta_j = b_j - c = \sqrt{d} h_{ij} + \varepsilon_{ij}$ , with  $|\varepsilon_{ij}| \leq d^{-1/5}$ , the tangent inequality from (L-A.A.6) gives

$$\begin{aligned} F_i(b) &\leq F_i(c\mathbf{1}) - \sum_{j \in V_i} T_{ij} h_{ij} + \sum_{j \in V_i} \left( \frac{e_{ij} \delta_j}{\sqrt{d}} - \frac{\mu_w \delta_j^2}{2} \right) + \frac{d^{-1/5}}{\sqrt{d}} \sum_{j \in V_i} T_{ij} \\ &\leq F_i(c\mathbf{1}) - \sum_{j \in V_i} T_{ij} h_{ij} + \sum_{j \in V_i} \frac{e_{ij}^2}{2\mu_w d} + \frac{d^{-1/5}}{\sqrt{d}} \sum_{j \in V_i} T_{ij}. \end{aligned} \quad (\text{L-A.A.10})$$

The second line is the scalar square completion  $xy - \mu_w y^2/2 \leq x^2/(2\mu_w)$ . Thus every boundary edge merges with its full coefficient  $T_{ij}$ ; only the explicit residual is paid.

**D.3. Mass identities.** Let  $\Delta_{ij} = T_{ij} - t_0$ . Backward induction in the first endpoint gives, for every  $j < q$ ,

$$A_0(r-j-1) + \theta(j-1) \leq \Delta_{jq} \leq \frac{A_0 + \rho\theta}{1-\rho}(r-j-1) + \theta(j-1). \quad (\text{L-A.A.11})$$

For  $S_i$  and  $R_i$  as in Section 14, this implies

$$S_i \geq a_-(r-2) \binom{k}{2}, \quad R_i \leq a_+(r-2)(k-1), \quad \frac{R_i}{S_i} \leq \frac{2a_+}{a_-k}. \quad (\text{L-A.A.12})$$

The exact accumulated sums are obtained by exchanging the order of summation. The future-degree part of (L-A.A.11) counts each ordered quadruple twice, while the residual part contributes

$$K_r := \sum_{i=1}^{r-1} (r-i)(i-1)^2 = \binom{r}{3} + 2 \binom{r}{4}. \quad (\text{L-A.A.13})$$

Consequently

$$\sum_{i=1}^{r-1} S_i \geq 2A_0 \binom{r}{4} + \theta K_r, \quad \sum_{1 \leq i < j \leq r} e_{ij}^2 = \theta^2 K_r. \quad (\text{L-A.A.14})$$

No asymptotic replacement is used in (L-A.A.13)–(L-A.A.14).

**D.4. Three-factor Hölder comparison.** Fix a red-admissible  $\mathcal{F}_{i-1}$ -history for target  $r$ , condition further on  $A_i^R$ , and write

$$X_j = -\frac{m(b_j)}{\sqrt{d}} + \frac{\zeta_j}{\sqrt{d}}, \quad u_j^0 = \frac{t_0}{d} \sum_{q \neq j} m(b_q), \quad \delta u_j = \frac{1}{d} \sum_{q \neq j} \Delta_{jq} m(b_q). \quad (\text{L-A.A.15})$$

The variables  $\zeta_j$  are conditionally independent, centered and 1-subgaussian. The centered exponent is exactly

$$\sum_j (u_j^0 + \delta u_j) \zeta_j - \frac{t_0}{d} \sum_{j < q} \zeta_j \zeta_q - \frac{1}{d} \sum_{j < q} \Delta_{jq} \zeta_j \zeta_q. \quad (\text{L-A.A.16})$$

Use conjugate exponents

$$Q_0 = \frac{d}{4t_0k}, \quad Q_1 = \frac{d}{4R_i}, \quad P^{-1} = 1 - Q_0^{-1} - Q_1^{-1}. \quad (\text{L-A.A.17})$$

They are all greater than one, with  $P \leq 2$ , because

$$\frac{4m_0}{D} + \frac{4a_+}{D^2} < \frac{1}{2}. \quad (\text{L-A.A.18})$$

Hölder gives one linear-CGF factor and two quadratic factors. The uniform  $t_0$  factor is the frozen HMS factor. For the increment, use

$$-\sum_{j < q} \Delta_{jq} \zeta_j \zeta_q \leq \frac{1}{2} \sum_j R_j \zeta_j^2, \quad \frac{Q_1 R_j}{2d} \leq \frac{1}{8}. \quad (\text{L-A.A.19})$$

The square-MGF estimate  $\log \mathbb{E} e^{x\zeta_j^2} \leq (16/3)x$  for  $0 \leq x \leq 1/8$  therefore gives

$$\frac{1}{Q_1} \sum_j \frac{16}{3} \frac{Q_1 R_j}{2d} = \frac{16}{3} \frac{S_i}{d}. \quad (\text{L-A.A.20})$$

For the linear factor, let  $P^0 = P_{k,d}^0$ . The source exact-CGF deficit on this identical history is

$$\mathfrak{d}_i(b) = \sum_j \left[ (u_j^0)^2 - \frac{1}{P^0} K_{b_j}(P^0 u_j^0) \right] \geq \mathfrak{d}_{k,w,d}. \quad (\text{L-A.A.21})$$

Thus it remains only to bound the change from  $(P^0, u^0)$  to  $(P, u^0 + \delta u)$ . Since  $0 \leq K_b'' \leq 1$ ,

$$\frac{K_b(P(u + \delta)) - K_b(Pu)}{P} \leq 2u\delta + \delta^2, \quad (\text{L-A.A.22})$$

$$0 \leq \frac{\partial}{\partial P} \frac{K_b(Pu)}{P} \leq u^2. \quad (\text{L-A.A.23})$$

Furthermore

$$P - P^0 \leq \frac{16R_i}{d}, \quad \sum_j \delta u_j \leq \frac{2M_w S_i}{d}, \quad \max_j \delta u_j \leq \frac{M_w R_i}{d}, \quad (\text{L-A.A.24})$$

and

$$\sum_j (u_j^0)^2 \leq \frac{2m_0^2 M_w^2 S_i}{a_- d}. \quad (\text{L-A.A.25})$$

Substitution in (L-A.A.22)–(L-A.A.24) gives, in order,

$$\frac{4m_0 M_w^2 S_i}{D} \frac{1}{d}, \quad \frac{2M_w^2 a_+ S_i}{D^2} \frac{1}{d}, \quad \frac{32m_0^2 M_w^2 a_+ S_i}{a_- D^2} \frac{1}{d}. \quad (\text{L-A.A.26})$$

These are precisely the three summands of  $\hat{L}_w(C) S_i/d$ . Combining them with (L-A.A.20) and the central deterministic gain  $-m_0^2 S_i/d$  gives

$$-\left(m_0^2 - \frac{16}{3} - \hat{L}_w(C)\right) \frac{S_i}{d} = -\hat{B}_w(C) \frac{S_i}{d}. \quad (\text{L-A.A.27})$$

Equation (L-A.A.21) is charged once, (L-A.A.27) pays only the new  $\Delta = T - t_0$  factor and the change of exponent, and (L-A.A.10) pays the residual gradient. This is the promised same-history separation.

**D.5. One-step conclusion and extraction.** Combining (L-A.A.10), (L-A.A.21), and (L-A.A.27) proves (L2.37) with the value  $\lambda_i$  in (L6.2). The exact weighted reverse-induction rule (S3), followed by (L-A.A.14), then gives the empty-state bound

$$P_{R,r}^* \leq \mathcal{H}_C(r) \exp \left[ -\mathcal{D}_{C,w,d}(r) - \frac{1}{d} \left\{ 2\hat{B}_w A_0 \binom{r}{4} + E_w K_r \right\} + J_{r,d} \right]. \quad (\text{L-A.A.28})$$

where

$$0 \leq J_{r,d} \leq \frac{m_0 + \theta/D}{1 - \rho} d^{-1/5} \binom{r}{2} = o_C(r^2). \quad (\text{L-A.A.29})$$

If extraction deletes  $u$  vertices from a retained set of size  $\ell$ , then the old exact-CGF restoration cost is at most

$$\mathcal{D}_{C,w,d}(\ell) - \mathcal{D}_{C,w,d}(\ell - u) \leq \frac{m_0^4}{D^2} \ell u \leq \ell u, \quad (\text{L-A.A.30})$$

and the loss in the new deterministic exponent is at most

$$\frac{1}{d} \left[ 2\hat{B}_w A_0 u \binom{\ell}{3} + E_w \left( u \binom{\ell}{2} + 2u \binom{\ell}{3} \right) \right] \leq \left[ \frac{\hat{B}_w A_0 + E_w}{3D^2} + \frac{E_w}{2D^2 \ell} \right] \ell u. \quad (\text{L-A.A.31})$$

For sufficiently large  $C$  and then sufficiently large  $\ell$ , the bracket in (L-A.A.31) is

$$O(1/\log C) + O_C(\ell^{-1}) < 1.$$

Together with the source-envelope restoration  $[\log(1/p) + 1]\ell u$ , the total cost is at most  $[\log(1/p) + 3]\ell u$ . The pinned failure factor therefore leaves  $\eta_C = 10 \log(10/p) - \log(1/p) - 3 > 0$ , and

$$\sum_{u=1}^{\ell} \binom{\ell}{u} e^{-\eta_C \ell u} = (1 + e^{-\eta_C \ell})^{\ell} - 1 = o(1). \quad (\text{L-A.A.32})$$

This proves that (L-A.A.28) survives perfect-sequence extraction with its entire old and new exponent intact.

## APPENDIX E. PINNED SOURCE INTERFACE AND QUANTIFIER LEDGER

This appendix records the exact external boundary of the lower-bound argument. It separates version-pinned conditional probability statements from the explicit weighted-extension rule and imports neither the first-moment conversion nor the Ramsey conclusion proved in Section 16.

**E.1. Content-addressed source snapshots.** The source files are Hunter–Milojević–Sudakov, arXiv:2512.17718v2, file `arXiv_version.tex`, SHA-256 `b72958ac35554eccb94dedab5800349d2c021af7d60f767125cb46998e0fd54a` and Lin–Niu, arXiv:2605.25843v2, file `off-diagonal-Ramsey-R.tex`, SHA-256

`dbf1bfa2c7603c81b2e56a97028b138ccf9a9299a3d1f20c0b08cf489136f429`

The repository manifest records the versioned retrieval URLs and independently checks these hashes.

**E.2. Source-line map for the source interface.**

| Pinned source lines      | Exact use in this paper                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HMS 145–170, 443–498     | The Gaussian graph, Bartlett representation, column exposure, edge events, perfect events, and the unweighted projection state. These are the probability-space and history definitions in (L2.24)–(L2.33).                                                                                                                                                              |
| HMS 955–971, 1192–1218   | Optimized red-perfectness and the exact cutoff approximation. Cauchy–Schwarz gives $\sqrt{d} h_{ij}  \leq \alpha_R^2 \ell / \sqrt{d} \leq \omega_0$ , while the diagonal approximation contributes $d^{-1/5}$ . This is (S1).                                                                                                                                            |
| HMS 330–354, 878–935     | The subgaussian definition, square-MGF lemma, pre-Cauchy centered-linear allocation, and centered quadratic estimate. These give the quantified statements (L2.35)–(L2.36), the uniform $t_0$ factor in (L5.12), and the square estimate in (L5.14); no final Cauchy coarsening is imported.                                                                             |
| Lin–Niu 359–493, 499–537 | Monotonicity of upper-truncated Gaussian variance and the exact positive-direction CGF. After replacing their upper cutoff by $-b$ , these give (L2.4)–(L2.6) and Lemma 11.1.                                                                                                                                                                                            |
| HMS 1192–1289            | The red law of total probability, cutoff conditioning, projection merge, and scalar reverse-induction allocation. Equations (L2.33), (L2.39), and (L2.40) record the algebra that the separately assumed weighted extension (S3) must preserve; (L2.36a) is the frozen scalar baseline. The nonuniform estimate is proved locally and is not attributed verbatim to HMS. |
| HMS 385–410 and 955–971  | For every specified retained set, the red-perfectness failure factor is $(p/10)^{10\ell u}$ . This is the sole external extraction input (L2.41); the restoration estimates and binomial sum are proved in (L6.5)–(L6.11).                                                                                                                                               |

| Pinned source lines                                     | Exact use in this paper                                                                                                                                                                                  |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HMS 977–1181,<br>1294–1312,<br>together with<br>385–410 | The unchanged blue reverse induction, its perfect bound, and the same greedy extraction argument, yielding the raw probability inequality (L2.42). The first-moment threshold (L7.4) is derived locally. |

The separately fetched Yang–Mao v1 member has SHA-256  
155b7104ec5b6935a576ae9f2b161976a966b0b46bd2b69153c0934ca688da2a  
agreeing with the repository manifest. It is not an input to the lower-bound component, so no Yang–Mao line range is invoked here; assigning one would misstate the lower component’s dependency boundary.

**E.3. Why the cutoff window is  $\omega_0$ .** On the event  $B_r^R$ ,

$$|b_{ij} - c| \leq \sqrt{d} |h_{ij}| + d^{-1/5} \leq \frac{\alpha_R^2 \ell}{\sqrt{d}} + d^{-1/5} \leq \omega_0 + d^{-1/5}. \quad (\text{L-B.B.1})$$

For every fixed  $w > \omega_0$ , choosing  $\ell \geq \ell_0(C, w)$  makes the last expression at most  $w$ . The proof is first carried out for this open range of  $w$ , and only after the fixed- $C$ ,  $\ell \rightarrow \infty$  limit is taken do we let  $w \downarrow \omega_0$ . No finite- $\ell$  history is claimed to lie in the closed boundary window. This also explains why  $p$ , rather than  $p_C$ , occurs in  $\alpha_R$  and  $\omega_0$ .

**E.4. The old exact-CGF entry is an input formula, not a conclusion.** At a step with  $k$  future vertices, define

$$b_j = b_{ij}, \quad M_j = \sum_{q \neq j} m(b_q), \quad u_j^0 = \frac{m_0 M_j}{\sqrt{d}}. \quad (\text{L-B.B.2})$$

Stopping the HMS centered-moment calculation before its final Cauchy coarsening gives the allocation  $\sum_j (u_j^0)^2$ . Applying the exact positive-direction CGF with  $P_{k,d}^0$  gives

$$\sum_j \frac{1}{P_{k,d}^0} K_{b_j}(P_{k,d}^0 u_j^0). \quad (\text{L-B.B.3})$$

Their difference is the same-history quantity  $\mathfrak{d}_i(b)$  in (L5.16). The cutoff monotonicities give (L5.17), and summing its deterministic lower bound gives (L2.15). With  $k/\ell \rightarrow x$  and  $\sqrt{d}/\ell \rightarrow D$ , the corresponding Riemann sum is (L2.17). Hence  $G_*(C)$  is defined and charged before the controlled residual is added; it is not imported through a final Ramsey theorem.

**E.5. Raw ledgers and the local crossing calculation.** The red interface before the new two deficits is the explicit envelope

$$\mathcal{H}_C(r) = p^{(r)} \exp \left[ -\vartheta_C \frac{a^3}{p^3 \sqrt{d}} \binom{r}{3} + \frac{a}{p} d^{-1/5} \binom{r}{2} + r \right]. \quad (\text{L-B.B.4})$$

The blue interface is the explicit probability estimate (L2.42). Dividing their logarithms by  $\ell^2$ , and only then applying  $\binom{n}{k} \leq n^k$ , gives the red and blue

thresholds (L7.2) and (L7.4). The comparison is not assumed: the source asymptotics imply

$$\frac{B_R(C)}{2} = \frac{1}{12} + o(1), \quad \frac{CB_B(C)}{2} = \frac{1}{2} + o(1), \quad (\text{L-B.B.5})$$

whereas

$$G_*(C) + \hat{H}_*(C) = O(1/\log C) = o(1). \quad (\text{L-B.B.6})$$

Thus the blue-minus-red gap is  $5/12 + o(1)$ , proving locally that red is the active constraint for sufficiently large fixed  $C$ .

**E.6. Order of quantifiers and proof status.** The absolute constant  $K$  in (L2.10) is fixed before selecting the large- $C$  threshold, but its numerical value is not provided by the pinned source. At theorem level the quantifiers therefore begin in the order

$$\text{fix } K, \quad \exists C_0 = C_0(K) \forall C \geq C_0 \exists \ell_0(C) \forall \ell \geq \ell_0(C), \quad (\text{L-B.B.7})$$

followed by  $\ell \rightarrow \infty$ . Within the proof, the auxiliary  $w$  is fixed in an admissible right-neighborhood of  $\omega_0$  during this limit and is then sent down to  $\omega_0$ . Only after that fixed- $C$  theorem is established is  $C \rightarrow \infty$  used to obtain the coefficient  $1/64$ . No explicit value of  $C_0(K)$  or  $\ell_0(C)$  is asserted.

The resulting theorem is source-relative: its local residual algebra, three-factor bridge, reverse-induction charge, extraction sum, first moments and red-blue crossing are proved here. Items (S1), (S2), (S4), and (S5) are version-pinned source hypotheses; (S3) is the separately stated weighted extension rule and is not attributed verbatim to either source. The arithmetic checker corroborates the finite formulas but does not replace these probabilistic interfaces.

## APPENDIX F. CONTENT-ADDRESSED ARTIFACT LEDGER

The digests in the two artifact tables below are SHA-256 identities of the compressed files. Their byte counts refer to those same compressed files and therefore permit a check before decompression. Digests explicitly introduced as uncompressed identities or as metadata-record identities have the separate meaning stated in their surrounding text.

Table 7: Budget-five proof pairs.

| branch | artifact | bytes    | SHA-256                                                              |
|--------|----------|----------|----------------------------------------------------------------------|
| 0      | CNF      | 837880   | d64f009f840597e266ccfb3b31c88f7e87eb760<br>d1fc551b7ee08a0854749ba66 |
| 0      | DRAT     | 13125926 | 60687241833df8ae957b3b4fcb02e6e9f7595c6<br>e3dcbee8b77f02dad5443f913 |
| 1      | CNF      | 1797965  | d6d321a4ac1aeec5e99c48843a8cda9395271e8<br>9555cea328e29bcda4e70ef15 |
| 1      | DRAT     | 37102450 | 33a9e037e19beeab8dd2af4dca5ce9010d4906b<br>39dc3e5cc8fd62823f518497a |

| branch | artifact | bytes    | SHA-256                                                              |
|--------|----------|----------|----------------------------------------------------------------------|
| 2      | CNF      | 1363549  | 8d8a543bb3cc17014941fee432673a1dc2b99c8<br>cb4885d4d91bba9005f3b10a7 |
| 2      | DRAT     | 21678734 | 159865da9ad23eea298ebd4e9406f2f37ecc601<br>5a593975e9d63e3b08a859897 |

The corresponding uncompressed identities, independently recomputed during review, are as follows:

**Branch 0 CNF:** c907dc96c1ff3378d345ead0ea22456eccef50f321cc53b6cf7dda467  
84d6389

**Branch 0 DRAT:** 517d076efd026eab6eeb89583c0226e713be968120344e6b82c3471  
e0efe6b5d

**Branch 1 CNF:** 5ba541f58ea42d7fb73ff51a962e98153eece0d4a8bf13c1bea3a86ed  
0d50429

**Branch 1 DRAT:** 84a2236b7692207f59c3d2dcb3521b2f2b947428470384ac643c34a  
7bfbf69

**Branch 2 CNF:** 441b70e023f4a9fa1837f16b1502ef3b54a2f00b6143fd5e7de0131ac  
95beebf

**Branch 2 DRAT:** 4ff3a43bda7774c09695417a9f9684e8a856fb46bf62225a08e746f  
bf37d1ad6

Table 8: Exact-six proof pairs.

| branch | artifact | bytes     | SHA-256                                                              |
|--------|----------|-----------|----------------------------------------------------------------------|
| 0      | CNF      | 17600735  | 47052de808b98598f2f144251a6ad73c1415dd0<br>0fef9e04b99ffe2f176229fbf |
| 0      | DRAT     | 656144450 | 4d948ab34f1d2475af6efe943e0a4899827c78b<br>96c2bc3835b5161f189c26bb5 |
| 1      | CNF      | 5052085   | a438c9fef4d99ae516829e4405da8c9f0793985<br>01361f9c5f69f46a494ffad14 |
| 1      | DRAT     | 433552903 | cfe694fd728903ee9ac9f08a66ade64faf7e687<br>63246d1d52ca29a85727494ec |
| 2      | CNF      | 1205208   | c51fed43eeca3becf64bcdfe747a4fdf3e516c<br>e0e6a02d7d867941b87a99c06  |
| 2      | DRAT     | 152784369 | c9648a4e38c17dcdf97d873095eda002abc689<br>b3b3bd93f02fc8a369198c4b9  |

For the exact-six formulas, the frozen cut-bank identities are

**Branch 0, 251771 masks:** 91b5709248ff641a315f5a0389b4f3fde3d38514f3b1a8  
b31b6cad31224f250b

**Branch 1, 63943 masks:** f8064f0f226659b4750382f781ba9a862bcc19eb49006eb  
8cdd2afc002ed227c

**Branch 2, 5422 masks:** aaa79a2f25d4c0d61c4467f1d0a721fd670a624e0391d96b  
2122723f97ba4598

The exact-six proof records have SHA-256 identities

**Branch 0:** 20729b2a818aaca35cdee6e9deeb5a0e  
6e4bbd6dc5ca11e5abb6dfe9409b035f

**Branch 1:** 4a3e7b4c729f696ff304c5f71621e018  
786b552146784a2141529a4c6f028666

**Branch 2:** ee0c73b80407a4dd48ca69bd1a666f39  
1bfd5729b9a56e062deef2a6f62018de

The source files use the following release-asset names:

r3\_18\_budget5\_branch\_0\_twostage.{cnf,drat}.gz  
r3\_18\_budget5\_branch\_1.{cnf,drat}.gz  
r3\_18\_budget5\_branch\_2.{cnf,drat}.gz  
r3\_18\_budget6\_branch\_0\_universal\_union.{cnf,drat}.gz  
r3\_18\_budget6\_branch\_1.{cnf,drat}.gz  
r3\_18\_budget6\_branch\_2.{cnf,drat}.gz

## APPENDIX G. INDEPENDENT VERIFICATION PROTOCOL

This appendix states a reviewer-oriented protocol. It is intentionally more specific than the convenience wrapper in the repository.

**G.1. Seed verification.** Parse the  $100 \times 100$  matrix, reject any non-binary, asymmetric, or non-zero-diagonal input, and verify its SHA-256. Count undirected edges by the strict upper triangle. Enumerate every vertex triple and record all triangles. Run the exact independent-set verifier at target 18. Repeat the structural checks for the 99-vertex principal submatrix and compare it byte-for-byte with its separately stored certificate. A successful outcome must report 827 edges and exactly the triangle  $\{97, 98, 99\}$  for  $H$ , and 809 edges, no triangle, and no independent 18-set for  $G_0$ .

**G.2. Formula reconstruction.** For each exact-six branch  $i$ :

1. Assign primary identifiers  $1, \dots, 4950$  to pairs in lexicographic order.
2. Generate the 161700 triangle clauses in lexicographic triple order and compare them to the DIMACS prefix.
3. Form the 826 residual deletion literals in the lexicographic order inherited from the primary pair list. Without calling PySAT's `CardEnc`, generate the four Sinz clause families in Lemma 21.1. Generate the equality block as at-most 821 on the complemented literals followed by at-most 5 on the original literals. Compare all 16420 clauses and the final variable identifier 13160.
4. Compare the fixed negative unit for  $f_i$ .
5. For each remaining clause, require 153 distinct, positive primary literals. Invert the pair map, recover exactly 18 vertices, and verify that the clause is the full  $\binom{18}{2}$  pair family of those vertices.



6. Compare the recovered ordered masks with the frozen cut bank and check that no unparsed clause remains.

The analogous budget-five reconstruction replaces equality at five by an at-most-four counter. It has 8238 total variables and 7394 counter clauses in each branch. The three hitting-bank sizes are 287, 15872, and 8704. The executable audit is `routes/finite/check_r3_18_seqcounter.py`; its schema-only mode is part of the lightweight tier, and the complete mode compares the six actual DIMACS blocks during `finite-heavy`.

**G.3. DRAT replay.** Build `drat-trim` from source commit `2e3b2dc0ecf938addbd779d42877b6ed69d9a985`. Verify that the source tree is clean. The repository helper

```
bash scripts/build_drat_trim.sh \
  /absolute/new/checker-directory
```

performs this source checkout and records the resulting binary hash. A fresh platform build need not reproduce the historical executable hash. For each of the twelve compressed assets in Tables 7 and 8, check compressed byte count and SHA-256, test gzip integrity, and decompress into a new temporary directory. Then invoke

```
/path/to/drat-trim formula.cnf proof.drat
```

and require both exit status zero and an exact output line `s VERIFIED`. The checker may print timing diagnostics after that status line. Delete the temporary raw files only after recording the uncompressed hashes.

For the complete package, the repository offers integrated semantic and proof replay:

```
.venv/bin/python reproduce.py finite-heavy \
  --artifact-dir /path/to/release-assets \
  --drat-trim /path/to/drat-trim
```

Time limits may need to be increased on slower hardware. A timeout is UNKNOWN, never a failed or successful proof.

**G.4. Logical audit checklist.** A reviewer can check the mathematical implication independently of the code:

1. the three  $f_i$  are precisely the edges of a seed triangle;
2. every triangle-free final graph omits at least one  $f_i$ ;
3. original nonedges are primary variables but not deletion literals;
4. every installed  $C_S$  is entailed by  $\alpha(F) < 18$ ;
5. omitting uninstalled  $C_S$  clauses makes the CNF weaker, so an UNSAT result remains sufficient;
6. the budget-five refutations force any putative budget-six candidate to have exactly six deletions;
7. after choosing one absent  $f_i$ , exactly five residual deletion literals are true; and
8. checked UNSAT of all three exact-six branches contradicts the model provided by any candidate graph.

## APPENDIX H. FROZEN EXACT-SEVEN FIRST-ROUND RECORD

All three endpoints are UNKNOWN. The branch records are content-addressed as follows.

Table 9: Exact-seven bounded-search records.

| branch | result JSON SHA-256                                                      | checkpoint SHA-256                                                       |
|--------|--------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 0      | 97e777da32b82a83b8e999c9af1f92<br>7c87b67689b8fb72f8b31ded07b39a<br>2f7f | 918dc1cc80860d97b3209f6d969a15<br>6b9df5fb8b4df7f0fa910e72fdc577<br>8098 |
| 1      | 81734a23298ea9c3364a047c3b7b10<br>910bcd44faf9ff55c3a251fd709f6a<br>c64f | 8491cbd5e073170b6f7a503efb3d6a<br>f6c077b6d72defbf9cd6feb6f06e85<br>fc7f |
| 2      | 92e2c948892d6f298e0ca3a5055cf3<br>1cf3eea361b6eee8197ba536922e43<br>7554 | 9e8ea23904378ec1bfc6e4c486fd32<br>f10ccc99c1646855c7589c21cc3c9b<br>9f5c |

The frozen-state checker, its unit tests, and the branch generator have respective SHA-256 identities

**checker:** 8c1f2629193de8b16e018f036ce462b8e18110b9f4f1c13b3162b25fe415d6f4

**tests:** 46b0324d67baaf66bae49027bf4ddbfe3aa08684d3cbe364326bc4400914c987

**generator:** 13a7cfeb564297fcad2ea2c6ca831a6bf4b4755e0223abcf9d1bd1178103d0d  
e

Each branch loaded the common universal bank in under six seconds and then spent approximately 300 seconds in its first solver call. The registered endpoints were

| branch | fixed absent | solver      | endpoint |
|--------|--------------|-------------|----------|
| 0      | (97, 98)     | CaDiCaL     | UNKNOWN  |
| 1      | (97, 99)     | Glucose 4.2 | UNKNOWN  |
| 2      | (98, 99)     | MapleChrono | UNKNOWN  |

The state checker verifies formula dimensions, the universal bank digest, the exact counter truth table, exclusion of original nonedges from the counter, and the three timeout records. It does not transform those timeout records into a satisfiability or unsatisfiability claim.

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